

Part I — Let the Machine Reveal Itself

A first-principles route to the tokamak

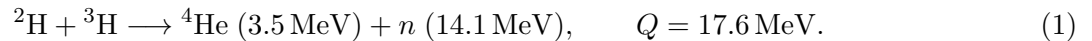
Jony (TUNEM)

Rebuilt opening, August 2026

Promise to the reader. We will not begin by displaying a tokamak and asking you to accept its parts. We will begin with a desired physical event—controlled deuterium–tritium fusion—and let each failure of the simplest attempted design force the next component into existence. Every load-bearing equation will either be derived where it appears or tied to an accessible source. An approximation will be named before it is used.

1. The first requirement: make nuclei approach

The reaction chosen by nearly every near-term fusion-power design is



The released energy is the decrease in rest mass multiplied by c^2 . The products share it in inverse proportion to their masses because momentum is conserved: the lighter neutron carries about four fifths and the alpha particle about one fifth. These numbers will later decide which energy remains in the plasma and which must be caught by a blanket. For now, the obstacle comes before the reward.

A deuteron and triton both carry charge $+e$. At separation r their electrostatic potential energy is

$$U(r) = \frac{e^2}{4\pi\epsilon_0 r}. \quad (2)$$

At $r = 2 \text{ fm}$, a nuclear length scale, Eq. (2) is about 0.72 MeV . A classical particle would need a comparable collision energy to reach that distance. Fusion plasmas operate at much lower thermal energies, typically $T \sim 10\text{--}20 \text{ keV}$, because quantum tunnelling gives some collisions a finite probability of crossing the remaining barrier. Temperature is being measured in energy units: T means $k_B T$. Thus

$$10 \text{ keV} = 10^4 \text{ eV} \quad \Longleftrightarrow \quad T_K = \frac{10^4 \text{ eV}}{k_B} = 1.16 \times 10^8 \text{ K}.$$

This does not mean every ion has the same energy. It means the distribution has a hot tail; fusion occurs disproportionately in that tail. A later section will derive the Gamow peak and the reaction rate. Here we need only the first unavoidable conclusion.

Design consequence. Useful D–T fusion requires matter at roughly one hundred million kelvin. Sustained contact with a material surface would cool and contaminate the plasma, while reactor heat flux would erode and damage the surface. The fuel must therefore be kept in vacuum and away from material walls.

Reaction data and the reactor-design context are reproduced in J. P. Freidberg, F. J. Mangiarotti, and J. Minervini, “Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?” (2015), open manuscript: https://library.psfc.mit.edu/catalog/reports/2010/15ja/15ja021/15ja021_full.pdf.

2. The handle nature gives us: charged particles

At 10 keV, the thermal energy is hundreds of times larger than the 13.6 eV binding energy of ground-state hydrogen. Deuterium and tritium do not remain neutral atoms: electrons are stripped from their nuclei. The result is a *plasma*, containing positive ions and negative electrons.

The plasma is not, as a whole, a charged ball. On length scales large compared with a small electrostatic screening length, its positive and negative charge densities nearly cancel: it is *quasineutral*. The useful fact is subtler. Its constituent particles carry charge, so electromagnetic fields can redirect them without a material surface.

For a particle of mass m , charge q , and velocity $\mathbf{v}$, the Lorentz equation is

$$m \frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (3)$$

Set $\mathbf{E} = 0$ temporarily and take a uniform field $\mathbf{B} = B\hat{\mathbf{z}}$. Dotting Eq. (3) with $\mathbf{v}$ gives

$$\frac{d}{dt} \left(\frac{1}{2} m |\mathbf{v}|^2 \right) = q \mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0.$$

The magnetic force changes direction but not kinetic energy. It is therefore a steering force, not a refrigerator and not a source of free plant power.

Write $w = v_x + iv_y$. The perpendicular components of Eq. (3) become

$$\dot{v}_x = \frac{qB}{m} v_y, \quad \dot{v}_y = -\frac{qB}{m} v_x, \quad \implies \quad \dot{w} = -i \frac{qB}{m} w.$$

Hence $w(t) = w(0)e^{-iqBt/m}$: the perpendicular velocity rotates. Integrating once, the particle follows a circle of radius

$$\omega_c = \frac{|q|B}{m}, \quad r_L = \frac{v_\perp}{\omega_c} = \frac{mv_\perp}{|q|B}. \quad (4)$$

Meanwhile the parallel component obeys $\dot{v}_z = 0$. The full orbit is a helix: a small circle across the field combined with free streaming along it.

For a 10 keV deuteron in 3 T, $v_\perp = \sqrt{2T/m_D} = 9.8 \times 10^5$ m/s and $r_L = 6.8$ mm. A magnetic field can therefore keep transverse excursions small compared with a metre-scale chamber. Equation (4), however, exposes the failure: it does nothing to stop motion along the field.

Design consequence. The “invisible wall” cannot be a wall in the ordinary sense. It must be a set of magnetic field lines that guide charged particles, and those lines must eventually have no open ends.

3. First attempted machine: a straight magnetic corridor

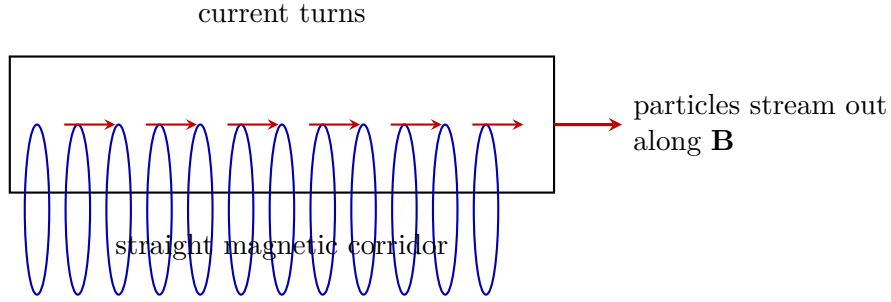
Before closing the ends, we must make a field. Electric current is its source. In a static situation Ampère's law is

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{linked}}. \quad (5)$$

Wind wire into many circular turns around a long cylinder. If there are n turns per unit length, take a rectangular Ampère loop with one long side of length L inside the winding and the other far outside. For an ideal long solenoid, the exterior field is negligible and the two short sides are perpendicular to $\mathbf{B}$. Equation (5) then gives

$$BL = \mu_0(nL)I, \quad \text{so} \quad B \simeq \mu_0 nI. \quad (6)$$

This is our first actual component: a current-carrying coil. Its job is not yet to heat or shape the fuel. Its job is to create the field that makes the gyro-orbits of Eq. (4).



The corridor controls cross-field motion, but Eq. (3) already told us that $v_{\parallel}$ remains free. At the deuteron speed above, a particle crosses a 10 m device in roughly

$$t_{\text{end}} \sim \frac{L}{v_{\parallel}} \sim \frac{10 \text{ m}}{10^6 \text{ m/s}} = 10 \mu\text{s}.$$

Making the straight corridor stronger reduces r_L but does not change this end-loss time. Magnetic mirrors can reflect some pitch angles, but particles within their loss cone still escape. We have learned something more valuable than a failed design: the problem is topological, not merely quantitative.

Design consequence. Do not try to plug the ends of the magnetic corridor. Remove the ends by joining them. The simplest closed version of a tube is a torus.

Model and limits. Equation (6) assumes a solenoid long compared with its radius. End fields and finite-coil corrections are deliberately omitted because the calculation is being used to expose, not conceal, end loss.

4. Closing the corridor creates the torus

Bend the straight magnetic corridor until its two ends meet. The field now runs the long way around a torus; this direction is called *toroidal*. A cross-section of the tube has radius a , while the distance from the symmetry axis to the centre of that cross-section is the major radius R_0 :

$$A \equiv \frac{R_0}{a} \quad (\text{aspect ratio}). \quad (7)$$

These parameters have appeared because the design has acquired two geometrically different lengths, not because a tokamak diagram handed them to us.

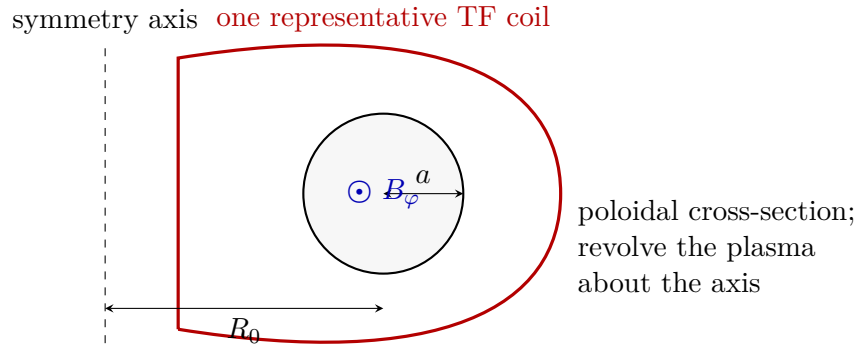
Primary Freidberg parameter — a (minor radius) and R_0 (major radius). Closing the straight corridor makes these two dimensions unavoidable: a measures the cross-sectional reach available to the plasma, while R_0 measures the radius of the closed path that removes end loss. Their ratio $A = R_0/a$ is useful, but is a derived supporting quantity rather than an additional primary parameter.

How do coils make a toroidal field? Arrange current-carrying loops around the torus so that, in the ideal limit of continuously distributed coils, the system is axisymmetric. Apply Ampère's law to a circle of radius R centred on the symmetry axis. The field is tangent to the circle and constant along it, while the loop links total current NI_c :

$$2\pi R B_\varphi(R) = \mu_0 N I_c, \quad \boxed{B_\varphi(R) = \frac{\mu_0 N I_c}{2\pi R}}. \quad (8)$$

Thus the toroidal field is intrinsically stronger on the inboard side and weaker on the outboard side. Across a circular plasma,

$$\frac{B_{\text{in}}}{B_{\text{out}}} = \frac{R_0 + a}{R_0 - a} = \frac{A + 1}{A - 1}. \quad (9)$$



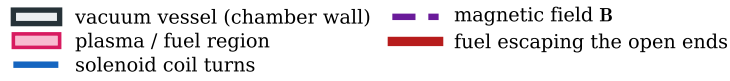
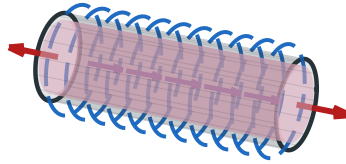
We have now removed the ends. We have not yet built a tokamak. The object presently on the page is only a toroidal magnetic bottle, and Eq. (8) has introduced a new asymmetry: the field strength varies across the tube. The next question is whether this closed field actually confines a particle.

Model and limits. Equation (8) is exact for the axisymmetric continuous-coil idealization. A finite number of real toroidal-field coils produces ripple and a small external stray field; those are later engineering corrections, not erased facts.

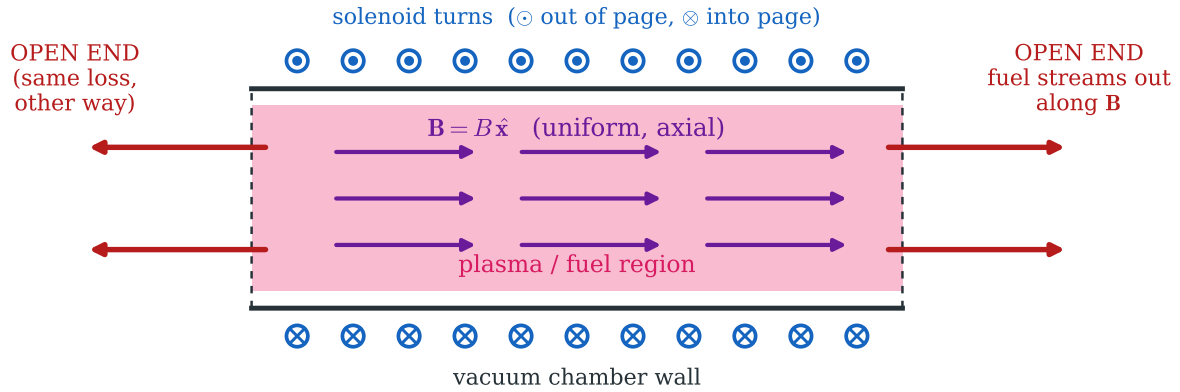
Stage 1 — the straight magnetic corridor

a solenoid makes $B_{\parallel}$; the ends are open, so the fuel leaves

perspective view



longitudinal section



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 1: The first machine state. A solenoid makes the field; the corridor has open ends, which is the failure the next section names.

5. The closed bottle still leaks: toroidal drifts

The gyro-orbit derivation assumed uniform $\mathbf{B}$. Equation (8) makes the field stronger on one side of every orbit than the other, and the field line itself is curved. We now calculate the leading effect, assuming $r_L/R_0 \ll 1$ so the field changes little during one gyro-orbit.

First derive a useful general result. Add a constant force $\mathbf{F} \perp \mathbf{B}$ to Eq. (3), with $\mathbf{E} = 0$. Seek a solution consisting of gyromotion plus a constant velocity $\mathbf{v}_F$. Requiring the constant terms to cancel gives

$$q\mathbf{v}_F \times \mathbf{B} + \mathbf{F} = 0 \quad \Longrightarrow \quad \boxed{\mathbf{v}_F = \frac{\mathbf{F} \times \mathbf{B}}{qB^2}}. \quad (10)$$

The torus supplies two effective forces. Conservation of the magnetic moment $\mu = mv_\perp^2/(2B)$ is not merely asserted: a gyro-orbit is a current loop, so

$$\mu = \left(\frac{q\omega_c}{2\pi}\right) (\pi r_L^2) = \frac{mv_\perp^2}{2B}.$$

When B changes little during one orbit, gyro-averaging makes μ an adiabatic invariant; the orbit therefore has potential energy μB and gradient force $\mathbf{F}_{\nabla B} = -\mu \nabla B$.¹ Following a field line curved with radius R also requires centripetal acceleration; in the frame of the guiding centre this enters as the outward curvature force $\mathbf{F}_c = (mv_\parallel^2/R)\hat{\mathbf{R}}$.

For Eq. (8), $\nabla B = -(B/R)\hat{\mathbf{R}}$. Therefore both effective forces point radially outward. With $\mathbf{B} = B\hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{R}} \times \hat{\boldsymbol{\varphi}} = \hat{\mathbf{Z}}$, Eq. (10) gives

$$\begin{aligned} \mathbf{v}_{\nabla B} &= \frac{mv_\perp^2}{2qBR} \hat{\mathbf{Z}}, \\ \mathbf{v}_c &= \frac{mv_\parallel^2}{qBR} \hat{\mathbf{Z}}, \\ \therefore \quad \boxed{\mathbf{v}_d &= \frac{m}{qBR} \left(v_\parallel^2 + \frac{1}{2} v_\perp^2 \right) \hat{\mathbf{Z}}}. \end{aligned} \quad (11)$$

Equation (11) is fatal to the design in its present form. Positive and negative charges drift in opposite vertical directions because the sign is set by q . The resulting charge separation creates a vertical electric field. Returning to Eq. (3), an electric force $q\mathbf{E}$ inserted into Eq. (10) gives

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}, \quad (12)$$

which is independent of charge and mass. Ions and electrons now move together. For the vertical charge-separation field produced here, that common drift is radially outward, toward the vessel wall.

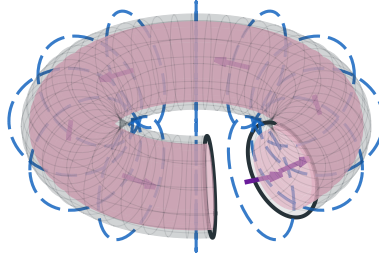
Design consequence. Closing the straight field into a torus removes end loss but creates systematic toroidal drifts. A purely toroidal magnetic field cannot confine the plasma. The field line must be made to visit both the top and bottom of the torus so that the same vertical drift alternates between outward and inward displacement.

¹A systematic open derivation of guiding-centre motion and its ordering assumptions is A. J. Brizard and T. S. Hahm, “Foundations of nonlinear gyrokinetic theory,” <https://arxiv.org/abs/physics/0409119>.

Stage 2 — the corridor is closed into a torus

the same tube bent end-to-end; TF coils now make $B_\varphi \propto 1/R$

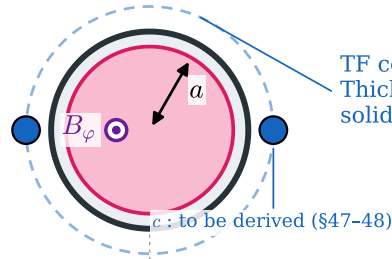
perspective view (vessel cut open)



vacuum vessel (chamber wall)
 toroidal-field (TF) coils
 plasma / fuel region
 magnetic field B

symmetry axis

poloidal cross-section (R - Z plane)



TF coil seen edge-on (it pierces this plane twice). Thickness c : TO BE DERIVED (§47--48). The coil is solid because §4 earned it; only c is still open.

c : to be derived (§47-48)

R_0 major radius

$\odot$: the vector points out of the page

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 2: The corridor closed into a torus. The straight solenoid turns have become toroidal-field coils around the same tube; R_0 , a and B_φ now exist because the geometry acquired two lengths.

6. A second field is forced upon us

A field line confined to one horizontal plane experiences the drift of Eq. (11) with the same radial consequence throughout its journey. Make the line wind the short way around the tube as well as the long way around the torus. It then passes successively through the top and bottom of the cross-section; a vertical displacement is outward on one half and inward on the other. The unbounded drift is replaced, to leading order, by a bounded orbit excursion.

The required short-way field is called *poloidal*. There are two broad ways to make it. External helical windings can do so, producing a stellarator. The tokamak makes the main poloidal field with a toroidal electric current carried by the plasma itself. On a circular cross-section, apply Ampère's law to a poloidal circle of radius r enclosing plasma current $I_{\text{enc}}(r)$:

$$2\pi r B_\theta(r) = \mu_0 I_{\text{enc}}(r), \quad \boxed{B_\theta(r) = \frac{\mu_0 I_{\text{enc}}(r)}{2\pi r}}. \quad (13)$$

The combined field

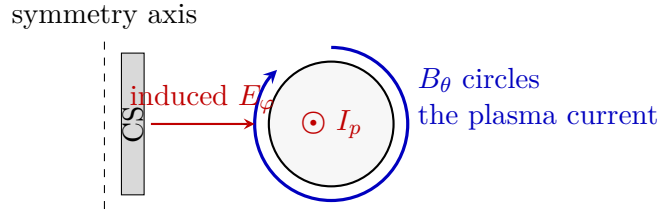
$$\mathbf{B} = B_\varphi \hat{\boldsymbol{\varphi}} + B_\theta \hat{\boldsymbol{\theta}}$$

is helical. A field line no longer remains at one height. The toroidal drift problem has dictated a plasma current, and the plasma current has dictated a helical field.

Where does that current come from? Place a solenoid through the central hole and vary its magnetic flux. Faraday's law,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}, \quad (14)$$

creates a toroidal electric field. The plasma acts as the one-turn secondary winding of a transformer; the induced electric field drives I_p . The central solenoid has therefore not been added because tokamaks traditionally possess one. It has been demanded by the decision to obtain the poloidal field through plasma current.



Design consequence. The machine revealed so far contains a vacuum vessel, toroidal-field coils, a toroidal plasma, and a means of driving plasma current. This is the minimal physical chain that creates the tokamak field. It is not yet a reactor: we have not shown pressure equilibrium, stability, adequate confinement time, heating, exhaust, breeding, magnet survival, or net electricity. Each of those failures will demand its own component.

One distinction will matter later. The plasma current produces the principal poloidal field that makes field lines helical. Separate external poloidal-field coils will be introduced only when radial force balance, position control, and shaping require them. The next derived parameter will be the pitch of the helix—the safety factor q —because the current that repairs drift also creates the machine's first major stability limit.

7. The helix has a pitch: the safety factor q

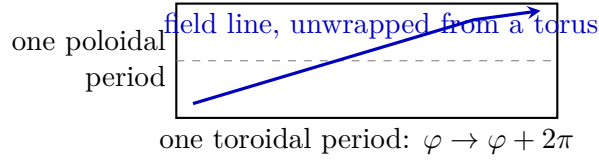
The combined field has two components, so a field line advances two ways at once. On a nearly circular magnetic surface of minor radius r , a short displacement has toroidal length $R d\varphi$ and poloidal length $r d\theta$. Because the displacement is parallel to $\mathbf{B}$, their ratio must equal the corresponding ratio of field components:

$$\frac{R d\varphi}{r d\theta} = \frac{B_\varphi}{B_\theta}. \quad (15)$$

At large aspect ratio the variation of R and B_φ around one cross-section is small enough that $R \simeq R_0$ and $B_\varphi \simeq B_0$. One poloidal circuit, $\Delta\theta = 2\pi$, then advances the field line toroidally by $2\pi q$, where

$$q(r) \equiv \frac{\Delta\varphi}{2\pi} \simeq \frac{r B_0}{R_0 B_\theta(r)}. \quad (16)$$

Thus q is simply the number of long-way circuits made per one short-way circuit. The word *safety* is historical; the stability reason for that name comes later.



Substitute the poloidal field already derived in Eq. (13):

$$q(r) \simeq \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\text{enc}}(r)}, \quad q_a \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p}. \quad (17)$$

The edge value q_a has not been chosen by a convention. It has emerged from geometry, Ampère's law, and the decision to use plasma current for the poloidal field. Reading the same relation backwards gives the equally important current relation

$$I_p \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 q_a}. \quad (18)$$

Any later lower bound on q_a is therefore an upper bound on plasma current. This connection is one of the load-bearing constraints in the later reactor audit.

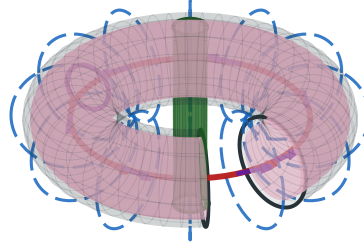
A rational value $q = m/n$, for integers m, n , closes after n poloidal and m toroidal turns. An irrational value never exactly closes and instead samples its surface.

Design consequence. The toroidal field, the plasma current, the size of the torus, and the helix pitch are not independent knobs. Equation (18) ties them together. The current that makes confinement possible has already spent part of the machine's future stability budget.

The shaped generalization and later stability use are given by Freidberg, Mangiarotti, and Minervini (2015), Eq. (30): https://library.psfc.mit.edu/catalog/reports/2010/15ja/15ja021/15ja021_full.pdf.

Stage 3 — the tokamak field: a central solenoid drives I_p
the plasma's own current supplies B_θ , so field lines become helical

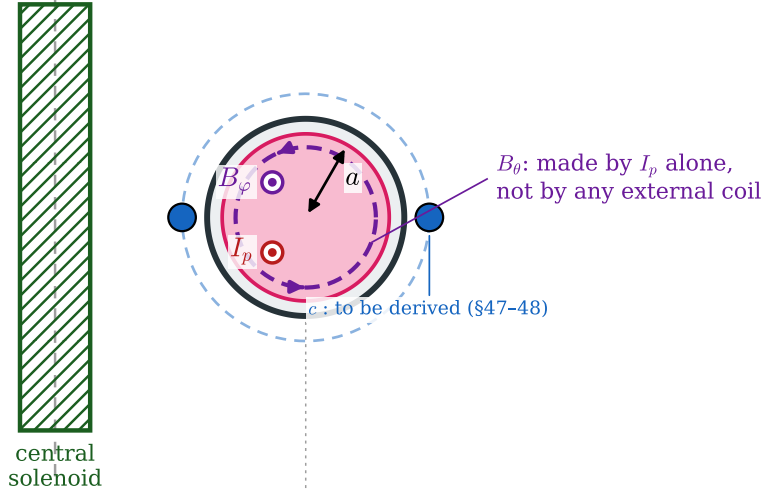
perspective view (vessel cut open)



- | | | | |
|--|------------------------------|--|-----------------------------|
| | vacuum vessel (chamber wall) | | central solenoid (CS) |
| | plasma / fuel region | | magnetic field $\mathbf{B}$ |
| | toroidal-field (TF) coils | | plasma current I_p |

symmetry axis

poloidal cross-section (R - Z plane)



R_0 major radius

$\odot$: the vector points out of the page

Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 3: The tokamak field. A central solenoid drives I_p , and I_p alone supplies B_θ — no external coil does this. Every component of the previous state is retained.

8. A hot plasma is not a passenger: it has pressure

Until now we have followed one particle in fields prescribed from outside. A fusion plasma contains enough particles that its collective push matters. For one ideal gas species, random impacts transfer momentum to a surface and give $p = nk_B T$. We are using energy units for temperature, so this is simply $p = nT$. Electrons and ions each contribute; quasineutrality gives $n_e \simeq n_i \equiv n$, hence

$$p = n_e T_e + n_i T_i \simeq 2nT \quad \text{when } T_e \simeq T_i \equiv T. \quad (19)$$

For orientation, $n = 10^{20} \text{ m}^{-3}$ and $T = 10 \text{ keV}$ give $p \simeq 3.2 \times 10^5 \text{ Pa}$, about three atmospheres. This pressure is inside a near-vacuum vessel, so it tries to expand the plasma.

At scales large compared with a gyro-orbit, it is useful to average the particles into one conducting fluid. Its momentum equation is

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \mathbf{J} \times \mathbf{B}. \quad (20)$$

Here $-\nabla p$ is the force due to pressure and $\mathbf{J} \times \mathbf{B}$ is the magnetic force density. A static equilibrium requires

$\mathbf{J} \times \mathbf{B} = \nabla p.$

(21)

This is a model declaration, not a universal law: it assumes slow evolution, negligible bulk flow, small gyro-orbits compared with the plasma size, and an approximately scalar pressure. Those assumptions will be checked rather than silently inherited when we use the equation quantitatively.

One consequence follows without solving anything. Dot Eq. (21) with $\mathbf{B}$:

$$\mathbf{B} \cdot \nabla p = \mathbf{B} \cdot (\mathbf{J} \times \mathbf{B}) = 0. \quad (22)$$

Pressure is constant along a field line. The invisible wall is therefore not a loose collection of lines: a static confined plasma requires nested magnetic surfaces on which pressure can live.

There is a natural way to compare plasma push with magnetic capability. Build the field of a solenoid slowly. With N turns, length L , area A , and flux linkage NBA , the source supplies $dW = I d(NBA)$. Since $B = \mu_0 NI/L$, integration gives

$$W = \frac{B^2}{2\mu_0} AL. \quad (23)$$

Thus $B^2/(2\mu_0)$ is the magnetic energy per unit volume and has the same units as pressure. Define

$\beta \equiv \frac{p}{B^2/(2\mu_0)}.$

(24)

It is the fraction of magnetic pressure-scale that the plasma is asking the field to hold. Raising B makes a given pressure easier to support; raising p makes the same magnet work harder.

Design consequence. A tokamak cannot be designed by field-line geometry alone. Its magnetic surfaces must also balance pressure. This forces a second task on the coil system: hold the plasma at a chosen radial position while its own pressure and current try to move it.

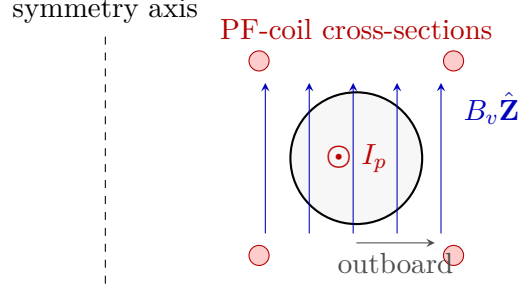
The static-MHD model and the use of β in reactor design are treated in the open Freidberg–Mangiarotti–Minervini manuscript cited on page 1. The field energy relation above is derived from the work needed to establish a solenoidal field.

9. Holding position forces external poloidal-field coils

The plasma current is a toroidal current loop. A current loop in a vertical magnetic field feels a radial force. This is not a slogan; apply the elementary force law $d\mathbf{F} = I d\boldsymbol{\ell} \times \mathbf{B}$ to a circular loop. For $d\boldsymbol{\ell} = R_0 d\varphi \hat{\boldsymbol{\varphi}}$ and $\mathbf{B}_v = B_v \hat{\mathbf{Z}}$,

$$d\mathbf{F} = I_p R_0 B_v d\varphi \hat{\mathbf{R}}, \quad \mathbf{F}_R = 2\pi R_0 I_p B_v \hat{\mathbf{R}}. \quad (25)$$

Changing the sign of B_v changes the force from outward to inward.



The plasma has several sources of radial imbalance: its pressure is higher in the interior; its own poloidal field has a curved, toroidal geometry; and the applied toroidal field varies as $1/R$. Equation (21) says that the sum must cancel, but it does *not* say that a free current ring automatically sits at the desired R_0 . Equation (25) gives the actuator that can correct the residual force.

This finally distinguishes two uses of a poloidal field. The field generated by I_p is the principal field that makes field lines helical and repairs drift. The external vertical component shown here is part of an applied *poloidal-field* (PF) system. Its first job is equilibrium and radial position. The same system will later acquire separate coils for shaping and fast feedback.

We deliberately do not yet quote one “required vertical field” formula. Its exact coefficient depends on the pressure profile, current profile, cross-section, and the plasma’s self-inductance. Giving a single number before those objects have been derived would conceal those assumptions. At this stage physics has rigorously forced the existence and the direction of the control field; the full equilibrium calculation will set its magnitude.

Design consequence. The first coils around the torus made B_φ . The central solenoid drove I_p . Pressure balance now forces a separate external PF-coil system: it must hold the plasma’s position rather than merely create field lines.

10. Flux surfaces turn coil currents into a plasma boundary

To say that coils “hold position” is still too vague. Axisymmetry lets us name the geometric object they control. In cylindrical coordinates (R, φ, Z) , with no φ dependence, $\nabla \cdot \mathbf{B} = 0$ becomes

$$\frac{\partial(RB_R)}{\partial R} + \frac{\partial(RB_Z)}{\partial Z} = 0. \quad (26)$$

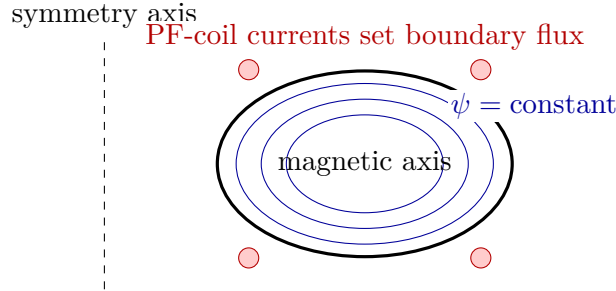
This is the two-dimensional statement that a flow has no sources or sinks. Therefore there is a scalar function $\psi(R, Z)$ whose derivatives generate the poloidal field:

$$B_R = -\frac{1}{R} \frac{\partial \psi}{\partial Z}, \quad B_Z = \frac{1}{R} \frac{\partial \psi}{\partial R}, \quad \mathbf{B}_p = \frac{\nabla \psi \times \hat{\varphi}}{R}. \quad (27)$$

Substituting these two derivatives into Eq. (26) verifies the claim in one line. More importantly,

$$\mathbf{B} \cdot \nabla \psi = 0. \quad (28)$$

The toroidal part is also perpendicular to $\nabla \psi$ because ψ does not depend on φ . Thus a field line stays on a surface $\psi = \text{constant}$. The quantity $2\pi\psi$ is the poloidal magnetic flux enclosed by the toroidal loop at (R, Z) .



The outer closed contour is not just the vessel wall. It is the last closed magnetic surface selected by the plasma current together with the PF-coil currents. The vessel must contain and protect this arrangement, but it cannot replace the magnetic boundary: a conducting wall cannot hold a 10 keV plasma in steady contact.

This is also why shape will matter. Moving PF-coil currents changes the boundary value of ψ , and therefore changes the cross-section of the nested surfaces. A circle is only the first solvable shape, not a physical law. Elongation and triangularity will be introduced only when a specific performance or stability problem demands them.

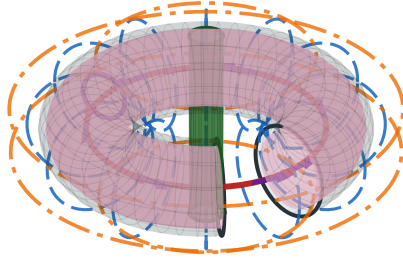
Design consequence. PF coils control flux, and flux contours are the plasma boundary. The emerging machine therefore needs a vacuum vessel outside a deliberately shaped, magnetically defined plasma—not a metal container that touches the fuel.

This axisymmetric flux representation is derived from $\nabla \cdot \mathbf{B} = 0$ above. For an openly available full equilibrium treatment, see Cerfon and Freidberg, “One size fits all” analytic solutions to the Grad–Shafranov equation (2010), [open public manuscript](#).

Stage 4 — equilibrium and position control

external PF coils apply a vertical field B_v to hold the current ring at R_0

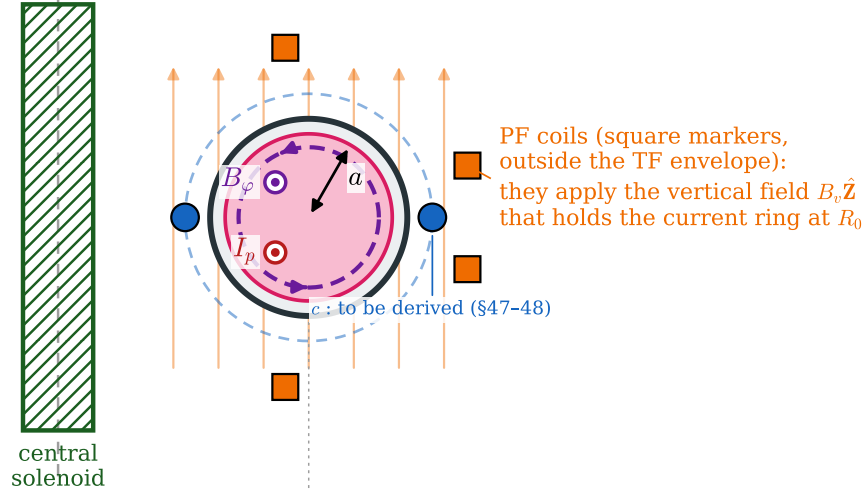
perspective view (vessel cut open)



- | | | | |
|--|------------------------------|--|---------------------------|
| | vacuum vessel (chamber wall) | | poloidal-field (PF) coils |
| | plasma / fuel region | | magnetic field B |
| | toroidal-field (TF) coils | | plasma current I_p |
| | central solenoid (CS) | | |

symmetry axis

poloidal cross-section (R - Z plane)



R_0 major radius

$\odot$: the vector points out of the page

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 4: Equilibrium and position control. External PF coils, drawn as square markers strictly outside the toroidal-field envelope, apply the vertical field that holds the current ring at R_0 .

11. The transformer has a clock

The central solenoid was introduced to drive the toroidal current. Faraday’s law now reveals a price. Around a toroidal loop,

$$V_{\text{loop}} \equiv \oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_{\text{link}}}{dt}. \quad (29)$$

Integrating through a discharge gives

$$\int V_{\text{loop}} dt = -\Delta\Phi_{\text{link}}. \quad (30)$$

The available linked-flux change has units of volt-seconds. A central solenoid can only swing through a finite magnetic flux before its current, field, and mechanical limits are reached. It can start and sustain a pulse, but it cannot supply a steady DC loop voltage forever. A volt-second is exactly a weber: $1 \text{ Vs} = 1 \text{ Wb}$. It is therefore a unit of linked magnetic flux—the transformer’s finite magnetic-flux budget—not a unit of stored energy.

The induced current initially supplies useful heat. The local electrical power density is $\mathbf{J} \cdot \mathbf{E}$; integrating over the plasma gives

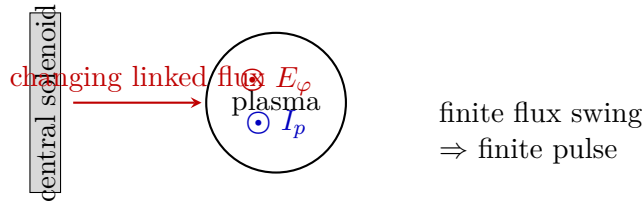
$$P_{\text{ohm}} = \int \mathbf{J} \cdot \mathbf{E} dV \simeq I_p V_{\text{loop}} = I_p^2 R_p. \quad (31)$$

The final form is the lumped-circuit approximation, with plasma resistance R_p . It is useful for the design argument, not a claim that the current density is uniform.

There is a second limit. In a fully ionized, collisional plasma the classical Spitzer resistivity falls approximately as

$$\eta \equiv \frac{E_{\parallel}}{J_{\parallel}} \propto T_e^{-3/2}. \quad (32)$$

Hot electrons collide less effectively, so at comparable current density the plasma becomes a better conductor and ohmic heating weakens. This is a pleasant fact for current flow and an inconvenient fact for trying to heat a reactor plasma by resistance.



Design consequence. The central solenoid has two jobs—start the plasma current and provide early ohmic heat—but neither is an indefinitely scalable heat source. The machine must acquire auxiliary heating, and eventually non-inductive current drive, if it is to operate beyond a transformer pulse.

The $T_e^{-3/2}$ Spitzer scaling and the power-density form ηJ^2 are available openly in the NRL Plasma Formulary: https://www.nrl.navy.mil/Portals/38/PDF%20Files/NRL_Plasma_Formulary_2019.pdf.

12. Reaching fusion temperature forces a heating system

The statement “make the fuel hot” on page 1 can now be made into an energy balance. Each Maxwellian species has thermal energy density $3n_s T_s/2$. For an approximately uniform D–T plasma with equal ion and electron temperatures,

$$W \simeq 3nTV, \quad P_{\text{loss}} \equiv \frac{W}{\tau_E}. \quad (33)$$

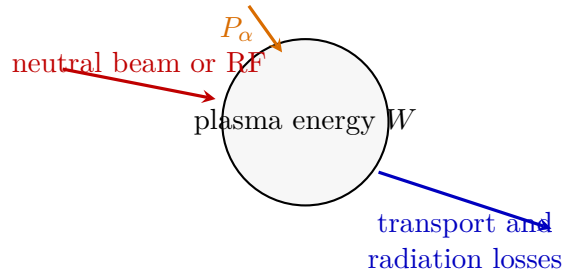
The first relation is energy stored in the plasma. The second *defines* the energy-confinement time τ_E : if all heating were removed, it is the time scale on which the stored energy would be lost at the present loss rate.

The bookkeeping law is therefore

$$\frac{dW}{dt} = P_{\text{aux}} + P_{\alpha} - P_{\text{loss}} - P_{\text{rad}}. \quad (34)$$

At startup P_{α} is small because fusion is small. Auxiliary power must raise W faster than transport and radiation remove it. Later, the 3.5 MeV alpha particles from Eq. (1) are charged and can be magnetically confined; their deposited energy is the desired self-heating term P_{α} . The 14.1 MeV neutrons are neutral and escape, eventually to be captured in a blanket.

Why not simply fire charged particles into the plasma? The same magnetic field that confines the fuel bends an incoming charged beam before it reaches the core. A neutral beam solves that entry problem: neutral atoms cross the field, ionize in the plasma, and then transfer their energy and momentum by collisions. Radio-frequency systems use another handle already present in Eq. (3): an oscillating electric field can transfer energy resonantly to selected charged-particle orbits. These are not ornaments on a completed tokamak. They are responses to the transformer and power-balance limits just derived.



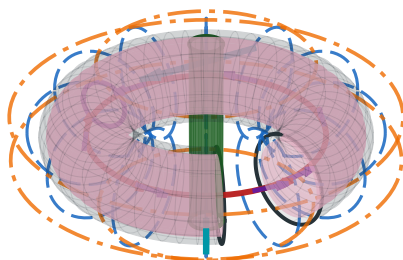
The next question is not yet whether the machine makes net electricity. It is more basic: how much pressure can a magnetic equilibrium and a stable plasma sustain, and how should the boundary be shaped to improve that answer? Those questions will force elongation, triangularity, feedback, and the first genuine stability limits. Only after the ordinary tokamak has been derived on those terms will it make sense to ask whether a spherical tokamak changes the bargain.

Design consequence. Twelve pages in, the machine has emerged component by component: a vacuum environment, toroidal-field coils, a plasma current and central solenoid, external PF coils, and auxiliary heating. What remains absent is just as important: the shaped boundary, exhaust, blanket, shielding, superconducting magnets, and the reactor’s energy ledger must each earn their place through the next physical failure.

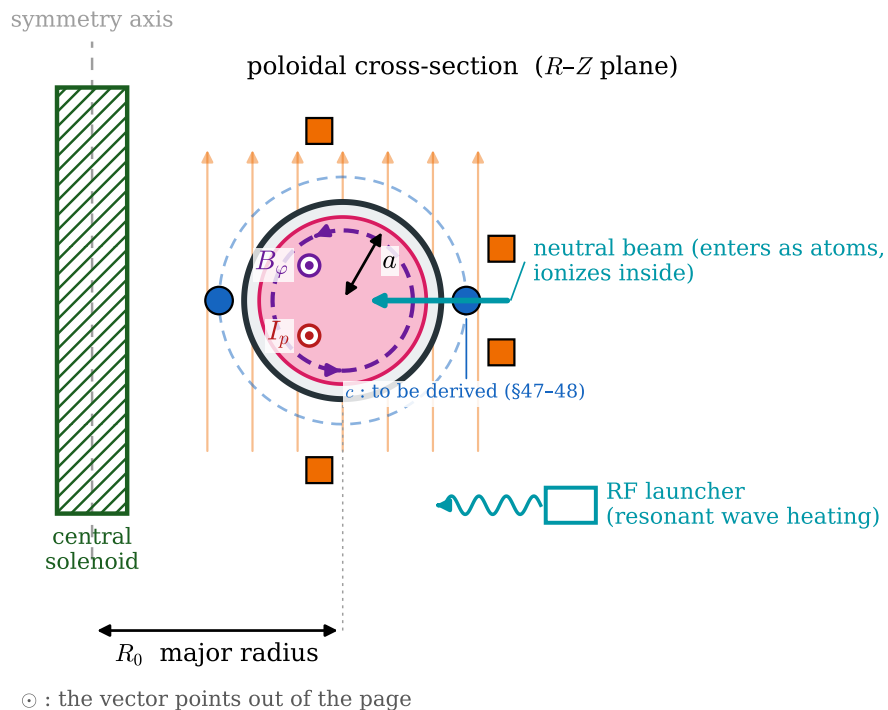
Stage 5 — auxiliary heating

a neutral beam and an RF launcher add power the transformer cannot

perspective view (vessel cut open)



- | | | | |
|---|------------------------------|---|---------------------------|
|  | vacuum vessel (chamber wall) |  | poloidal-field (PF) coils |
|  | plasma / fuel region |  | magnetic field B |
|  | toroidal-field (TF) coils |  | plasma current I_p |
|  | central solenoid (CS) |  | auxiliary heating ports |



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 5: Auxiliary heating. A tangential neutral beam and a radial RF launcher add the power the transformer cannot. No blanket, divertor or shield appears: none has been derived.

13. More room is useful; a taller cross-section supplies it

The circular cross-section used so far was deliberately austere. It makes the geometry easy to see, but it is not the only allowed shape. Suppose the plasma boundary is an ellipse with horizontal semi-axis a and vertical semi-axis κa . The dimensionless number

$$\kappa \equiv \frac{\text{vertical semi-axis}}{\text{horizontal semi-axis}} \quad (35)$$

is called *elongation*. A circle has $\kappa = 1$; a vertically stretched plasma has $\kappa > 1$. Parameterize the boundary by

$$R(\theta) = R_0 + a \cos \theta, \quad Z(\theta) = \kappa a \sin \theta.$$

The cross-sectional area is $\pi(\kappa a)a = \pi\kappa a^2$. Rotating that area once around the major radius R_0 gives, to leading order in a/R_0 ,

$$V = 2\pi R_0 (\pi\kappa a^2) = 2\pi^2 R_0 a^2 \kappa. \quad (36)$$

Thus, at fixed R_0 and a , elongation buys volume linearly. More volume can hold more fuel and more fusion power; later we will state exactly which other quantities must be held fixed before making that comparison.

Primary Freidberg parameter — κ (elongation). Equation (35) changes the plasma volume in Eq. (36) and enters the shaped safety-factor and current limits derived below. It is not decorative geometry: it is a design lever with a stability cost.

The price is immediate. A round set of poloidal-field coils naturally supports a round boundary. An elliptical boundary needs an intentionally nonuniform external field: some coils pull or push more strongly above and below the midplane than at the outboard side. The PF-coil system derived on page 9 therefore acquires a second job. It does not merely keep the plasma centered; it helps choose the boundary shape. This is the first time an earlier component has been asked to do more than solve the failure that first called it into existence.

PLACE IMAGE HERE — Cumulative construction state 5: elongated conventional tokamak. Use the existing toroidal vacuum vessel, TF coils, central solenoid, plasma-current arrow, and external PF coils. Change only the plasma cross-section from circular to vertically elongated, label its semi-axes a and κa , and show that the upper/lower shaping coils are part of the same PF system. This is a construction-state image, not a reset physics sketch. Generate it reproducibly with the project figure workflow.

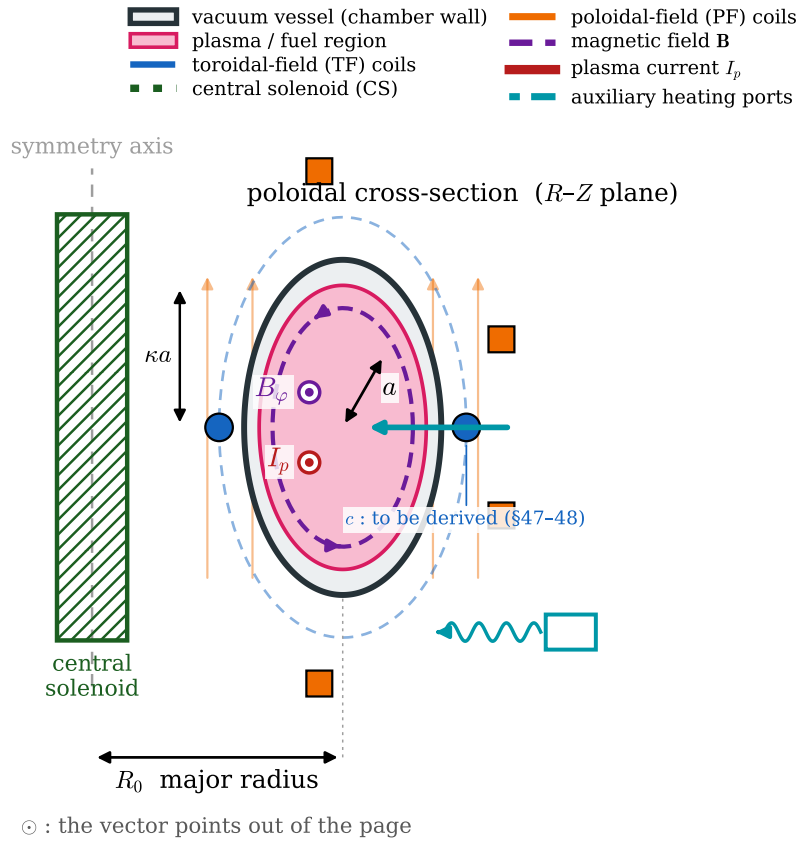
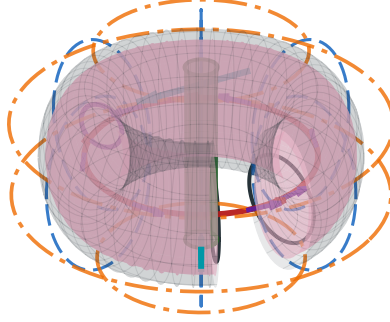
Model and limits. Equation (36) treats the cross-section as an ellipse and the torus as thin enough that its circumference is $2\pi R_0$. Real equilibria also use triangularity and have shifted magnetic surfaces. Those refinements come only after the basic geometric benefit and cost of κ are clear.

Design consequence. If a reactor needs more plasma volume without simply increasing every linear dimension, it has a reason to elongate the cross-section. That choice requires shaping PF coils and raises a new question: is the taller plasma passively stable?

Stage 6 — the boundary is elongated ($\kappa > 1$)

a taller cross-section buys area, hence I_p at fixed q_ ; the coil set follows the same boundary*

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 6: Machine state: elongation. A taller cross-section supplies volume without growing every linear dimension, and the coil set follows the same boundary. Every component of the previous state is retained.

14. A taller plasma can tip: shaping requires feedback

Imagine displacing an elongated plasma a small distance z upward. The magnetic forces from the surrounding coils change. Near a chosen equilibrium height, any smooth vertical force can be expanded as

$$F_Z(z) = F_Z(0) + \left. \frac{\partial F_Z}{\partial z} \right|_0 z + \dots \quad (37)$$

At equilibrium $F_Z(0) = 0$. Newton's law then becomes

$$M\ddot{z} = K_z z, \quad K_z \equiv \left. \frac{\partial F_Z}{\partial z} \right|_0. \quad (38)$$

If $K_z < 0$, Eq. (38) is an oscillator: a displacement creates a restoring force. If $K_z > 0$, its solutions include $z \propto e^{\sqrt{K_z/M}t}$: a small upward displacement grows instead of returning. This elementary sign test is the core of a vertical instability. Elongation generally makes this control problem harder, because a tall current-carrying column has a stronger tendency to move as a whole.

This does *not* make elongation forbidden. It says that its equilibrium cannot be trusted to remain in place by itself. A real machine measures its position with magnetic sensors, compares it with a requested position, and changes currents in selected PF coils. The control system must act before the growing displacement reaches the wall. The causal chain is therefore precise: more useful shape $\rightarrow$ weaker passive vertical stability $\rightarrow$ sensors, power supplies, and fast controlled coils.

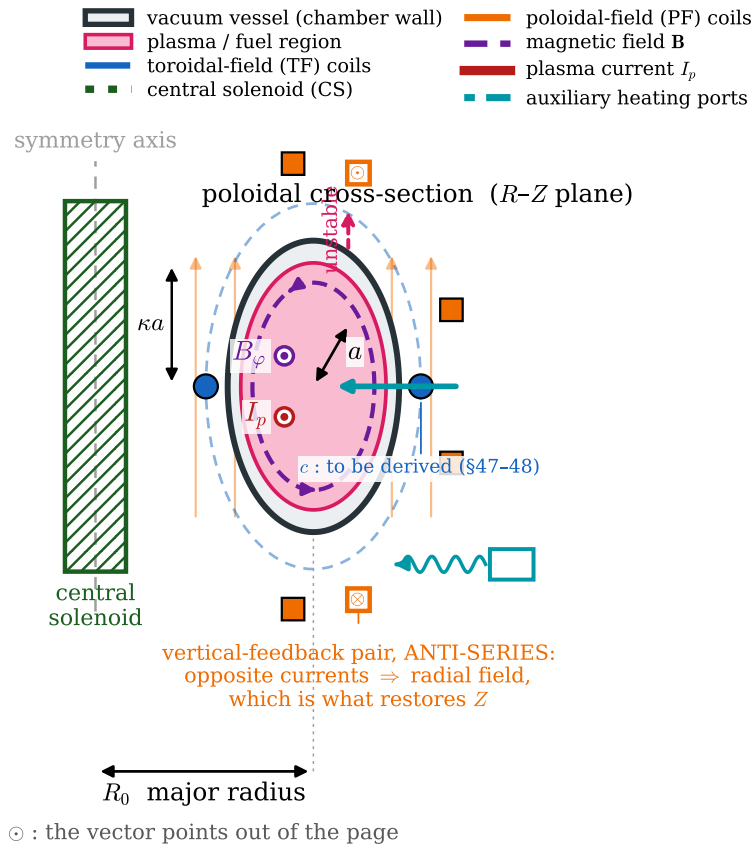
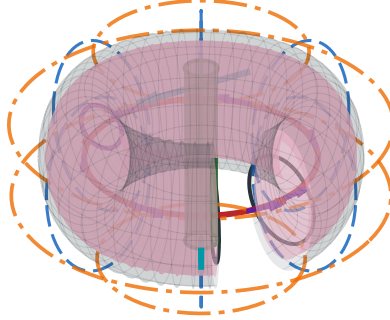
PLACE IMAGE HERE — Cumulative construction state 5B: add the control loop. Start with state 5. Add small magnetic-position sensors outside the vessel, arrows from them to a controller box, and arrows from the controller to two fast vertical-control PF coils. Draw a faint displaced plasma outline and label it $+z$; keep all previously derived components visible. The purpose is to show an engineering system forced by Eq. (38), not to imply that the controller itself is a magnetic component.

Model and limits. Equation (38) is a one-coordinate linear model. Actual vertical control couples plasma shape, conducting structures, coil circuits, and sensor delays. It is used here only to establish the sign logic that makes feedback necessary. For an openly available treatment of elongation and vertical stability, see E. Cerfon, J. P. Freidberg, and J. P. Vesey, “Tokamak elongation: how much is too much? Part 1,” <https://www.cambridge.org/core/services/aop-cambridge-core/content/view/1ABC7975053E19C0A93D9952CBB53102/S0022377815001270a.pdf>.

Design consequence. A shaped tokamak is not merely coils plus plasma. It includes a closed feedback system. The allowed elongation is set jointly by the gain in volume, the available PF-coil authority, nearby conducting structures, and the speed and reliability of active control.

Stage 7 — elongation is unstable, so it must be fed back
an anti-series coil pair supplies the radial field that opposes vertical displacement

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 7: Machine state: vertical control. The pair is wired *anti-series* ($\odot$ above, $\otimes$ below): opposite currents make the combined field radial on the midplane, which is what restores Z . A same-sign pair would only move the plasma in R .

15. The same geometry changes how much plasma current is allowed

We already derived a safety factor for a simple circular plasma. It counted, in effect, how far a field line moves toroidally while making one poloidal circuit. The local version at minor radius r is

$$q(r) = \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\text{enc}}(r)}, \quad I_{\text{enc}}(r) = 2\pi \int_0^r j_\varphi(r') r' dr'. \quad (39)$$

Here j_φ is the toroidal current density and I_{enc} is the current inside the selected magnetic surface. The first relation follows from the same field-line ratio used before, $q \simeq r B_\varphi / (R_0 B_\theta)$; the second is the area integral of current density, and Ampère's law gives $B_\theta \simeq \mu_0 I_{\text{enc}} / (2\pi r)$.

An elongated boundary alters this simple geometry. A widely used large-aspect-ratio, elongated-plasma estimate is

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p} \frac{1 + \kappa^2}{2}. \quad (40)$$

The first fraction is the circular edge estimate. The factor $(1 + \kappa^2)/2$ records that one poloidal circuit is no longer circular. Equation (40) is not being introduced as a mysterious correction: it is the next-order geometry that becomes necessary once we deliberately choose $\kappa \neq 1$.

For the simple global kink constraint used in the reactor-design model, require $q_* > q_K$, where $q_K \simeq 2$. Solving Eq. (40) for I_p gives a current ceiling,

$$I_p < \frac{2\pi a^2 B_0}{\mu_0 R_0 q_K} \frac{1 + \kappa^2}{2}. \quad (41)$$

The direction of the result is instructive. More toroidal field B_0 , more minor cross-section, or more elongation permits more current under this particular limit; larger major radius R_0 lowers the permitted current when the other variables are held fixed.

Primary Freidberg parameter — q_* and I_p . Equation (40) is the shaped global safety factor used by Freidberg's reactor model. Equation (41) turns its stability requirement into a quantitative current constraint. Together they couple geometry (R_0, a, κ) , field (B_0) , and the plasma current (I_p) .

Model and limits. The estimate in Eq. (40), and the $q_K \simeq 2$ limit, are the large-aspect-ratio design closure adopted in Freidberg, Mangiarotti, and Minervini (2015), Eqs. (30) and (39), open manuscript cited on page 1. A full stability boundary depends on profiles, wall response, and feedback. We use the explicit model before asking it to carry any design conclusion.

Design consequence. Elongation is valuable twice in this reduced model: it increases volume and relaxes a particular current limit. But it also created the feedback demand on the preceding page. A good parameter cannot be optimized one equation at a time.

16. A single safety factor is not enough: the current profile matters

Equation (39) contains more information than the edge number q_* . If two plasmas have the same total I_p but different functions $j_\varphi(r)$, then $I_{\text{enc}}(r)$ and therefore $q(r)$ differ throughout the interior. The machine cannot treat its current as a single knob when the plasma's magnetic geometry depends on where that current flows.

The reason is already visible in the field-line picture. When $q = m/n$ for integers m and n , a field line closes after m poloidal and n toroidal circuits in the idealized description. Such rational surfaces are not automatically harmful, but perturbations can accumulate coherently there. That is why later stability discussions specify a *profile* such as $q(0)$ and $q(a)$, rather than only a total current.

There is also a practical lesson. The transformer acts on current induction, while auxiliary systems can add or reshape current through other mechanisms. The sources of current and the resulting profile must both be accounted for in any steady-state claim. At this stage we need not choose a detailed profile. We need to preserve the variable that the later stability and bootstrap-current arguments will require.

Here bootstrap current means a pressure-gradient-driven share of the toroidal plasma current. Its orbit-and-collision mechanism is derived on pages 36–37, before any numerical bootstrap fraction is used.

Supporting quantity — $q(r)$ and $j_\varphi(r)$. The radial safety-factor profile in Eq. (39) carries the current-distribution information that a single q_* loses. Later closure arguments will need it to distinguish total current from the fraction supplied by the plasma itself or by external drive.

PLACE IMAGE HERE — Cumulative construction state 6: shape and profile-control hardware. Start with the elongated, feedback-controlled machine. Add only a clear callout that the central solenoid and auxiliary current-drive/heating ports influence the toroidal-current profile $j_\varphi(r)$, alongside a small inset of nested flux surfaces labelled $q(r)$. Do not draw a spherical tokamak or a finished power plant. This image should make the distinction between total I_p and profile $j_\varphi(r)$ legible.

Design consequence. The next design calculations must carry both a global current I_p and a profile $q(r)$. That added bookkeeping is not mathematical fussiness: it records which field lines and which stability limits the plasma actually contains.

Stage 7a — total current is not the same as its profile

the solenoid and the launchers do not just set I_p ; they set where the current sits, $j_\varphi(r)$, and that is what fixes $q(r)$

perspective view (vessel cut open)

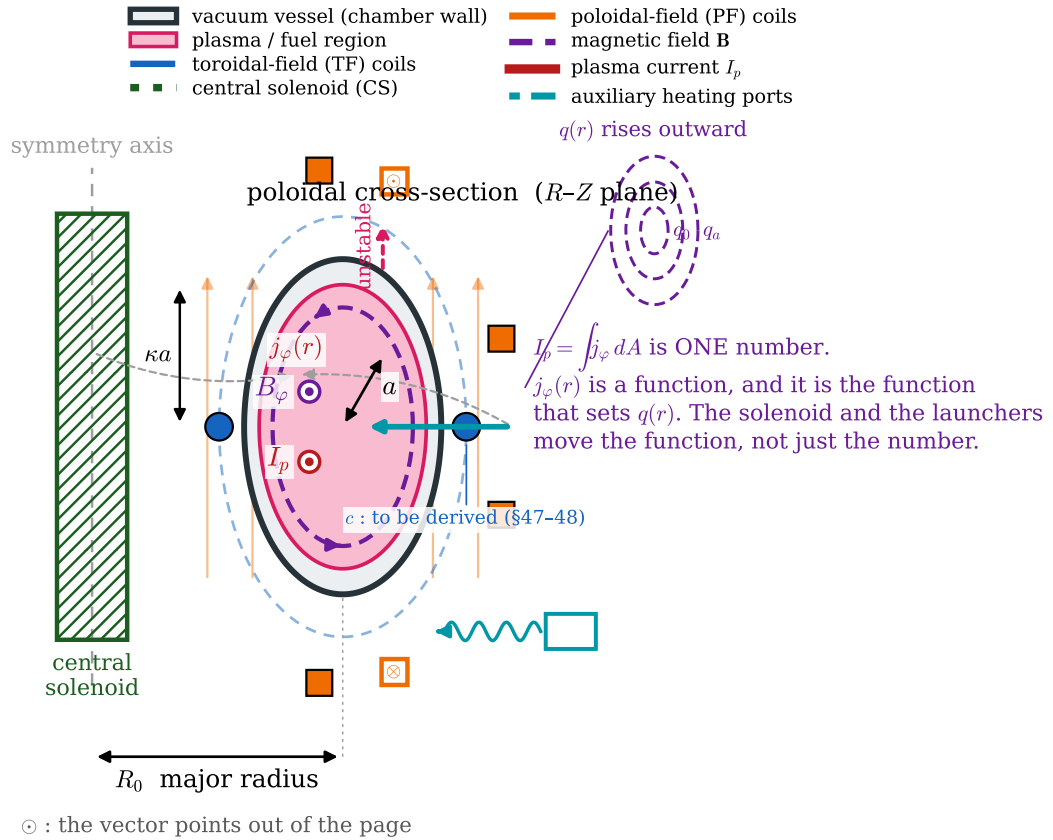
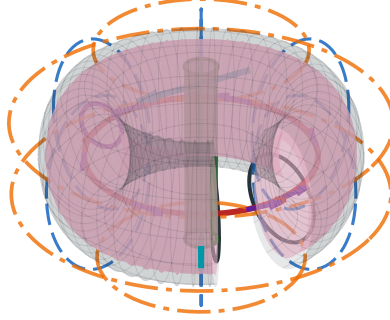


Figure 8: Machine state: the current profile. Total current and its distribution $j_\varphi(r)$ are different quantities, and it is the profile that fixes $q(r)$. No new hardware appears — this is an annotation layer.

17. Why field strength and volume dominate the first performance estimate

A fusion reaction needs one deuterium ion and one tritium ion. In a roughly equal D–T mixture, the number of reaction opportunities per unit volume is proportional to n^2 , where n is a representative fuel density. At a fixed useful temperature, the reactivity $\langle\sigma v\rangle$ is also fixed, so

$$P_{\text{fus}} \propto n^2 V. \quad (42)$$

This is a scaling statement, not yet a reactor-power calculation: the missing proportionality constant contains the reaction energy and $\langle\sigma v\rangle(T)$.

For a fully ionized, equal-temperature D–T plasma, ions and electrons both contribute to pressure, so $p \simeq 2nT$. The beta definition from page 8 gives $p = \beta B_0^2/(2\mu_0)$ in this reduced scaling model. Therefore, at fixed T ,

$$n \propto \beta B_0^2, \quad P_{\text{fus}} \propto \beta^2 B_0^4 V \propto \beta^2 B_0^4 R_0 a^2 \kappa. \quad (43)$$

The fourth power of field is the memorable point. If all the stated assumptions hold, raising B_0 slightly has a much larger effect on this estimate than raising it by the same percentage appears to have at first glance. That is a motivation, not permission, to choose an arbitrarily large field: coils, stresses, superconductors, plasma limits, and the central stack will later impose the corresponding prices.

Primary Freidberg parameter — B_0 and β . Equation (43) is the first bridge from plasma equilibrium to reactor output. It explains why the toroidal field B_0 , normalized pressure β , and shaped volume $2\pi^2 R_0 a^2 \kappa$ recur throughout Freidberg’s design closure. The equation is a conditional scaling law; temperature, confinement, and stability have not been eliminated.

Model and limits. This page holds T , fuel mix, and $\langle\sigma v\rangle$ fixed, identifies the confining field with B_0 , and ignores profile factors. The actual D–T reactivity varies strongly with temperature. The purpose is to derive the dependence that remains once a target operating temperature has been chosen, not to replace a reactivity model.

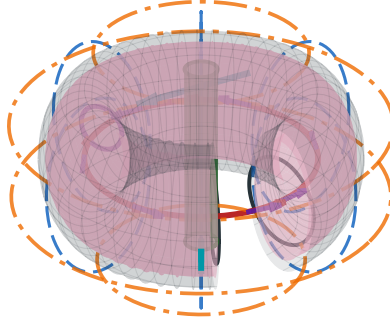
PLACE IMAGE HERE — Cumulative construction state 7: performance quantities attached to the partly revealed machine. Use the stage-6 conventional tokamak. Add unobtrusive labels for R_0 , a , κ , B_0 , and I_p on the actual machine components, plus a small side panel showing the derived chain $P_{\text{fus}} \propto n^2 V \propto \beta^2 B_0^4 R_0 a^2 \kappa$. The figure must retain all previously introduced components and make clear that this is a scaling law, not a claim of net electrical power.

Design consequence. The machine now has a reason to seek strong field, high allowed pressure, and useful volume. The remainder of the thesis must show whether these desires can coexist with current limits, confinement, materials, coil technology, and power balance.

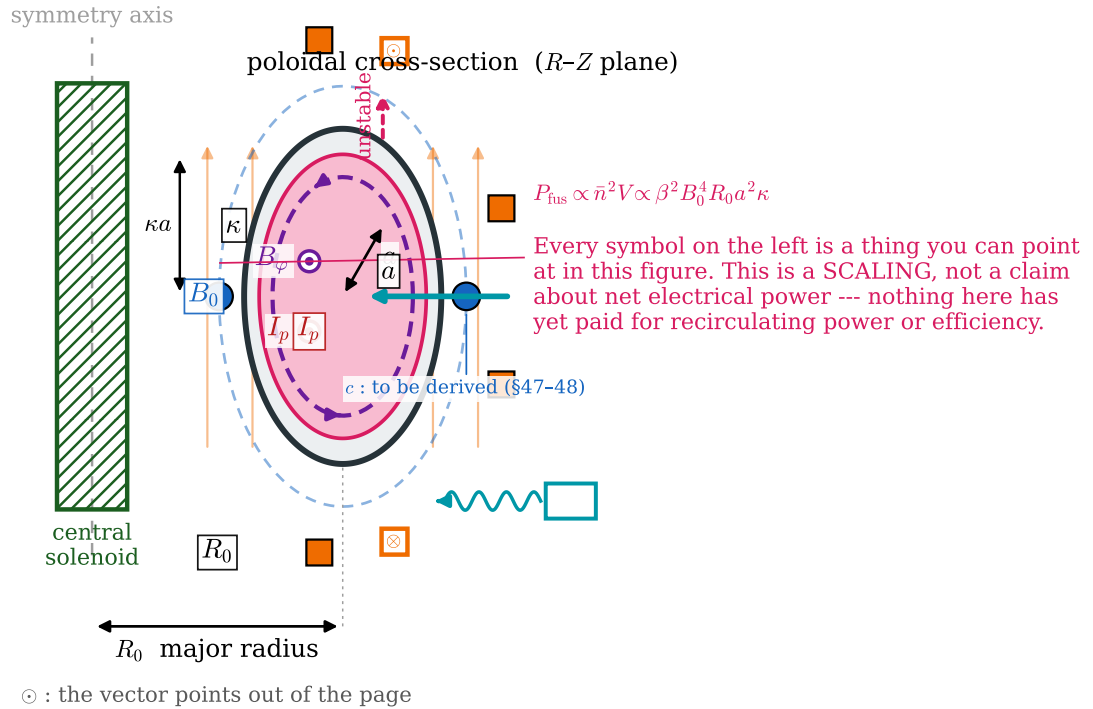
Stage 7b — the performance quantities live on the machine

every symbol in the fusion-power scaling is a feature you can point at

perspective view (vessel cut open)



- | | |
|------------------------------|-----------------------------|
| vacuum vessel (chamber wall) | poloidal-field (PF) coils |
| plasma / fuel region | magnetic field $\mathbf{B}$ |
| toroidal-field (TF) coils | plasma current I_p |
| central solenoid (CS) | auxiliary heating ports |



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 9: Machine state: the performance quantities, attached to the machine. Every symbol in the fusion-power scaling is a feature that can be pointed at.

18. A ledger before optimization: what has actually been derived?

These pages have built a ledger of demands rather than selected a tokamak by declaration. It prevents improving one formula while silently violating another.

Quantity	What it measures	Why it cannot be chosen alone
$R_0, a, A = R_0/a$	Overall and cross-sectional size	Set geometry, volume, and aspect ratio.
B_0	Toroidal confining field	Shrinks gyro-orbits; appears as B_0^4 in the conditional power scaling.
I_p	Toroidal plasma current	Creates poloidal field, but is bounded by the safety-factor constraint.
$q(r), q_*$	Field-line pitch and its shaped global measure	Depend on current distribution as well as total current.
κ	Vertical elongation	Increases volume and the reduced I_p allowance; requires shaping and feedback.
n, T, p, β	Fuel state and magnetic-pressure fraction	Set reaction opportunity and challenge confinement and stability.
τ_E	Energy-confinement time	Decides whether heating can outrun losses; it remains to be derived and modeled.

The drawing now has earned components: vacuum and magnetic field for hot fuel; a torus to remove open ends; a solenoid and plasma current for the poloidal field; PF coils for position and shape; feedback for vertical stability; and heating/current-drive ports because the transformer has a finite clock.

PLACE IMAGE HERE — Construction-atlas checkpoint. Make a one-page cumulative annotated cross-section and toroidal cutaway of the conventional tokamak derived so far: vacuum vessel; plasma; TF coils and B_φ ; central solenoid and induced I_p ; PF/shaping coils; elongated boundary with $a, \kappa a$, and R_0 ; magnetic sensors and feedback; heating/current-drive port. Number each component by the page where its physical need was derived. Leave out divertor, blanket, shielding, superconducting magnets, and all spherical-tokamak geometry: those have not yet earned their place.

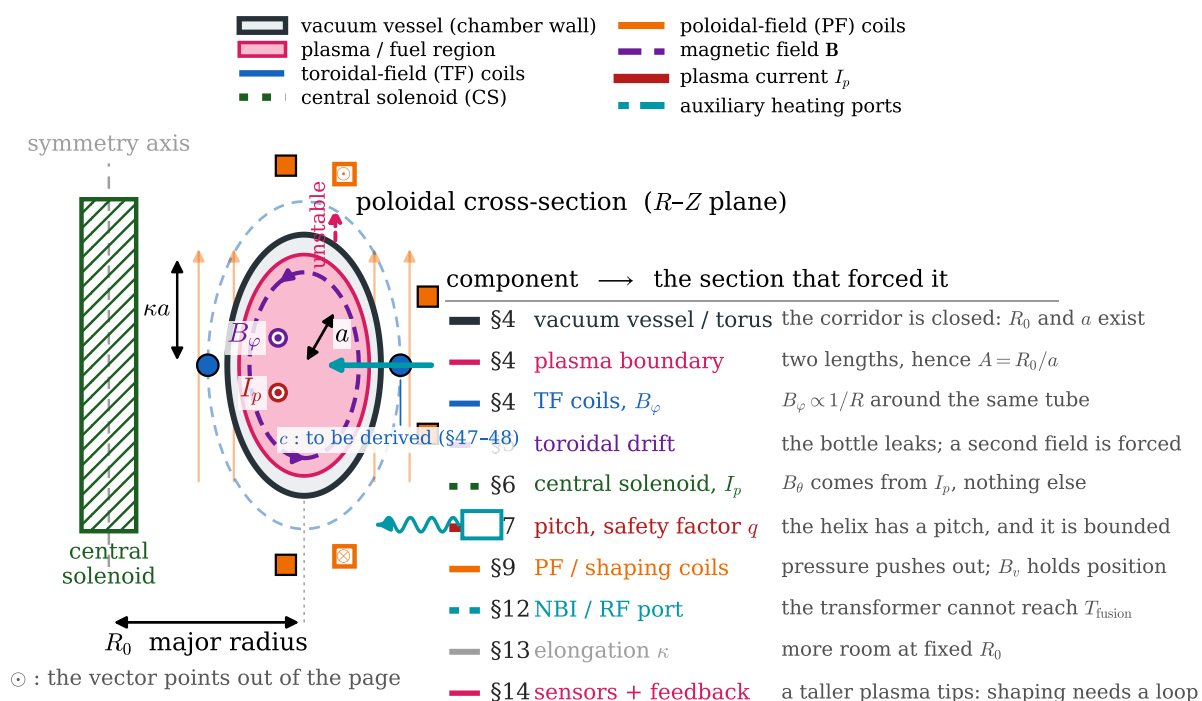
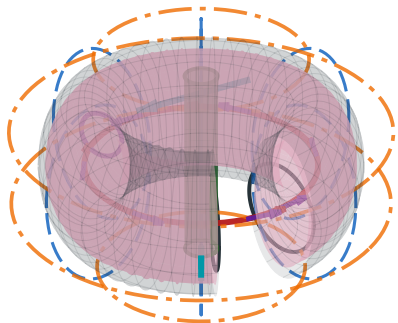
Supporting quantity — first coupled ledger. The variables in this ledger are the minimum state carried forward into the Freidberg-style closure problem. A proposed parameter set is acceptable only when the equations that constrain *all* of them are checked together.

Design consequence. The ordinary tokamak is now visible, but only as the accumulated answer to specific physical failures. The next section can add the remaining reactor subsystems in the same discipline: name the failure, derive the demand, then add the component and its variables to the ledger.

Atlas checkpoint — what has actually been derived so far

every component on the machine at this point, numbered by the section that forced it. Nothing later appears.

perspective view (vessel cut open)



NOT here, because not yet derived: triangularity, alpha heating, the neutron boundary, blanket, shield, superconductors, divertor, and every spherical-tokamak feature. A component enters this table only after a section has forced it.

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 10: Atlas checkpoint. Every component derived so far, numbered by the section that forced it. Nothing later than §18 appears: a part enters this table only after a section has made it necessary.

19. The next shape parameter: a D rather than an ellipse

Elongation says how tall the cross-section is. It does not say whether its upper point lies above the geometric centreline or leans toward the central hole. To name that second feature without drawing a machine first, take the boundary

$$R(\theta) = R_0 + a \cos(\theta + x \sin \theta), \quad Z(\theta) = \kappa a \sin \theta, \quad x = \arcsin \delta. \quad (44)$$

This is a useful *coordinate chart* for a family of closed curves, not a law of nature. At the top, $\theta = \pi/2$, Eq. (44) gives

$$R_{\text{top}} = R_0 + a \cos(\pi/2 + x) = R_0 - a \sin x = R_0 - \delta a.$$

Thus

$$\delta = \frac{R_0 - R_{\text{top}}}{a} \quad (45)$$

is the normalized inward displacement of the top point. Positive δ makes the familiar D-like cross-section; $\delta = 0$ returns the ellipse of page 13.

For small δ , we can see exactly which new geometric pattern it adds. Use $\cos(\theta + x \sin \theta) = \cos \theta - x \sin^2 \theta + O(x^2)$ and $\sin^2 \theta = (1 - \cos 2\theta)/2$. Since $x = \delta + O(\delta^3)$,

$$R(\theta) = R_0 - \frac{a\delta}{2} + a \cos \theta + \frac{a\delta}{2} \cos 2\theta + O(\delta^2). \quad (46)$$

The first cosine describes the ordinary left-right radius. Triangularity is, to leading order, a second angular harmonic plus a small centering shift. That is why it needs more than the simple shaping action which made the ellipse.

Supporting quantity — δ (*triangularity*). Equation (45) defines the D-shape parameter carried in reactor equilibria. Equation (46) shows that it is an independently adjustable geometric feature, not another name for elongation. Its stability benefit, when present, must be established separately from this definition.

PLACE IMAGE HERE — Cumulative construction state 8: add triangularity.

Start with the elongated, feedback-controlled conventional tokamak. Replace the ellipse in the cross-sectional inset with two labelled boundaries, $\delta = 0$ and $\delta > 0$, keeping the same a , κa , PF coils, and central solenoid. Mark the inward shift δa of the top point. The image must make clear that triangularity changes plasma shape, not the shape of the toroidal-field coils.

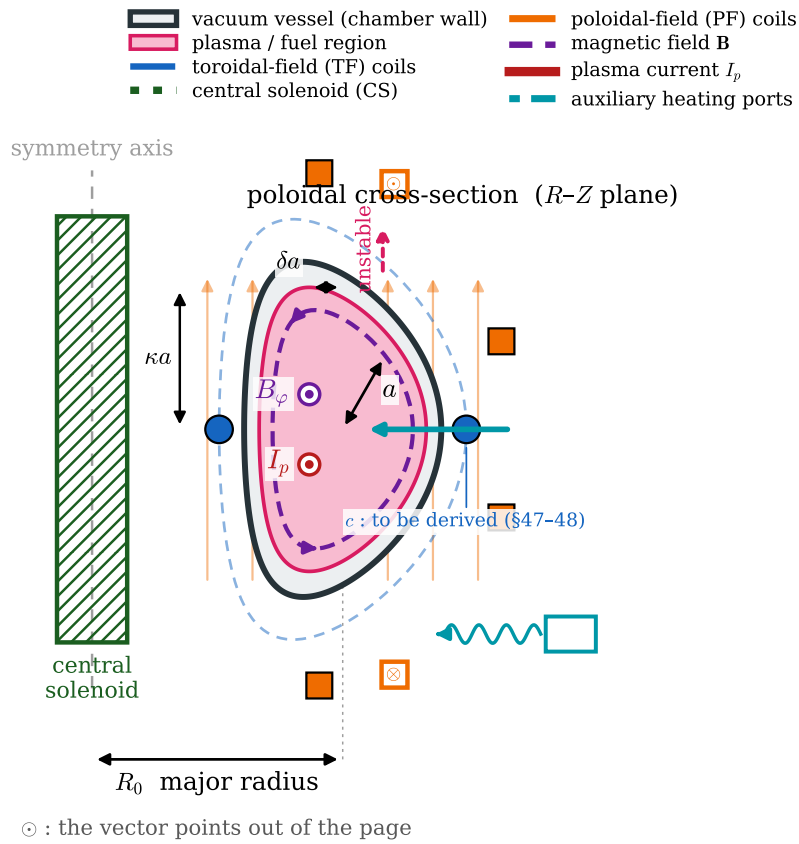
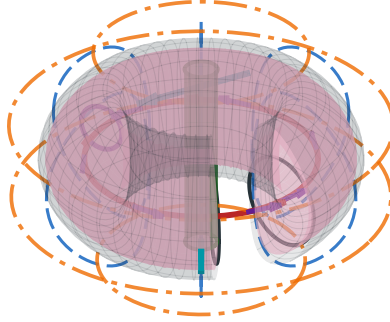
Model and limits. Equation (44) is the Miller local-equilibrium shape parameterization. Its role here is definitional and geometric; it is not derived from Maxwell's equations. A freely available analytic-equilibrium treatment by Cerfon and Freidberg is cited on page 10.

Design consequence. A D-shaped boundary creates a new demand on the PF system: it must produce another independently controllable field pattern. The machine has not gained a new species of coil, but the coil set has gained another job.

Stage 8 — triangularity ($\delta > 0$) is added

the outboard corners are pulled apart; the machine is now the shaped conventional tokamak

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 11: Machine state: triangularity. A D rather than an ellipse. No new species of coil appears — consistent with the text, the existing PF set has simply gained another job.

20. Why more shape means more carefully placed PF coils

Outside the plasma there is no plasma current density. In that region, magnetic field lines from PF coils satisfy the source-free magnetostatic equations

$$\nabla \times \mathbf{B} = 0, \quad \nabla \cdot \mathbf{B} = 0.$$

The first equation permits a scalar magnetic potential, $\mathbf{B} = -\nabla\Phi$; inserting it into the second gives $\nabla^2\Phi = 0$. This is Laplace's equation. Near the plasma centre, its smooth solutions can be organized by angular pattern.

The first three patterns have an immediate physical reading. A nearly uniform vertical field translates the plasma radially. A field whose horizontal component grows with height and whose vertical component grows with radius is a quadrupole: it stretches or squeezes the boundary into an ellipse. The next pattern, a hexapole, has the angular structure needed for the leading $\cos 2\theta$ term in Eq. (46). These are not three unrelated inventions. They are the first terms in one local expansion of the externally applied field.

A simple two-dimensional example makes the hierarchy visible. The harmonic functions

$$\Phi_1 = C_1 r \sin \theta, \quad \Phi_2 = C_2 r^2 \sin 2\theta, \quad \Phi_3 = C_3 r^3 \sin 3\theta$$

all solve $\nabla^2\Phi = 0$ in polar coordinates. Taking $-\nabla\Phi$ turns their successive angular structures into the uniform, quadrupole, and hexapole-like field patterns just described. A coil system chooses the coefficients C_1, C_2, C_3 by its locations and currents. The exact tokamak geometry is toroidal rather than this flat example, but the local lesson survives: more detailed boundary shape requires more independent field moments.

PLACE IMAGE HERE — Physics sketch: the external-field harmonic ladder.

Make a separate three-panel line drawing, not a cumulative machine view. Panel 1: uniform vertical field translating a circular plasma. Panel 2: quadrupole field stretching it into an ellipse. Panel 3: hexapole-like shaping field producing a D-like boundary. Label the patterns C_1, C_2, C_3 and state that real PF-coil geometry is toroidal. Generate from a reproducible script.

Design consequence. The phrase shape control hides a finite-dimensional control problem. A reactor needs enough independently powered PF circuits, appropriately located, to control position, elongation, and triangularity while the plasma current and pressure change.

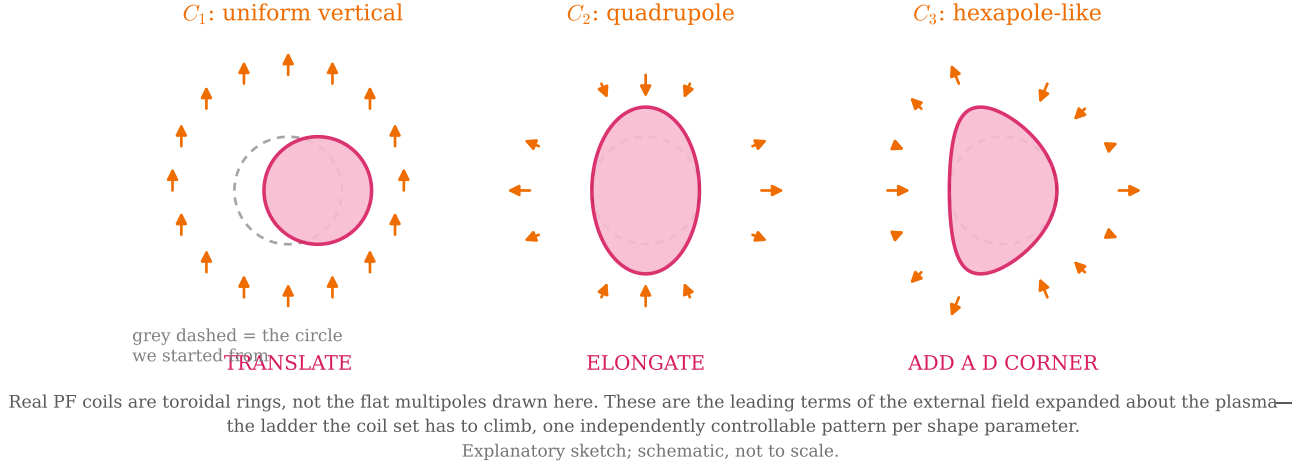


Figure 12: Physics sketch: the external-field harmonic ladder. *Idealized schematic; not a fabrication drawing.*

21. The plasma as a fluid: the equilibrium equation

The single-particle Lorentz force was enough to explain gyro-orbits and field-guided motion. To decide the shape of a plasma containing an enormous number of particles, we add their forces per unit volume. Let ρ_m be mass density, $\mathbf{u}$ the bulk velocity, p the scalar pressure, and $\mathbf{j}$ the current density. Momentum conservation for a small fluid element is

$$\rho_m \frac{d\mathbf{u}}{dt} = \mathbf{j} \times \mathbf{B} - \nabla p. \quad (47)$$

The magnetic term is simply the Lorentz force $q\mathbf{v} \times \mathbf{B}$ summed over charged particles per unit volume. The pressure term points from high pressure toward low pressure: a small box has slightly greater inward force on its high-pressure face than on its low-pressure face.

For the slowly changing equilibrium that defines the intended plasma boundary, set the bulk acceleration to zero. Equation (47) becomes

$$\nabla p = \mathbf{j} \times \mathbf{B}. \quad (48)$$

This one equation has an exceptionally concrete reading. At every location, magnetic force density must exactly push back against the local pressure gradient. If it does not, the plasma moves; the supposed boundary is not an equilibrium.

Dot Eq. (48) with $\mathbf{B}$. Since $\mathbf{B} \cdot (\mathbf{j} \times \mathbf{B}) = 0$, we obtain

$$\mathbf{B} \cdot \nabla p = 0.$$

Pressure is therefore constant along each magnetic field line in this equilibrium model. When field lines densely cover a nested magnetic surface, pressure is constant on that surface too. This is the precise reason flux surfaces became more than a drawing convention on page 10: they are also surfaces of constant pressure.

Supporting quantity — $p(\psi)$. In an axisymmetric equilibrium, pressure becomes a function of the flux-surface label ψ . This is one of the inputs to the Grad–Shafranov equation and later controls β , fusion power density, and pressure-driven stability.

Model and limits. Equation (47) assumes a single fluid with isotropic pressure and neglects viscosity, kinetic orbit effects, and rapid flows. Those effects matter in detailed equilibrium reconstruction; this model is the minimum one needed to derive the ordinary tokamak equilibrium equation.

Design consequence. A plasma boundary is not a material container. It is a self-consistent solution of pressure, plasma current, and externally produced field. The next pages derive the equation that tests that consistency.

22. A flux function compresses the axisymmetric geometry

Use cylindrical coordinates (R, φ, Z) and assume axisymmetry: $\partial/\partial\varphi = 0$. The toroidal component of the magnetic field is written as $B_\varphi = F(R, Z)/R$. For the poloidal part, define a scalar $\psi(R, Z)$ by

$$\mathbf{B}_p = \frac{1}{R} \nabla \psi \times \hat{\varphi}. \quad (49)$$

Writing out the cross product gives

$$B_R = -\frac{1}{R} \frac{\partial \psi}{\partial Z}, \quad B_Z = \frac{1}{R} \frac{\partial \psi}{\partial R}.$$

Substitution into $\nabla \cdot \mathbf{B}_p = R^{-1} \partial_R(RB_R) + \partial_Z B_Z$ shows the two mixed derivatives cancel. Thus Eq. (49) automatically satisfies $\nabla \cdot \mathbf{B}_p = 0$.

A curve of constant ψ has tangent displacement $d\ell$ satisfying $\nabla \psi \cdot d\ell = 0$. Equation (49) also gives $\mathbf{B}_p \cdot \nabla \psi = 0$. Hence the poloidal field is tangent to the curves $\psi = \text{constant}$; rotating such a curve around the symmetry axis creates one magnetic surface. The symbol ψ is therefore a surface label. Up to a conventional factor of 2π , it is also the poloidal magnetic flux enclosed by that surface.

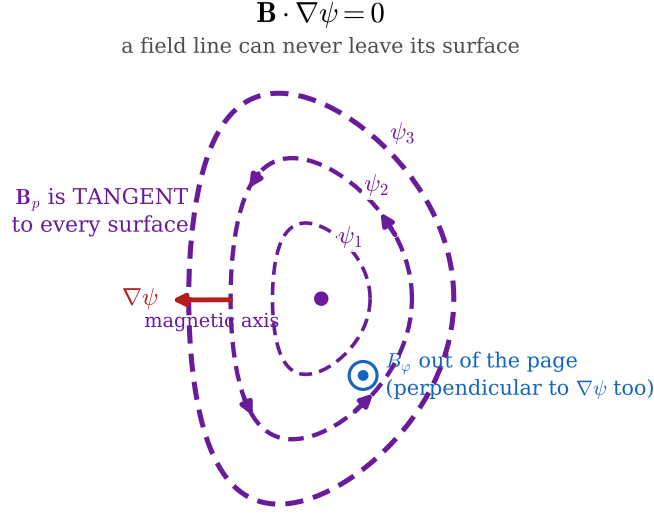
Putting the pieces together,

$$\mathbf{B} = \frac{1}{R} \nabla \psi \times \hat{\varphi} + \frac{F}{R} \hat{\varphi}. \quad (50)$$

This is not an approximation. It is a representation of any sufficiently smooth, axisymmetric, divergence-free magnetic field away from the coordinate axis.

PLACE IMAGE HERE — Physics sketch: reading ψ . Draw an axisymmetric poloidal cross-section with nested curves labelled ψ_1, ψ_2, ψ_3 , arrows tangent to the curves for $\mathbf{B}_p$, and a curled toroidal arrow for B_φ . Include a small arrow for $\nabla \psi$ perpendicular to the surfaces. Do not show hardware; this is the mathematical map that makes flux surfaces intelligible.

Design consequence. The entire two-dimensional equilibrium can now be described by two scalar functions, $\psi(R, Z)$ and $F(R, Z)$. This is the compression that makes a calculable plasma boundary possible.



Explanatory sketch. No hardware is drawn: this is the mathematical map that makes flux surfaces intelligible, not a picture of a machine.

Figure 13: Physics sketch: reading ψ . *Idealized schematic; not a fabrication drawing.*

23. Deriving the Grad–Shafranov equation, one force at a time

Ampere’s law without displacement current is $\mu_0 \mathbf{j} = \nabla \times \mathbf{B}$. Insert Eq. (50). The toroidal component comes only from the curl of $\mathbf{B}_p$:

$$j_\varphi = -\frac{1}{\mu_0 R} \Delta^* \psi, \quad \Delta^* \psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2}. \quad (51)$$

This follows directly by differentiating B_R and B_Z written on page 22: $(\nabla \times \mathbf{B}_p)_\varphi = \partial_Z B_R - \partial_R B_Z$.

The toroidal component of the equilibrium force in Eq. (48) must vanish because p is axisymmetric. It gives $\mathbf{B}_p \cdot \nabla F = 0$, so F is constant along the poloidal field and may be written $F = F(\psi)$. The same argument on page 21 gave $p = p(\psi)$. With those two results, direct cross products reduce the force balance to a single direction, normal to the surfaces:

$$\mathbf{j} \times \mathbf{B} = -\frac{1}{\mu_0 R^2} \left(\Delta^* \psi + F \frac{dF}{d\psi} \right) \nabla \psi, \quad \nabla p = \frac{dp}{d\psi} \nabla \psi.$$

The common vector $\nabla \psi$ can be cancelled wherever it is nonzero. The result is

$$\Delta^* \psi = -\mu_0 R^2 \frac{dp}{d\psi} - F \frac{dF}{d\psi}. \quad (52)$$

This is the Grad–Shafranov equation. It has no hidden tokamak magic: the first term is the pressure-gradient demand and the second represents toroidal-field/current effects. A boundary shape and the two functions $p(\psi)$ and $F(\psi)$ determine the equilibrium problem; external coils supply the boundary conditions.

Supporting quantity — ψ , $p(\psi)$, and $F(\psi)$. These are the minimal axisymmetric-equilibrium data beneath the reactor variables. They determine whether a proposed I_p , shape, and pressure can share nested magnetic surfaces.

Model and limits. The cancellation leading to Eq. (52) fails at magnetic nulls such as an X-point, where $\nabla\psi = 0$; the equation itself remains the standard equilibrium equation but the local geometric argument must be treated with care there.

24. What the equilibrium equation says before it is solved

Equation (52) is a differential equation, not a ready-made shape. To solve it, one specifies a vacuum-vessel or plasma-boundary condition for ψ , the external-coil currents that contribute to that condition, and profiles for $p(\psi)$ and $F(\psi)$. The solution then reports the nested surfaces, the plasma current, and the position of the magnetic axis. This is why a tokamak needs both a physics model and a coil-control model.

The equation also explains an effect visible in real cross-sections: the magnetic axis usually shifts outward from the geometric centre. Pressure gradients and the plasma's own poloidal field produce an outward hoop tendency; the poloidal-field system counters it. The shift is called the Shafranov shift. Its exact size depends on the profiles, so it must not be inserted as a universal number. The robust conclusion is only the causal one: increasing pressure or current changes the equilibrium shape, and fixed coil currents will no longer hold the intended boundary.

This is also the cleanest place to separate two kinds of D. A D-shaped *plasma* boundary is selected through δ and the equilibrium/stability tradeoffs. A D-shaped *toroidal-field coil*, when used, is a mechanical solution for reducing bending stress in a conductor exposed to a field that varies approximately as $1/R$. They can appear in the same machine, but they solve different equations and neither should be used to explain the other.

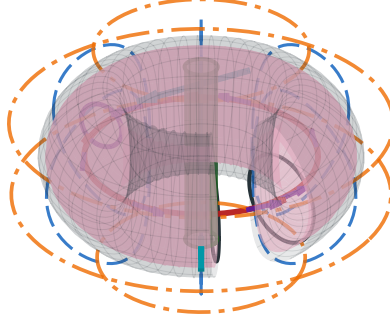
PLACE IMAGE HERE — Cumulative construction state 9: equilibrium is a moving target. Use the stage-8 conventional tokamak cross-section with shaped PF coils. Draw two nested sets of flux surfaces: a low-pressure reference centered near the geometric axis and a higher-pressure/current case shifted slightly outward. Label ψ , $p(\psi)$, and the PF-coil currents. Show the same hardware in both cases; the message is that operating state changes the demanded coil settings.

Design consequence. The physical boundary cannot be designed once and forgotten. Every planned operating point requires an equilibrium and a control solution. This is why later performance calculations must report profiles and margins, not merely a list of nominal dimensions.

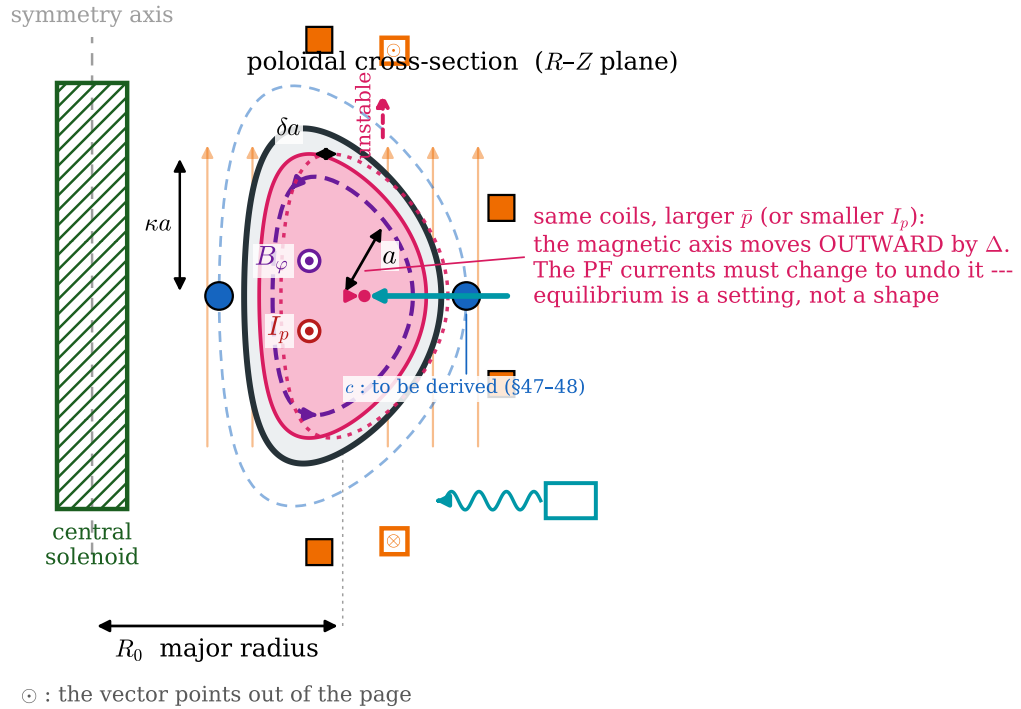
Stage 9 — the same machine, a different equilibrium

raise $p(\psi)$ or lower I_p and the axis shifts outward; the PF currents must change to put it back

perspective view (vessel cut open)



- | | |
|------------------------------|-----------------------------|
| vacuum vessel (chamber wall) | poloidal-field (PF) coils |
| plasma / fuel region | magnetic field $\mathbf{B}$ |
| toroidal-field (TF) coils | plasma current I_p |
| central solenoid (CS) | auxiliary heating ports |



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 14: Machine state: the same machine, a different equilibrium. Raise $\bar{p}$ or lower I_p and the magnetic axis shifts outward; the PF currents must change to undo it. The shift drawn is illustrative — its size follows from Grad–Shafranov.

25. From individual collisions to a fusion-power density

Consider a small volume dV containing $n_D dV$ deuterons and $n_T dV$ tritons. Choose one deuteron. During a short time dt , tritons with relative speed v sweep through the effective collision tube of area $\sigma(v)$ and length $v dt$. The expected number of its reactions is

$$n_T \sigma(v) v dt.$$

Particles in a thermal plasma do not all have the same relative speed. Average this expression over the normalized relative-speed distribution and multiply by the number $n_D dV$ of deuterons. Dividing by $dV dt$ gives the reaction rate per unit volume:

$$\mathcal{R}_{DT} = n_D n_T \langle \sigma v \rangle, \quad \langle \sigma v \rangle = \int_0^\infty \sigma(v) v f_{\text{rel}}(v) dv. \quad (53)$$

The brackets are an average, not a new force law. All difficult nuclear physics is contained in the measured cross section $\sigma(v)$; all thermal statistics is contained in $f_{\text{rel}}(v)$.

If each reaction releases total energy $E_f = 17.6 \text{ MeV}$, then

$$p_{\text{fus}} = E_f \mathcal{R}_{DT} = E_f n_D n_T \langle \sigma v \rangle, \quad P_{\text{fus}} = p_{\text{fus}} V. \quad (54)$$

This is the exact starting point behind the proportional sign on page 17. It says why a fusion plant cannot be discussed using temperature alone: density, volume, and the sharply temperature-dependent reactivity all matter.

Supporting quantity — $\langle \sigma v \rangle(T)$. The D–T reactivity turns a chosen temperature into a fusion-rate coefficient in Eq. (53). It is a measured nuclear input, not a free reactor-design variable and not something this thesis should replace with an unsupported universal formula.

Model and limits. Equation (53) assumes independent binary collisions and a thermal distribution. At reactor density, D–T fusion remains dilute enough for that description; collective plasma physics controls confinement, not the short-range nuclear reaction once two ions meet.

Design consequence. Fusion output grows with a product of densities. The next question is therefore a design question: for a fixed amount of fuel, which mixture makes that product largest?

26. Why the fuel mixture is nearly fifty–fifty

Let the total fuel-ion density be fixed:

$$n_i = n_D + n_T.$$

Write $n_D = xn_i$. Then $n_T = (1 - x)n_i$, and the density product in Eq. (54) is

$$n_D n_T = x(1 - x)n_i^2. \quad (55)$$

Complete the square:

$$x(1 - x) = x - x^2 = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2.$$

The squared term is nonnegative, so the largest possible value is $1/4$, achieved at $x = 1/2$. Thus

$$n_D = n_T = \frac{n_i}{2} \implies p_{\text{fus}} = \frac{1}{4}n_i^2 \langle \sigma v \rangle E_f. \quad (56)$$

This is an unusually clean design conclusion: equal fuel fractions maximize fusion opportunities at fixed total ion density and temperature.

The pressure used in magnetic confinement also includes electrons. A quasi-neutral 50–50 D–T mixture has electron density $n_e = n_i$. If the ions and electrons share a temperature T measured in energy units, their ideal-gas pressures add:

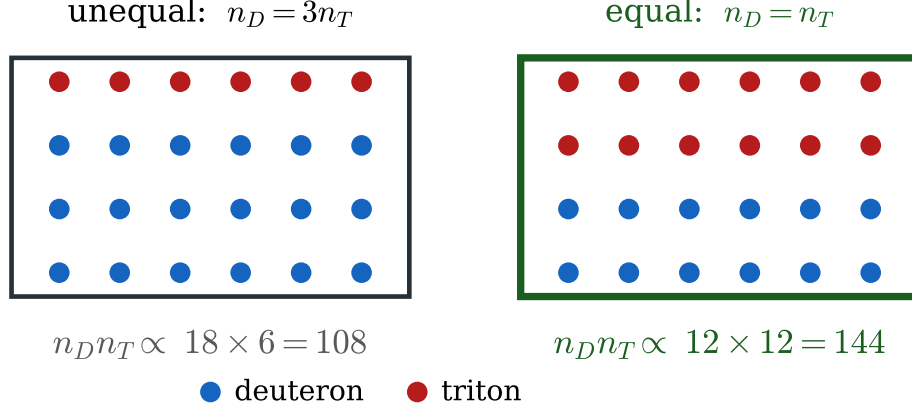
$$p = (n_i + n_e)T = 2n_i T. \quad (57)$$

Combining Eqs. (56) and (57) explains the scaling chain on page 17, but with its factors visible: at fixed T , $n_i = p/(2T)$ and hence $p_{\text{fus}} \propto p^2$.

Primary Freidberg parameter — T , n , and p . For the equal 50–50 mixture used here, n_i is the primary fuel-density variable n , and $p = 2nT$. A claimed gain in density at fixed magnetic field must therefore be checked against the pressure and density limits separately.

PLACE IMAGE HERE — Physics sketch: counting fusion pairs. Make a small conceptual graphic with two boxes containing the same total number of fuel ions: one unequal D–T mixture, one equal mixture. Show the pair count proportional to $n_D n_T$ and the completed-square maximum at $n_D = n_T$. This is a reset explanatory sketch, not a machine drawing.

Design consequence. The density needed for power also creates pressure. The magnetic field must therefore confine not only charged particles one at a time, but the collective pressure of the mixture.



Same 24 ions in both boxes. The reaction rate goes as the number of D-T PAIRS, $n_D n_T$, and with $n_D + n_T$ fixed that product is a downward parabola: complete the square,

$$n_D n_T = \frac{1}{4}(n_D + n_T)^2 - \frac{1}{4}(n_D - n_T)^2, \text{ which is largest exactly when } n_D = n_T.$$

Explanatory sketch; schematic, not to scale.

Figure 15: Physics sketch: counting fusion pairs. *Idealized schematic; not a fabrication drawing.*

27. Where the fusion energy goes decides what the machine must add

The D-T products are an alpha particle and a neutron. In the center-of-mass frame they leave with equal and opposite momentum, so their momentum magnitudes are equal: $m_\alpha v_\alpha = m_n v_n$. Their kinetic energies are $E_\alpha = p^2/(2m_\alpha)$ and $E_n = p^2/(2m_n)$. Therefore

$$\frac{E_n}{E_\alpha} = \frac{m_\alpha}{m_n} \simeq 4, \quad E_\alpha = \frac{1}{5}E_f \simeq 3.5 \text{ MeV}, \quad E_n = \frac{4}{5}E_f \simeq 14.1 \text{ MeV}.$$

The alpha is charged. It remains magnetically confined long enough to slow down by collisions and can heat the plasma. The neutron has no electric charge, so magnetic fields do not bend it; it escapes the plasma and deposits its energy only after meeting material outside the vacuum region.

Define alpha heating by $P_\alpha = (E_\alpha/E_f)P_{\text{fus}}$ before loss channels. Similarly, $P_n = (E_n/E_f)P_{\text{fus}}$ is neutron power. This is not a minor accounting choice. A burning plasma needs the alpha fraction to replace energy lost from the plasma. A power plant needs the neutron fraction to be captured in a future blanket, which will earn its place later by this physical necessity.

Supporting quantity — P_α and P_n . The split $P_\alpha = P_{\text{fus}}/5$ and $P_n = 4P_{\text{fus}}/5$ separates plasma self-heating from energy that leaves the magnetic bottle. It is the bridge between the plasma power balance and the later reactor-around-the-plasma discussion.

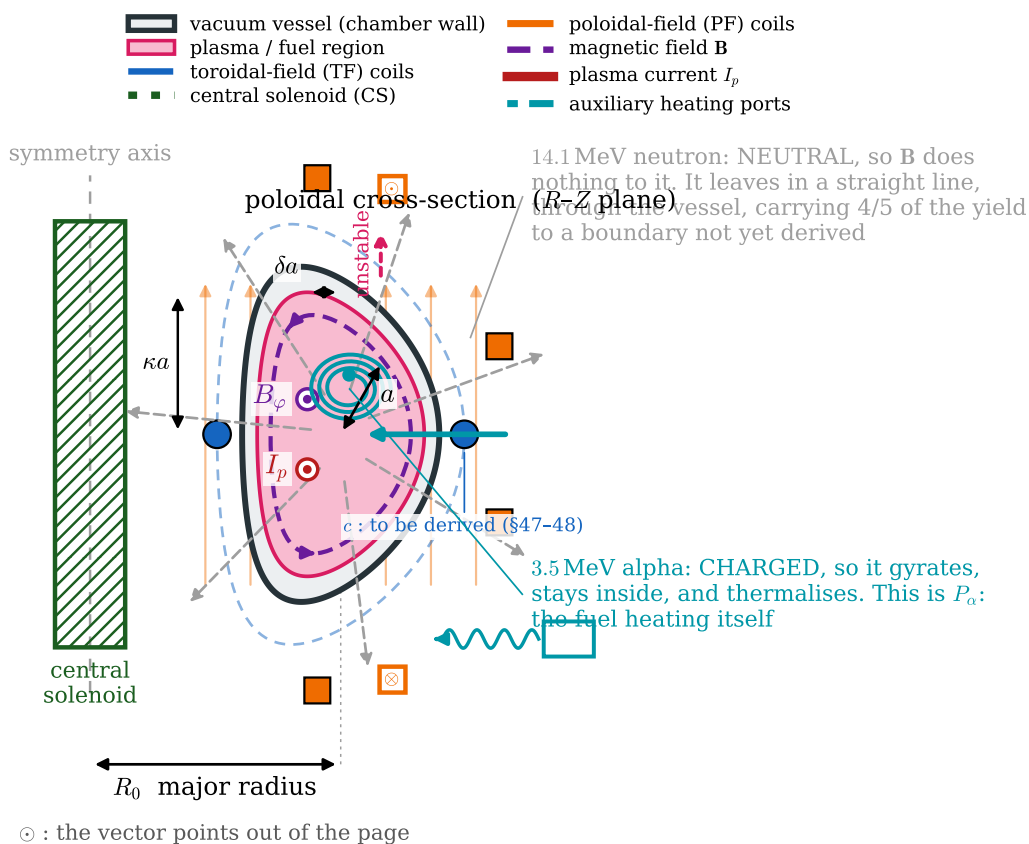
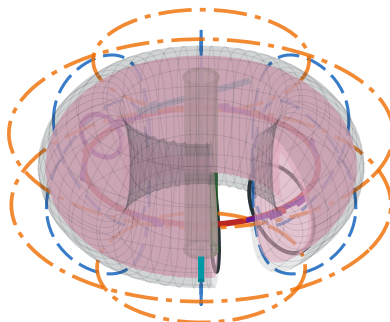
PLACE IMAGE HERE — Cumulative construction state 10: alpha heating and neutron escape. Use the existing conventional tokamak cross-section, without adding a blanket. Inside the plasma, show a short spiral alpha trajectory and label 3.5 MeV self-heating. Draw straight neutral-neutron arrows crossing the vacuum region and stopping at a clearly labelled future boundary outside the current derived machine. State explicitly that the blanket is not yet drawn because it has not yet been derived.

Design consequence. A magnetic field can confine the alpha energy that helps sustain the plasma, but it cannot confine neutron energy. That one charge difference will later force the first wall, blanket, shielding, and coolant into the machine.

Stage 10 — the fuel begins to heat itself

pha is charged and stays; the 14.1 MeV neutron is not, and leaves through a boundary this figure deliberately does

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 16: Machine state: one charge difference. The alpha is charged, so it gyrates and stays. The neutron is not, so $\mathbf{B}$ does nothing to it and it leaves in a straight line — toward a boundary this figure deliberately does not yet draw.

28. Stored energy and the meaning of confinement time

Before asking whether alpha heating is enough, we need a careful measure of loss. For an ideal gas with three translational degrees of freedom, the internal energy density is $3nT/2$. In the equal-temperature D–T plasma there are n_i ions and $n_e = n_i$ electrons, so the total thermal energy is

$$W = \left(\frac{3}{2}n_iT + \frac{3}{2}n_eT \right) V = 3n_iTV. \quad (58)$$

The energy-confinement time is defined, rather than guessed, by

$$\tau_E \equiv \frac{W}{P_{\text{loss}}}, \quad P_{\text{loss}} = \frac{W}{\tau_E}. \quad (59)$$

It answers a plain question: if all heating stopped and the loss rate stayed at its instantaneous value, how long would the present thermal energy take to leave?

This definition does not say that plasma energy literally decays as one perfect exponential. It packages all conduction, turbulent transport, radiation, and other losses into a measured or modelled net rate. That is exactly why it is useful in a design equation: the detailed mechanisms may be complicated, but the power balance must still know their sum.

Using Eq. (57), Eq. (58) can also be written $W = (3/2)pV$. Thus a high-pressure plasma stores more energy at a given volume, but it also demands proportionally more heating power if τ_E is fixed.

Primary Freidberg parameter — τ_E . Energy-confinement time is the variable that converts the plasma state (n, T, V) into an energy-loss power. It is a core Freidberg input, but unlike geometry it is not obtained from equilibrium alone.

Design consequence. The reactor must satisfy two equations simultaneously: fusion must create enough alpha heating, and transport must make $P_{\text{loss}} = W/\tau_E$ small enough. The next page combines them without skipping the cancellation.

29. The Lawson condition follows from one inequality

Ignition means that alpha heating alone can replace the plasma losses:

$$P_\alpha \geq P_{\text{loss}}. \quad (60)$$

Use the equal-mixture fusion result, Eq. (56), but retain only the alpha energy E_α :

$$P_\alpha = \frac{1}{4}n_i^2\langle\sigma v\rangle E_\alpha V.$$

Use Eq. (58) and Eq. (59) for the loss:

$$P_{\text{loss}} = \frac{3n_iTV}{\tau_E}.$$

Substitution into Eq. (60) gives

$$\frac{1}{4}n_i^2\langle\sigma v\rangle E_\alpha V \geq \frac{3n_iTV}{\tau_E}.$$

Cancel the positive common factor n_iV , then multiply by $4\tau_E$:

$$\boxed{n_i\tau_E \geq \frac{12T}{\langle\sigma v\rangle E_\alpha}}. \quad (61)$$

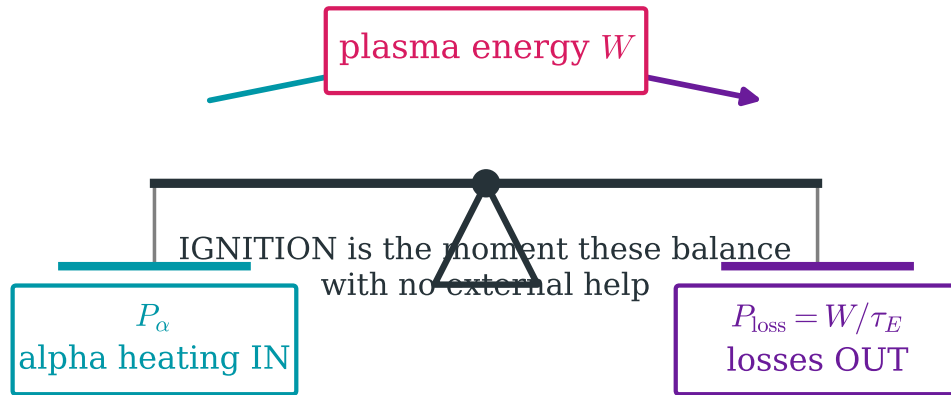
Every factor has a visible origin: $1/4$ from the equal fuel mix, 3 from ions plus electrons and three translational degrees of freedom, and E_α rather than E_f because neutrons do not heat the confined plasma.

Multiplying Eq. (61) by T produces the common triple product $n_iT\tau_E$. Its numerical threshold cannot be chosen without an openly stated reactivity curve and temperature. The expression above is more fundamental than a single memorized number because it keeps that dependence visible.

Supporting quantity — $nT\tau_E$. The triple product is not a third independent quantity. It condenses the density, temperature, and confinement requirement for self-heating. Equation (61) is the version used when the temperature dependence of D–T reactivity matters.

PLACE IMAGE HERE — Physics sketch: the ignition balance. Draw a two-pan balance. On the left, alpha heating P_α enters the plasma-energy box. On the right, $P_{\text{loss}} = W/\tau_E$ leaves it. Under the balance, show the three algebra steps from Eq. (60) to Eq. (61). This is a reset explanatory sketch, not a machine view.

Design consequence. Ignition does not mean high temperature alone. A reactor needs a particular combination of density, temperature, and confinement. Improving any one of them can relax the burden on the other two, but only within their own limits.



$$\frac{1}{4}n^2\langle\sigma v\rangle E_\alpha = \frac{3nT}{\tau_E}$$

$$\Rightarrow n\tau_E = \frac{12T}{\langle\sigma v\rangle E_\alpha} \quad \Rightarrow \quad nT\tau_E \gtrsim \text{const}$$

three steps, no new physics: cancel one power of n , solve for $n\tau_E$, then multiply by T to reach the form the reactor literature quotes

Explanatory sketch; schematic, not to scale.

Figure 17: Physics sketch: the ignition balance. *Idealized schematic; not a fabrication drawing.*

30. Gain is a ratio, not a synonym for ignition

A plasma can make useful fusion power without being ignited if external heating supplies the missing energy. Define the plasma gain

$$Q_{\text{plasma}} \equiv \frac{P_{\text{fus}}}{P_{\text{aux}}}, \quad (62)$$

where P_{aux} is the heating delivered to the plasma by neutral beams, radio frequency systems, or other actuators. In steady state, a stripped-down energy balance is

$$P_{\alpha} + P_{\text{aux}} = P_{\text{loss}}. \quad (63)$$

As P_{aux} tends to zero while the equality remains possible, Q_{plasma} tends to infinity: that limiting case is ignition. At finite Q_{plasma} , the plant must supply auxiliary power and also pay the inefficiency of producing and delivering it.

The distinction protects the argument from a common mistake. A large Q_{plasma} is a plasma metric. Net electric power is a plant metric that also depends on neutron-energy capture, thermal conversion, magnet refrigeration, pumps, controls, and all other recirculating loads. We will not declare a plant successful from Eq. (62); we have not yet derived the components that would make that claim meaningful.

Supporting quantity — Q_{plasma} . This ratio records how much fusion power is obtained per unit of auxiliary power delivered to the plasma. It enters the later Freidberg power ledger but cannot by itself establish net electricity.

Design consequence. The current ordinary-tokamak drawing has now earned auxiliary heating but not yet a power plant. The next physical obstacle is that τ_E is not supplied by the equilibrium equations: it is set by transport.

31. Why transport is the unresolved clock

A particle in a uniform magnetic field circles with a characteristic gyroradius ρ . A collision changes the direction of its velocity; the center of the next gyro-orbit is displaced by an amount of order ρ . If collisions occur at rate ν , then in time t there are approximately $N = \nu t$ independent steps. Random-walk addition gives

$$\langle \Delta r^2 \rangle \sim N \rho^2 = \nu t \rho^2.$$

By the definition $\langle \Delta r^2 \rangle \sim 2Dt$ in one radial direction, the classical cross-field diffusion estimate is

$$D_{\text{cl}} \sim \frac{\rho^2 \nu}{2}. \quad (64)$$

A container of minor radius a would then have a diffusion time of order

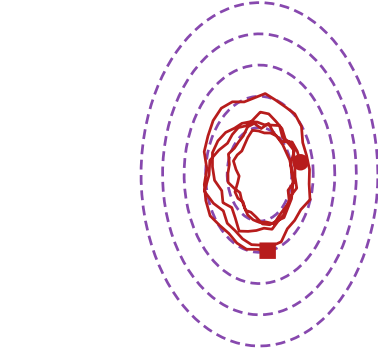
$$\tau_{\text{diff}} \sim \frac{a^2}{D_{\text{cl}}}. \quad (65)$$

This is a transparent reason strong field helps: $\rho \propto 1/B$, so the classical diffusion coefficient falls roughly as $1/B^2$ if the collision rate is held fixed.

Unfortunately, measured tokamak energy transport is frequently much faster than this naive collisional estimate. Fluctuations and collective turbulence create correlated motions across flux surfaces. The derivation above is therefore not a promised confinement law. It is the baseline that tells us why the problem is surprising and why an empirical confinement time must later enter the reactor calculation.

PLACE IMAGE HERE — Physics sketch: random-walk transport across flux surfaces. Draw nested flux surfaces with a guiding center taking many small, collision-induced radial steps of size about ρ . Beside it, show the derivation $N = \nu t$, $\langle \Delta r^2 \rangle \sim N \rho^2$, and $D_{\text{cl}} \sim \rho^2 \nu / 2$. Label turbulence as an additional, not-yet-modelled transport channel.

Design consequence. Magnetic geometry suppresses simple cross-field diffusion, but it does not guarantee the required τ_E . A reactor design must carry a confinement model and its uncertainty as explicitly as it carries B_0 or a .



one guiding centre, 150 collisions:
each kick moves it about one gyro-radius ρ

the estimate that follows

$$N = \nu t \quad \text{collisions in time } t$$

$$\langle \Delta r^2 \rangle \sim N \rho^2 = \nu t \rho^2$$

$$D_{\text{cl}} \sim \rho^2 \nu / 2$$

a diffusion coefficient built from ONE step size and
ONE step rate --- this is the CLASSICAL estimate

Real tokamaks lose heat faster than this. Toroidal geometry
adds a neoclassical enhancement, and TURBULENCE adds a
further channel that this picture does not model at all.

Explanatory sketch; schematic, not to scale.

Figure 18: Physics sketch: random-walk transport across flux surfaces. *Idealized schematic; not a fabrication drawing.*

32. An empirical confinement law must be treated as data, not destiny

For reactor studies, the working confinement time is often represented as

$$\tau_E = H \tau_{E,\text{ref}}(I_p, B_0, n, P_{\text{loss}}, R_0, a, \kappa, \dots). \quad (66)$$

Here $\tau_{E,\text{ref}}$ is a regression fitted to a specified database of experiments, and H is the ratio of the observed confinement to that reference. This equation is deliberately written without pretending that the exponents in a particular regression are fundamental constants. The role of the formula is to make an extrapolation auditable: which database, operating regime, units, heating mode, and shape range were used?

The distinction between a derived equation and a fit matters especially here. Equation (64) followed from a stated random-walk model. Equation (66) is a bookkeeping identity once a reference scaling is chosen; its predictive content lives in the data and the applicability range of $\tau_{E,\text{ref}}$. Neither should be disguised as the other.

A practical design calculation must therefore report at least three things: the selected reference law, the assumed H , and a sensitivity range. If the operating point requires H well above what comparable experiments have obtained, that is not a proof of impossibility, but it is an explicit research risk rather than hidden optimism.

Supporting quantity — H . The confinement multiplier H makes the assumption behind an empirical τ_E explicit. Freidberg-style closure cannot treat it as a harmless fitted detail, because Eq. (61) makes confinement directly load-bearing.

Model and limits. No universal first-principles turbulence calculation is used in this thesis opening. When a particular confinement regression is introduced in the Freidberg reconstruction, its published coefficient, units, database, and fit range will be quoted alongside the calculation.

Design consequence. A high-field, well-shaped plasma may still fail the energy balance if its actual transport is too fast. Confinement is not an optional correction added after geometry; it closes the same power balance.

33. Pressure cannot rise freely: the origin of a beta limit

Before the next two named constraints appear, keep their evidence status separate. The pressure-versus-line-bending argument below is a derived scaling. The practical normalized-beta boundary introduced on page 34 is an empirical/model closure. The Greenwald relation on page 35 is an empirical operating boundary. Their physical motivation is useful; none of their numerical coefficients follows from this local force-balance estimate alone.

The beta on page 8 compares plasma pressure with magnetic pressure:

$$\beta = \frac{p}{B_0^2/(2\mu_0)}.$$

Why should there be an upper limit? Imagine a bulge of a pressure-carrying flux tube. The pressure force trying to enlarge it scales as p times its cross-sectional area. Bending the field lines that surround it costs magnetic energy; the restoring scale is set by B_θ^2/μ_0 , where B_θ is the poloidal field that ties the plasma current to the geometry.

Ampere's-law scale at the edge is

$$B_\theta \sim \frac{\mu_0 I_p}{2\pi a}.$$

Equating pressure and bending *scales*, rather than claiming an exact stability threshold, gives

$$p_{\max} \sim \frac{B_\theta^2}{\mu_0} \sim \frac{\mu_0 I_p^2}{4\pi^2 a^2}.$$

Divide by $B_0^2/(2\mu_0)$ to obtain the dimensionless structural form

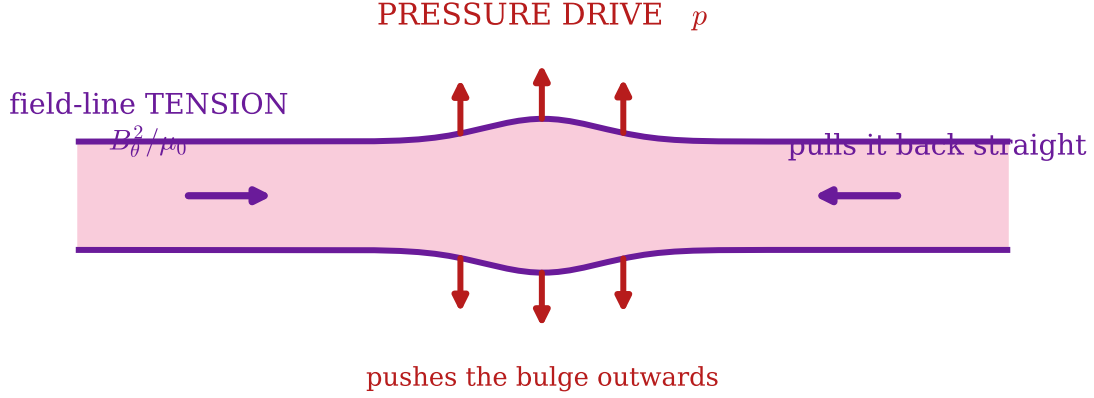
$$\beta_{\max} \propto \left(\frac{I_p}{aB_0} \right)^2 \quad \text{in this crude local balance.} \quad (67)$$

A full ideal-MHD stability calculation changes this simplified power and introduces shape and profile dependence. The important accessible result is the mechanism: pressure pushes outward, and magnetic line bending supplies the restoring budget.

PLACE IMAGE HERE — Physics sketch: pressure drive versus line bending.

Draw a small bulged segment of a flux tube. Use outward arrows labelled pressure drive p and inward tension arrows labelled poloidal-field bending B_θ^2/μ_0 . Include the edge-field estimate $B_\theta \sim \mu_0 I_p/(2\pi a)$, and state that this is a scaling argument rather than the final MHD stability calculation.

Design consequence. The variables desired for fusion power, p and B_0 , cannot be optimized separately from I_p , shape, and stability. The next page records the standard practical normalization of this coupled limit.



edge estimate: $B_\theta \sim \mu_0 I_p / (2\pi a)$, so the restoring term goes as I_p^2 while the drive goes as p
 This is a SCALING argument for which way each term pushes. It is not the MHD stability calculation, and it does not by itself give a numerical beta limit.

Explanatory sketch; schematic, not to scale.

Figure 19: Physics sketch: pressure drive versus line bending. *Idealized schematic; not a fabrication drawing.*

34. The normalized beta used in reactor design

Detailed ideal-MHD calculations and experiments are commonly summarized by a normalized beta. With β written as a fraction rather than a percentage, define

$$\beta_N \equiv \frac{\beta a(\text{m}) B_0(\text{T})}{I_p(\text{MA})}. \quad (68)$$

This is a convention with stated units, not a dimensionless number formed from pure SI variables. It is useful because comparable tokamak equilibria show an approximate operating boundary of the form

$$\beta \lesssim \beta_{N,\text{max}} \frac{I_p(\text{MA})}{a(\text{m}) B_0(\text{T})}. \quad (69)$$

The coefficient $\beta_{N,\text{max}}$ is empirical/model-dependent; it must be imported from a specified stability study or design paper, never derived by dimensional analysis alone.

Equation (69) is nevertheless more than a fit-shaped ornament. It preserves the direction demanded by the pressure-versus-bending argument: larger allowable current raises allowable pressure, while a larger aB_0 at fixed current lowers this particular normalized beta. Combining it with the current ceiling on page 15 makes the coupling visible: the kink limit restricts I_p , which then restricts the pressure that can be claimed for fusion power.

Supporting quantity — β_N . Normalized beta is the portable stability-performance variable used to turn an MHD pressure limit into a reactor constraint. Its numerical limit is a sourced assumption; its coupling to I_p , a , and B_0 is explicit in Eq. (69).

Design consequence. The safe operating pressure is bounded twice: directly by pressure-driven stability and indirectly by the current that kink stability permits. A credible design must check both bounds at the same point.

35. Density has a separate experimental limit

It would be tempting to use Eq. (57) and conclude that a chosen pressure can always be obtained by trading density against temperature. Tokamak operation does not permit an unlimited trade. A widely used experimental reference is the Greenwald density

$$n_G = \frac{I_p(\text{MA})}{\pi a^2(\text{m}^2)} \times 10^{20} \text{ m}^{-3}. \quad (70)$$

Operational scenarios usually report the fraction $f_G = n_e/n_G$, rather than merely a raw density.

There is an important methodological point here. Equation (70) is not derived from the preceding equilibrium equations. It summarizes observations across tokamak discharges, and the physical mechanism behind its most general form is not settled well enough to pretend otherwise. The equation belongs in the thesis because it constrains designs; it belongs under an empirical label because its coefficient is not a theorem.

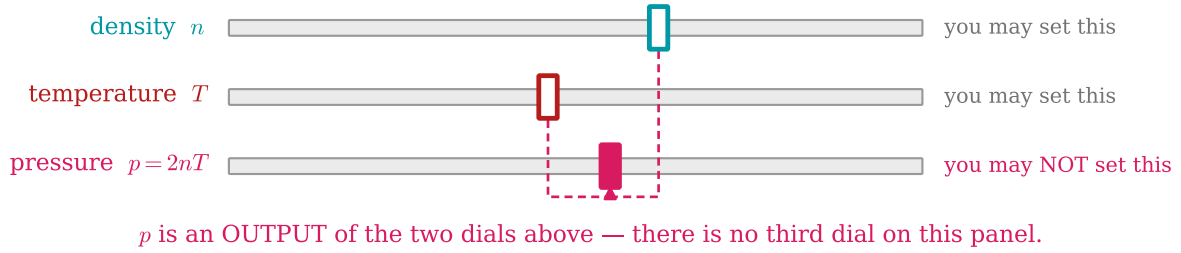
The formula still carries useful transparent geometry. At fixed current, a smaller cross-section raises the reference density as $1/a^2$. At fixed a , more current raises it. But that same current is limited by q_* and the same density contributes to pressure and radiation. The quantity f_G makes this web testable without claiming that one line of algebra has solved the underlying edge physics.

Supporting quantity — n_G and f_G . The Greenwald density and its operating fraction are empirical constraints on n_e . They are load-bearing whenever the fusion-power calculation seeks higher density, and their extrapolation must be treated as an assumption with a range.

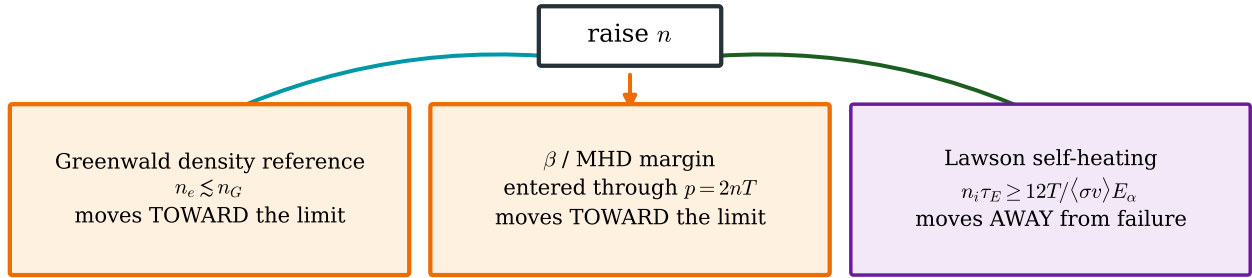
PLACE IMAGE HERE — Constraint sketch: pressure is not one knob. Draw a three-axis conceptual plot or linked sliders for density n , temperature T , and pressure $p = 2nT$. Overlay separate labels for the Greenwald reference n_G , the beta limit, and the Lawson requirement. The point is to show that increasing n affects several constraints at once; do not imply a precise universal operating boundary.

Design consequence. The density needed by the fusion rate has its own operational ceiling. A reactor must find a region that satisfies density, pressure, current, and confinement constraints simultaneously.

the control panel a reactor actually has



and n is not a private dial either



Two of the three are $\lesssim$, not $=$ — empirical or model-dependent margins. Only the third is derived.

The design question is therefore not "what is the right pressure?" but "which (n, T) pair satisfies all three at once?" — and the three do not share an optimum. No numerical axis is drawn here on purpose: there is no universal (n, T) boundary to draw. §41 turns this into an intersection problem.

Explanatory sketch; schematic, not to scale. Slider positions are illustrative and carry no numerical value.

Figure 20: Constraint sketch: pressure is not one knob. *Idealized schematic; not a fabrication drawing.*

36. Toroidal geometry traps some particles

The toroidal field is stronger on the inboard side because approximately $B_\varphi \propto 1/R$. A particle moving into stronger field converts parallel motion into perpendicular motion while conserving, to lowest adiabatic order,

$$\mu = \frac{mv_\perp^2}{2B} \quad \text{and} \quad \frac{1}{2}m(v_\parallel^2 + v_\perp^2).$$

Let the particle start where the field is $B_{\min}$ and let α be its pitch angle there, so $v_\perp = v \sin \alpha$. At the turning point $v_\parallel = 0$. Conservation of μ gives

$$\frac{v^2 \sin^2 \alpha}{B_{\min}} = \frac{v^2}{B_{\max}}, \quad \sin^2 \alpha = \frac{B_{\min}}{B_{\max}}.$$

Particles with larger perpendicular fraction reach this condition before the high-field side and bounce back. For isotropically distributed velocity directions, the trapped fraction is therefore

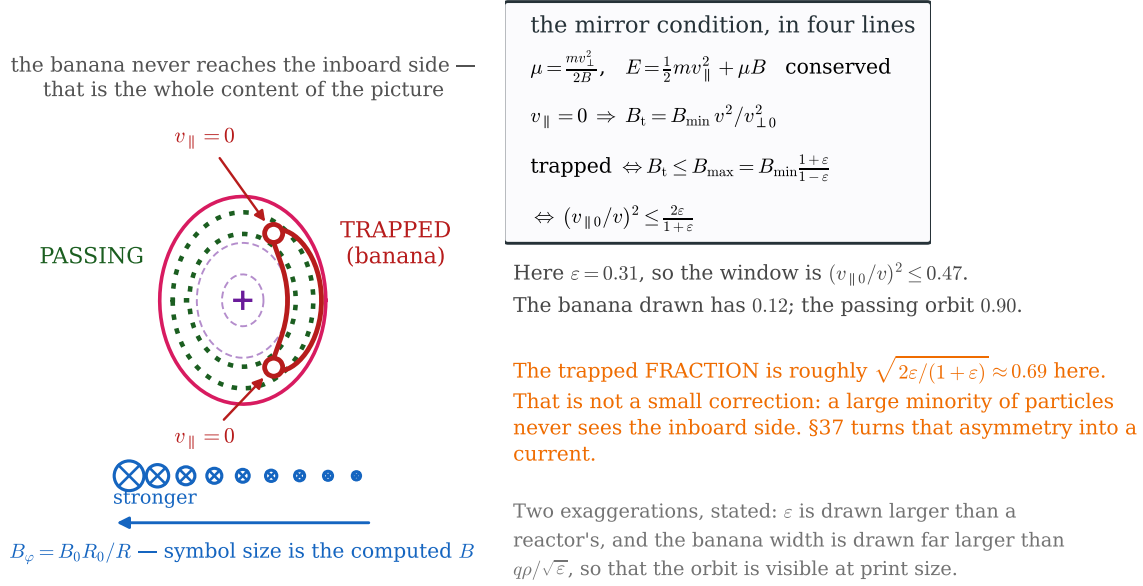
$$f_t = \sqrt{1 - \frac{B_{\min}}{B_{\max}}}. \quad (71)$$

This is a geometric consequence of the same $1/R$ toroidal field that was introduced to close the open magnetic corridor. It is neither uniformly good nor uniformly bad: trapping changes transport, and a pressure gradient acting on trapped orbits can also produce useful self-generated current.

Supporting quantity — f_t . The trapped-particle fraction in Eq. (71) is a geometry-dependent input to neoclassical transport and bootstrap-current reasoning. Here neoclassical means transport calculated from Coulomb collisions together with the torus-specific guiding-center orbits; it is more than the cylindrical classical-collision estimate, but it is still a stated kinetic model. This prepares the final missing current variable without invoking any spherical-tokamak comparison.

PLACE IMAGE HERE — Physics sketch: a trapped orbit. Show a poloidal cross-section with stronger field on the inboard side. Draw one passing orbit and one trapped guiding-center path, called a banana orbit because its poloidal projection has that shape, marking two turning points where $v_\parallel = 0$. Include a small inset deriving the mirror condition from conservation of μ and energy.

Design consequence. The torus creates more than a closed corridor. Its unavoidable field variation creates an orbit population whose asymmetry can help supply plasma current, but only through a kinetic mechanism that must be introduced honestly.



Explanatory sketch. The orbit shape is computed from $v_{\parallel}(\theta)$; the inverse aspect ratio and the banana width are exaggerated for legibility.

Figure 21: Physics sketch: a trapped orbit. *Idealized schematic; not a fabrication drawing.*

37. A pressure gradient can generate some of the required current

A trapped particle does not sample the two sides of the torus symmetrically. Its guiding-center drifts and collisions depend on energy and pitch angle. When pressure falls outward, there are more energetic particles on the inner side of a local radial interval than on the outer side. The resulting imbalance in trapped-particle motion is communicated by collisions to passing electrons. The outcome is a net toroidal current called the *bootstrap current*.

The direction and detailed coefficient require kinetic theory, because they depend on collisions, geometry, temperature gradients, and the current profile. What can be stated before importing such a coefficient is the structure:

$$j_{\text{bs}} \propto -\frac{dp}{dr} \quad \text{and} \quad I_{\text{bs}} = \int j_{\text{bs}} dA. \quad (72)$$

No pressure gradient means no bootstrap drive in this simplified statement. A steeper gradient can produce more self-driven current, but it also moves the plasma closer to pressure-driven stability limits. The word bootstrap should therefore not be heard as free current. It is a coupling between the profile needed for confinement and the current needed for magnetic geometry.

Define the bootstrap fraction

$$f_{\text{bs}} \equiv \frac{I_{\text{bs}}}{I_p}. \quad (73)$$

A later kinetic model can supply a coefficient and a profile-dependent prediction for f_{bs} . At this stage, the variable is already essential because it reduces the current that must be supplied by a transformer or external current drive.

Primary Freidberg parameter — f_B (bootstrap fraction). Here f_B is written $f_{bs} = I_{bs}/I_p$ in Eq. (73). It is the self-generated share of the required plasma current. Its numerical value must come from a cited kinetic model or experiment, not from the proportionality sign alone.

Model and limits. Equation (72) is a mechanism and sign statement, not a complete bootstrap-current formula. The later Freidberg reconstruction will identify exactly which neoclassical coefficient it uses and will keep its profile assumptions visible.

Design consequence. The pressure profile can help sustain the magnetic geometry, but it cannot be assigned independently of the stability and confinement requirements that created the profile in the first place.

38. The full current ledger

The toroidal plasma current can have several sources. Write the accounting identity

$$I_p = I_{\text{ohm}} + I_{\text{CD}} + I_{\text{bs}}. \quad (74)$$

The first term is transformer-induced ohmic current; it is available while the central solenoid changes linked flux. The second is externally driven non-inductive current, for example from waves or beams transferring directed momentum. The third is the pressure-gradient-driven bootstrap current from page 37.

This equation is not a performance claim. It is a definition of what must be accounted for. For a long pulse after transformer action has faded, $I_{\text{ohm}} \simeq 0$, and Eq. (74) reduces to

$$I_{\text{CD}} = (1 - f_{\text{bs}})I_p. \quad (75)$$

The larger the bootstrap fraction, the smaller the externally driven current requirement. But current drive consumes power and its efficiency depends on density, temperature, magnetic field, launch geometry, and the actuator itself. A design may not set f_{bs} near one merely to make the accounting easy.

Supporting quantity — I_{CD} . Equation (75) makes the unself-driven current explicit. It is the quantity later translated into heating and recirculating-power demand; it links plasma profiles to plant power balance.

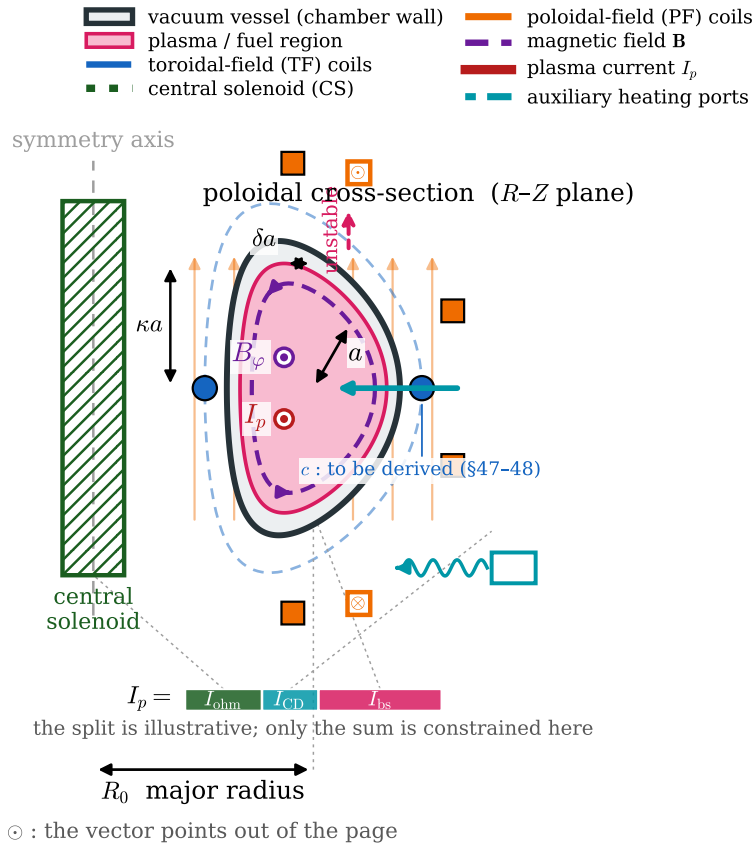
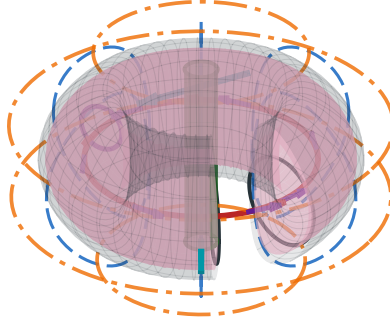
PLACE IMAGE HERE — Cumulative construction state 11: the current ledger on the machine. Start with the shaped conventional tokamak. Keep the central solenoid, heating/current-drive ports, and profile inset. Add a compact ledger beside the plasma: $I_p = I_{\text{ohm}} + I_{\text{CD}} + I_{\text{bs}}$. Use arrows from the solenoid, an auxiliary port, and the pressure-gradient/profile inset to the three terms. Do not draw engineering components that have not yet been derived.

Design consequence. Steady operation is a three-way closure: the current required by q_* must be consistent with the bootstrap current predicted by profiles and the external current-drive power the plant can afford.

Stage 11 — the plasma current has three sources

$I_p = I_{\text{ohm}} + I_{\text{CD}} + I_{\text{bs}}$: the transformer, the launchers, and the pressure gradient itself

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 22: Machine state: the full current ledger, $I_p = I_{\text{ohm}} + I_{\text{CD}} + I_{\text{bs}}$, drawn as a stacked bar because the three terms add. The split shown is illustrative; only the sum is constrained at this point.

39. Pulsed operation is also a legitimate design choice

The central solenoid is a transformer primary. Faraday's law gives the loop voltage around the torus:

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_{\text{link}}}{dt}.$$

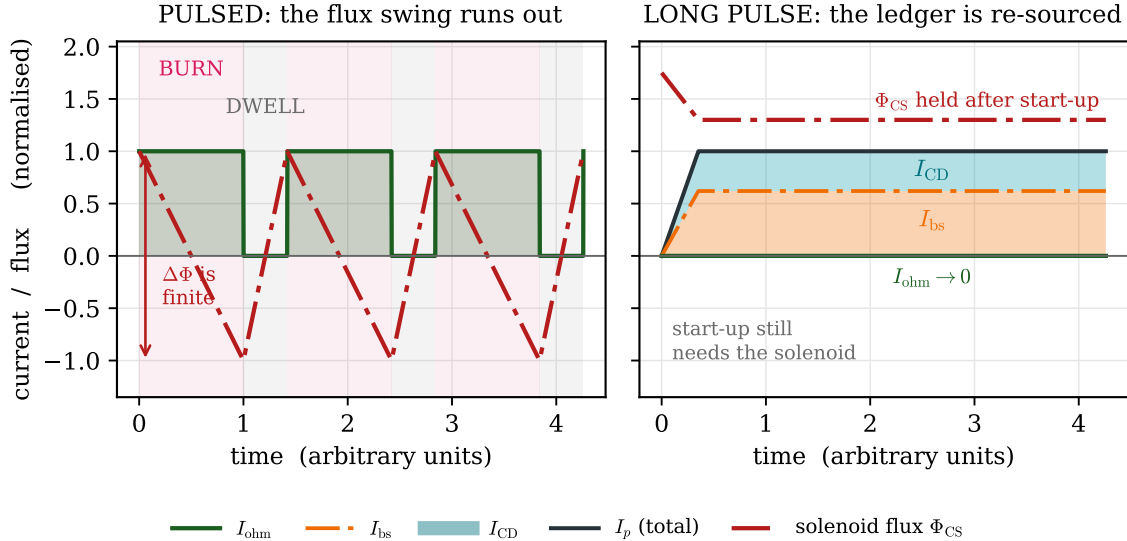
For a torus of major radius R_0 , a representative toroidal electric field satisfies $2\pi R_0 E_\varphi = -d\Phi_{\text{link}}/dt$. The solenoid has a finite usable flux swing $\Delta\Phi_{\text{link}}$, so it can induce current only for a finite time. This is the physical origin of the transformer clock introduced on page 11.

There are two coherent responses. One is a pulsed plant: use transformer current during a burn, then recharge the solenoid between pulses. The other is a long-pulse or steady-state plant: replace the lost ohmic contribution with the current-drive and bootstrap terms in Eq. (74). Neither option is free. Pulsing creates thermal cycling and dwell-time questions; non-inductive operation creates actuator power and accessibility questions. The correct choice belongs to an integrated design, not to a slogan about steady state.

Supporting quantity — $\Delta\Phi_{\text{link}}$. The available linked-flux swing is the pulsed-operation resource. It sets how much transformer action the central solenoid can provide before the current ledger must rely on non-inductive sources.

PLACE IMAGE HERE — Physics sketch: two ways to close the current ledger.

Make a two-panel conceptual diagram. Left: a pulsed timeline with solenoid flux decreasing during a burn and recharging during dwell. Right: a long-pulse balance with $I_{\text{CD}} + I_{\text{bs}}$ replacing I_{ohm} . Keep it a clean accounting sketch, not a claim that either architecture is automatically preferable.



Left, the only arithmetic on the figure: $V_{\text{loop}} = -d\Phi/dt$ and $I_{\text{ohm}} = V_{\text{loop}}/R_p$ give $t_{\text{burn}} = \Delta\Phi/(I_{\text{ohm}}R_p)$ — a finite swing buys a finite burn. (Internal inductance is neglected; start-up consumes part of $\Delta\Phi$ before the burn begins.) Right, the $I_{\text{bs}}/I_{\text{CD}}$ split drawn is ILLUSTRATIVE: only the sum is constrained here. Neither panel claims its architecture is preferable— each states a different price.

Figure 23: Physics sketch: two ways to close the current ledger. *Idealized schematic; not a fabrication drawing.*

Design consequence. The central solenoid has earned a place in the ordinary tokamak, but it also creates a choice between pulse duration, stored flux, and non-inductive current-drive burden. That choice will become an explicit Freidberg assumption.

40. The kink limit returns as a constraint on every current source

The current ledger does not allow arbitrary total current. A helical disturbance can be labelled by integers (m, n) through a displacement pattern $\xi \propto \cos(m\theta - n\varphi)$. Along an unperturbed field line, a poloidal turn changes θ by 2π while the toroidal angle changes by $2\pi q$. The phase change is therefore $2\pi(m - nq)$. When $q = m/n$, the disturbance remains in phase after a circuit; field-line bending has the least geometric leverage to average it away.

This is the accessible reason safety factor is a stability variable rather than only a geometric ratio. The full ideal-MHD energy principle determines the precise stability boundary. Reactor studies commonly use a specified global constraint

$$q_* > q_K, \tag{76}$$

with q_K supplied by the chosen stability model. Substituting the shaped safety-factor formula, Eq. (40), reproduces the current ceiling already derived in Eq. (41). Every term in the current ledger must fit under that same ceiling; bootstrap current helps meet a required I_p but does not abolish the stability constraint on I_p itself.

Supporting quantity — q_K . The kink threshold is a stability-model input applied to the shaped safety factor. It converts an otherwise desirable current into an upper bound and must be stated whenever a reactor calculation quotes I_p .

Model and limits. This page derives the resonance logic, not the full numerical value of q_K . The later reconstruction will distinguish the analytical Kruskal–Shafranov criterion from the practical design threshold used in the 2015 paper.

Design consequence. Current-drive accounting and kink stability are the same problem viewed from two directions: one asks where I_p comes from; the other asks how much total I_p the plasma can safely carry.

41. The operating point is an intersection, not a target number

We can now state the first coupled operating-space problem without claiming to have solved it. A candidate state must satisfy, at minimum,

$$q_* > q_K \quad \text{current and kink margin,} \quad (77)$$

$$\beta \lesssim \beta_{N,\max} \frac{I_p}{aB_0} \quad \text{pressure and MHD margin,} \quad (78)$$

$$n_e \lesssim n_G \quad \text{density reference,} \quad (79)$$

$$n_i \tau_E \geq \frac{12T}{\langle \sigma v \rangle E_\alpha} \quad \text{self-heating requirement,} \quad (80)$$

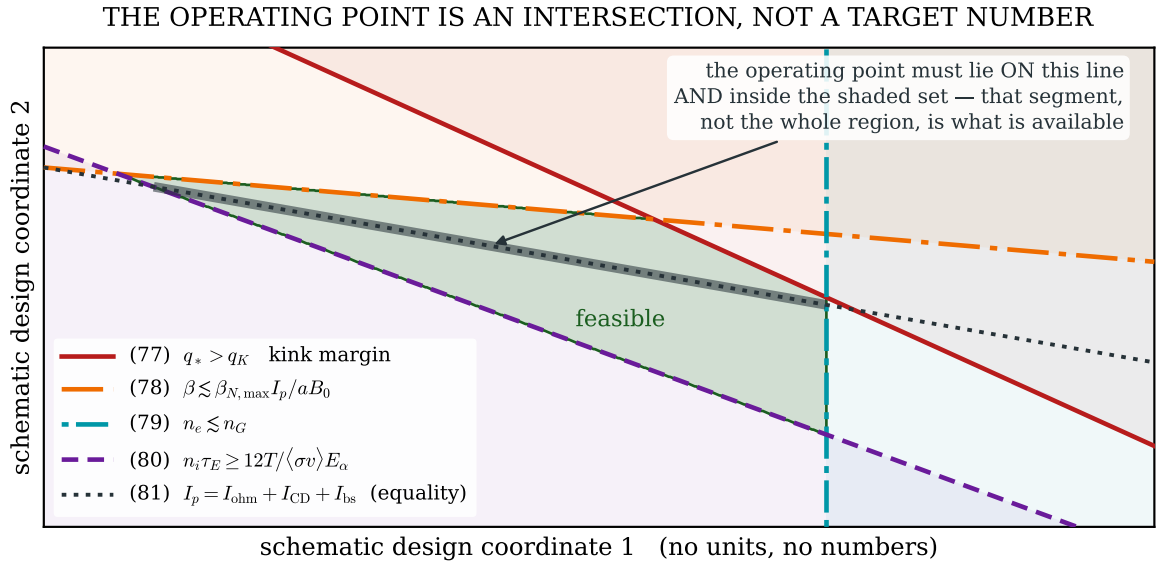
$$I_p = I_{\text{ohm}} + I_{\text{CD}} + I_{\text{bs}} \quad \text{current accounting.} \quad (81)$$

The notation $\lesssim$ in the second and third lines is deliberate. Those limits are modelled or empirical margins, not exact equalities. The last line is an identity, but its terms depend on operating mode and profiles. The first and fourth lines are inequalities derived from explicit physical balances, with a stability threshold or reactivity curve still to be specified.

This set is why the thesis refuses to present a tokamak as a collection of parts. Increasing density might help Eq. (80) but challenge Eq. (79); increasing current can improve one normalized beta bound but challenge Eq. (77); improving confinement changes needed heating and current drive. The correct question is whether the inequalities overlap with usable margin.

PLACE IMAGE HERE — Constraint-map sketch: the ordinary tokamak operating window. Create a schematic, clearly labelled overlap diagram rather than a numerically calibrated plot. Show five boundaries corresponding to Eqs. (77)–(81) and one small shaded feasible region. Include a warning that the plot is conceptual until a named confinement scaling, stability model, and profile model are supplied.

Design consequence. The next stage is not to add more variables randomly. It is to take the exact set just derived and reconstruct, line by line, how Freidberg’s 2015 reference design attempts to close it.



CONCEPTUAL. The axes are two schematic design coordinates, not physical variables, and no boundary position here is calibrated. The five conditions do not share a common two-dimensional space — q_* and the ledger are not functions of (n, T) — so plotting them on real axes would be a lie of convenience. What IS asserted is the topology: four inequalities bound a small region, and the fifth is an equality inside it. The picture stays conceptual until a NAMED confinement scaling, stability model and profile model are supplied.

Figure 24: Constraint-map sketch: the ordinary tokamak operating window. *Idealized schematic; not a fabrication drawing.*

42. The primary parameter set is not complete yet

The conventional tokamak has appeared in a causal order. Fusion temperature required a nonmaterial container. Charged particles made magnetic steering possible. Open field lines forced the torus and TF coils; drifts and pressure forced the plasma current, central solenoid, PF coils, and shaped equilibrium; transformer limits and power balance forced heating, current drive, and the distinction between pulsed and non-inductive operation. Shape was added only after its geometric and control price was visible.

The canonical target is Freidberg’s Table-1 set,

$$a, R_0, \kappa, b, c, T, n, p, \tau_E, B_0, \beta, I, q_*, f_B.$$

This draft has derived the physical role of $a, R_0, \kappa, T, n, p, \tau_E, B_0, \beta, I, q_*$, and f_B . It has deliberately also introduced supporting quantities—for example $A, \delta, q(r), \beta_N, H, I_{CD}$, and Q_{plasma} —because they make the causal argument intelligible. They are not additional primary Freidberg parameters or automatically independent design knobs. The two missing Table-1 hardware parameters are b (blanket-region thickness) and c (TF-magnet thickness). They must earn their place before numerical Freidberg closure begins.

Supporting quantity — remaining Part-I task. The next physical chain is neutron escape to first wall, blanket, shielding, and then the TF magnet. It must explain why the blanket-region thickness b and TF-magnet thickness c are indispensable primary parameters before the 2015 numerical reactor argument is reconstructed.

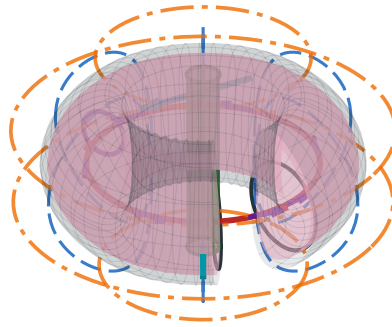
PLACE IMAGE HERE — Construction-atlas checkpoint: before the nuclear island. Update the cumulative conventional-tokamak atlas through state 11. Add a margin legend linking each physical component to its derived variable: TF coils to B_0 ; solenoid to $\Delta\Phi_{\text{link}}$ and I_{ohm} ; PF coils to κ, δ, ψ ; plasma to p, n, T, I_p, q ; heating/current-drive ports to P_{aux}, I_{CD} ; and feedback to vertical control. Leave a labelled future radial-space budget for first wall, blanket region b , shield, and TF-magnet thickness c ; do not draw them as completed components yet. Exclude spherical-tokamak geometry.

Design consequence. Before calculating a reactor, the next pages must derive the nuclear island and the toroidal-field-magnet thickness. Only after b and c have joined the same causal parameter set can the thesis begin Freidberg’s numerical engineering-and-nuclear-physics closure.

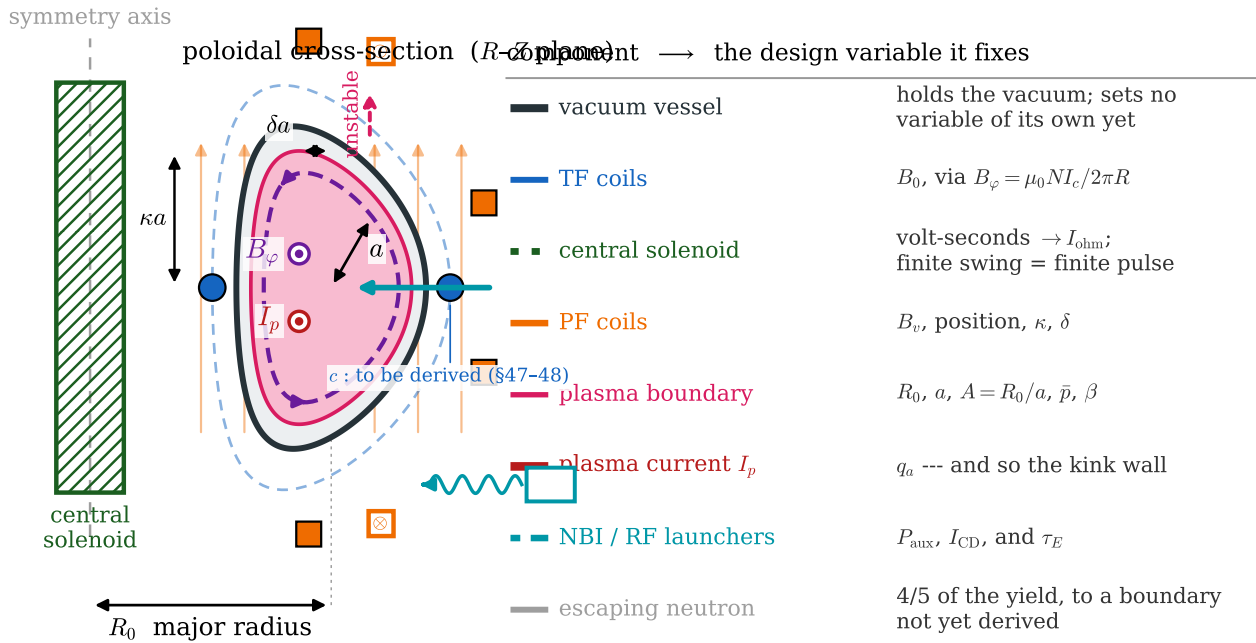
Atlas — the conventional tokamak, and what each part buys

every component derived so far, and the design variable it puts into the Part II audit. No later reactor hardware.

perspective view (vessel cut open)



- vacuum vessel (chamber wall)
- poloidal-field (PF) coils
- plasma / fuel region
- magnetic field $\mathbf{B}$
- toroidal-field (TF) coils
- plasma current I_p
- central solenoid (CS)
- auxiliary heating ports



⊙ : the vector points out of the page

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 25: Atlas. Every component derived so far, and the design variable each one puts into the audit that follows. No later reactor hardware appears.

43. A magnetic bottle does not catch the neutron

The D–T reaction in Eq. (1) has two products with very different fates. The charged alpha particle remains magnetically guided and can transfer energy back to the plasma. The neutron has no electric charge. Its Lorentz force is exactly zero:

$$\mathbf{F}_n = 0 \cdot (\mathbf{E} + \mathbf{v}_n \times \mathbf{B}) = 0.$$

It leaves the magnetic surfaces in a nearly straight line and reaches the first material it encounters. This is not an engineering inconvenience added after the plasma is designed. It is an unavoidable consequence of choosing the D–T reaction.

The neutron carries 14.1 MeV, approximately four fifths of the fusion energy. A power-producing machine must therefore do four jobs outside the plasma: survive the first impact, slow and capture neutrons, breed replacement tritium, and protect the magnets and vacuum boundary from the remaining radiation. These jobs force a radial region between plasma and TF coil. We call its total inboard/outboard thickness b .

Primary Freidberg parameter — b (blanket-region thickness). The D–T neutron is neutral, so magnetic confinement cannot retain it. A reactor must reserve radial space for a first wall, neutron multiplier or moderator where used, tritium-breeding material, shielding, and a vacuum boundary. The parameter b records the total required radial build of this nuclear island. It is not an arbitrary clearance and it is not, by itself, a complete blanket design.

PLACE IMAGE HERE — Reset physics sketch: charged alpha versus neutral neutron. At manuscript state 12, show the same D–T reaction at the plasma edge. The 3.5 MeV alpha follows a curved magnetic guiding path back into the plasma; the 14.1 MeV neutron travels straight across magnetic field lines toward a not-yet-detailed radial material stack. Label "magnetic field does not act on a neutral." Keep all previously earned conventional-tokamak components visible only where needed for orientation.

This brief asks for the same D–T reaction that cumulative state 10 already shows in Figure 16 — charged alpha retained, neutral neutron leaving toward a boundary not yet drawn. Under the standing rule that redundancy is fixed rather than expanded, no second figure is drawn; the two quantities this brief would add (the mass-ratio energy split and the alpha gyro-radius) belong in the prose or in that figure's caption.

Design consequence. A fusion reactor is not merely a plasma plus magnets. The neutron forces a nuclear island, and that island consumes radial space precisely where a toroidal-field coil would prefer to be close to the plasma.

Model and limits. We use an intentionally simple, one-dimensional neutron model to derive the meaning and scale of b . Real blanket design requires material-specific transport, heat removal, structural design, activation analysis, and tritium-accounting margins. Those are later engineering calculations, not facts we may silently hide in one number.

44. Why the nuclear island has a thickness

First consider the slowing-down part of the problem. In a homogeneous material with target number density N and an effective interaction cross section σ_s , the mean distance between relevant collisions is

$$\lambda_s = \frac{1}{N\sigma_s}. \quad (82)$$

This follows directly from probability. In a thin distance dx , a neutron has collision probability $N\sigma_s dx = dx/\lambda_s$. Thus the chance of no collision decreases exponentially with distance.

For a transparent first model, suppose each collision removes a fixed fraction ξ of the neutron energy on average. Over many collisions, the average energy loss per distance can be represented by

$$\frac{dE}{dx} = -\frac{\xi E}{\lambda_s}. \quad (83)$$

Separating variables gives

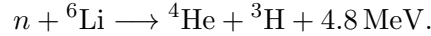
$$\frac{dE}{E} = -\frac{\xi dx}{\lambda_s} \implies E(x) = E_0 \exp\left(-\frac{\xi x}{\lambda_s}\right).$$

To reduce a 14.1 MeV neutron from E_0 to a chosen capture-friendly energy E_c , the associated slowing-down distance is therefore

$$b_{\text{slow}} = \frac{\lambda_s}{\xi} \ln\left(\frac{E_0}{E_c}\right). \quad (84)$$

The logarithm matters: a large energy ratio does not require a proportional thickness, but it does require several interaction lengths. Equation (84) already shows why a thin wall cannot be a reactor solution.

Now consider breeding. One useful reaction is



The emitted tritium replaces fuel consumed by D-T fusion. If slow neutrons are absorbed with macroscopic probability per length $\Sigma_B = N_6\sigma_B$, their surviving flux Γ obeys

$$\frac{d\Gamma}{dx} = -\Sigma_B \Gamma, \quad \Gamma(x) = \Gamma(0)e^{-x/\lambda_B}, \quad \lambda_B = \frac{1}{\Sigma_B}.$$

Capturing a chosen fraction f_{cap} then requires

$$b_{\text{breed}} = -\lambda_B \ln(1 - f_{\text{cap}}). \quad (85)$$

Both λ_s and λ_B depend on material and neutron energy. Thus Eqs. (84)–(85) derive a physical requirement, not a material-independent numerical answer.

Supporting quantity — neutron mean free path. The quantities λ_s and λ_B are supporting transport lengths, not Primary Freidberg parameters. They explain why the primary radial-build parameter b cannot be taken arbitrarily small.

45. From microscopic collisions to the reactor dimension b

The nuclear island is a sequence of functions, not a single substance. At the level needed for a first reactor layout, write

$$b = b_{\text{FW}} + b_{\text{mult}} + b_{\text{slow}} + b_{\text{breed}} + b_{\text{shield}} + b_{\text{vac}}. \quad (86)$$

Here b_{FW} is the first-wall allowance; b_{mult} represents any neutron-multiplying region; b_{slow} and b_{breed} are motivated by the preceding transport calculation; b_{shield} protects coils and surrounding structures; and b_{vac} is the vacuum-boundary allowance. Some concepts combine or reorder these functions. Equation (86) is deliberately a functional ledger, not a claim that every reactor has six sharply separated layers.

The first wall also sees a power-density demand. In the large-aspect-ratio circular approximation, a torus with major radius R_0 and minor radius a has surface area

$$S_{\text{wall}} \simeq (2\pi R_0)(2\pi a) = 4\pi^2 R_0 a.$$

If P_n is the neutron power and W_n is an allowed average neutron wall loading,

$$\frac{P_n}{4\pi^2 R_0 a} \leq W_n \quad \implies \quad R_0 a \geq \frac{P_n}{4\pi^2 W_n}. \quad (87)$$

Elongation and detailed surface shape modify this estimate; they do not remove its message. At fixed fusion-neutron power, a smaller first-wall area means a harsher wall loading. Nuclear performance therefore couples the two plasma dimensions R_0 and a even before we make a plasma-transport calculation.

The more immediate geometric consequence of b is radial crowding. On the inboard side, the blanket lies between the plasma edge and the TF coil. It displaces the coil toward the symmetry axis, where a toroidal field is stronger. The next section turns that qualitative statement into an equation.

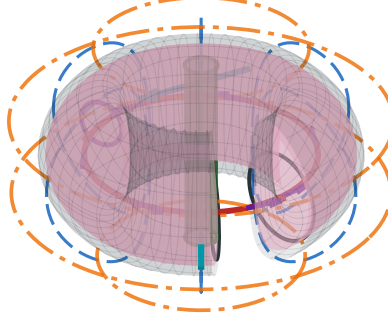
PLACE IMAGE HERE — Cumulative construction state 12: the earned nuclear island. Update the atlas with the plasma, vacuum region, first wall, blanket/breeding region, shield, and vacuum boundary as a radial stack. Use a bracket labelled b for the complete first-wall-to-vacuum-boundary build; do not label a single inner layer as b . Show one neutral-neutron path depositing energy and one tritium-breeding reaction. Retain the already-earned TF coil as a solid component, but annotate its radial bracket as c : thickness derived next, rather than assigning it a thickness. Exclude spherical-tokamak geometry and a divertor.

Design consequence. The blanket thickness b has become a geometric actor. It sets a minimum separation between the fusion plasma and the field-producing magnet; it cannot be optimized independently of R_0 , a , B_0 , and the eventual coil thickness c .

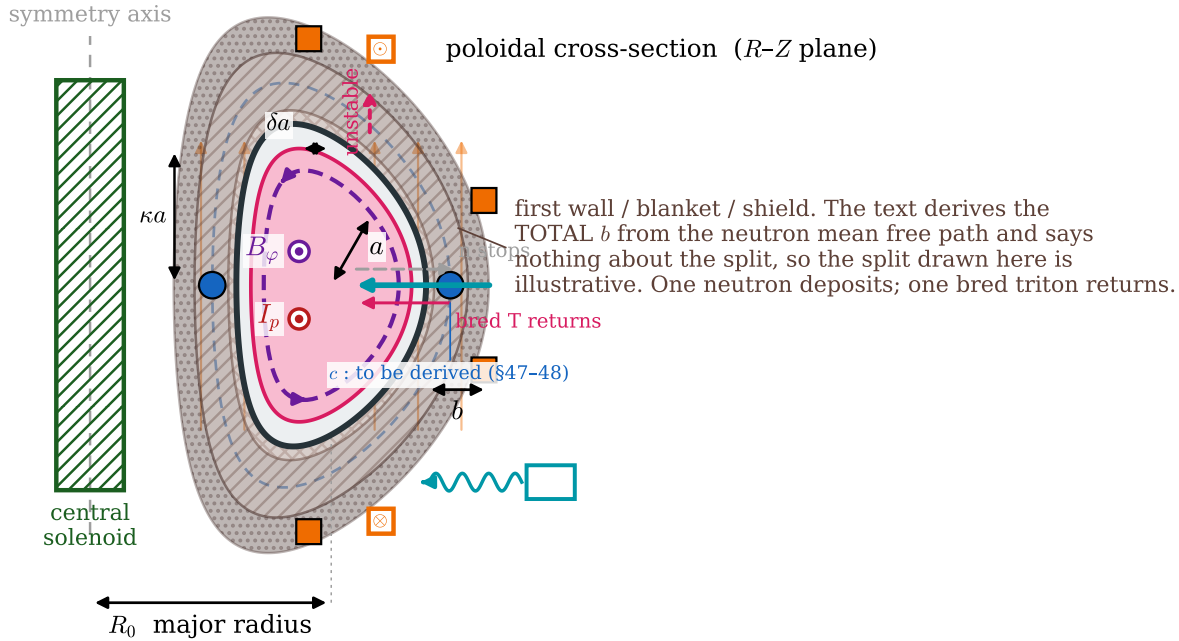
Stage 13 — the earned nuclear island

the neutron that left at stage 10 has to stop somewhere: first wall, blanket, shield. Their total thickness is b .

perspective view (vessel cut open)



- | | |
|------------------------------|---------------------------|
| vacuum vessel (chamber wall) | poloidal-field (PF) coils |
| plasma / fuel region | magnetic field B |
| toroidal-field (TF) coils | plasma current I_p |
| central solenoid (CS) | auxiliary heating ports |



⊙ : the vector points out of the page

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 26: Machine state: the earned nuclear island. The neutron that left has to stop somewhere — first wall, blanket, shield. Their total thickness is b . No layer split is asserted beyond what the text derives.

46. The field gets stronger as the coil moves inward

The continuous-coil model already gave

$$B_\varphi(R) = \frac{\mu_0 N I_c}{2\pi R}.$$

At the plasma centre $R = R_0$, write this as B_0 . The inboard edge of the plasma is at $R_0 - a$. After the nuclear island and an inboard TF-coil build have been added, the innermost high-field location lies approximately at

$$R_{\text{in}} = R_0 - a - b - c. \quad (88)$$

The sign is important: moving inward means decreasing R , which increases the toroidal field for fixed coil current.

Using $B_\varphi(R)R = B_0 R_0$, the maximum field in this simplified radial build is

$$B_{\text{max}} = \frac{B_0 R_0}{R_0 - a - b - c}. \quad (89)$$

Equivalently,

$$B_0 = B_{\text{max}} \left(1 - \frac{a + b + c}{R_0} \right). \quad (90)$$

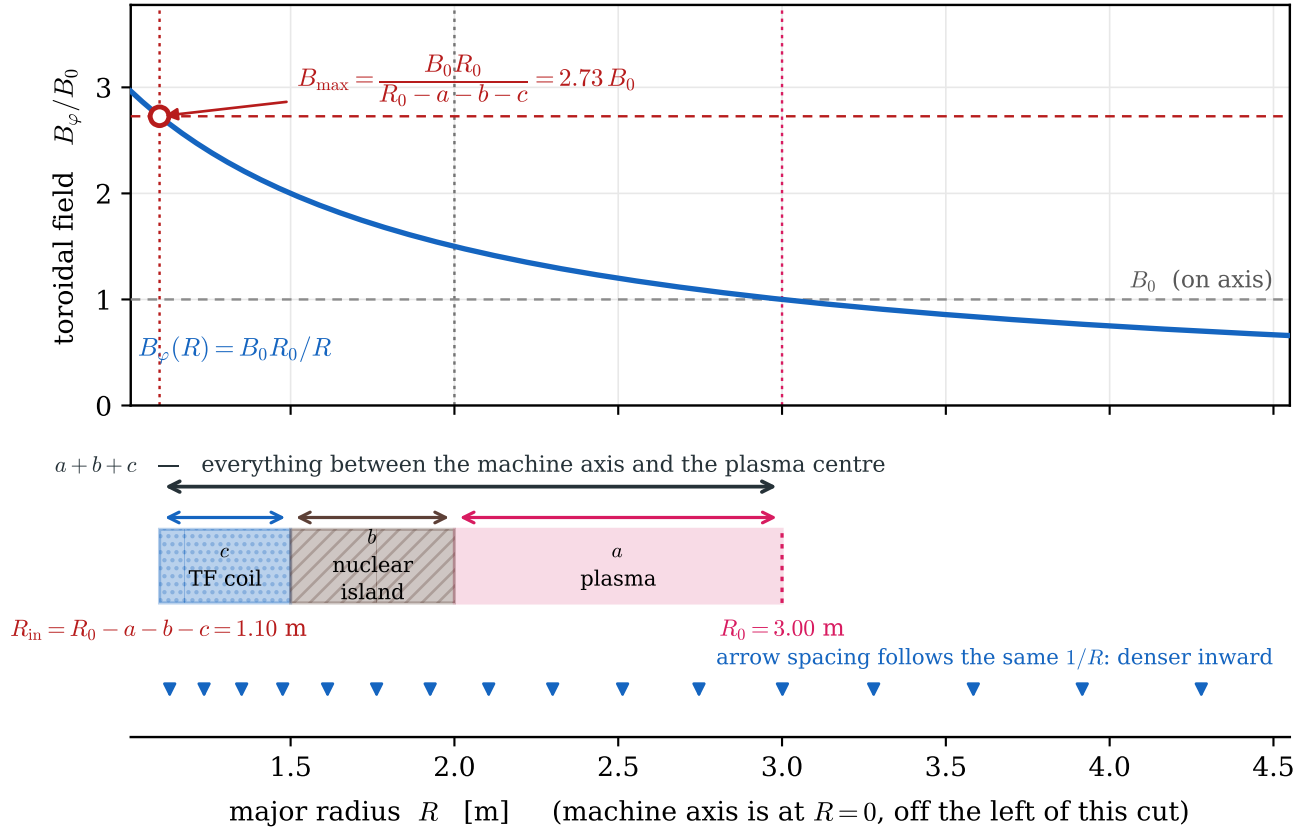
This is the first compact expression of the radial-build problem. A larger plasma minor radius, a thicker nuclear island, or a thicker TF coil leaves less inboard radius. At fixed material limit B_{max} , each reduces the attainable central field B_0 .

Supporting quantity — inboard radial-build fraction. It is often convenient to define $\epsilon_{\text{build}} = (a+b+c)/R_0$. It is a derived bookkeeping ratio, not another primary parameter. Equation (90) says simply $B_0 = B_{\text{max}}(1 - \epsilon_{\text{build}})$.

The equation also explains why a tokamak cannot be designed by choosing a desirable B_0 and adding the blanket and magnets afterwards. Those supposedly external components feed back directly into the field available to the plasma.

PLACE IMAGE HERE — Reset geometry sketch: why inboard build limits central field. Draw a horizontal midplane cut through a conventional tokamak. Mark, from the plasma centre inward: a , the full nuclear-island bracket b , the TF-coil bracket c , and $R_{\text{in}} = R_0 - a - b - c$. Show toroidal-field arrows with increasing density inward. Place Eqs. (89) and (90) beside the sketch.

THE INBOARD BUILD SETS THE FIELD: every millimetre inboard is spent twice



Rearranged, Eq. (90): $B_0 = B_{\max}(1 - (a+b+c)/R_0) = 2.73 \times (1 - 1.90/3.00) = 1.00 B_0$ — the identity, evaluated.

The consequence is the one the primer draws out: at a FIXED material limit $B_{\max}$, a larger plasma, a thicker nuclear island or a thicker coil each reduce the attainable B_0 . The three compete for the same inboard radius, and none of them is free.

a , b , c here are ILLUSTRATIVE: b is not derived until the nuclear-island section and c not until the coil-stress section.

Figure 27: Reset geometry sketch: why inboard build limits central field. *Idealized schematic; not a fabrication drawing.*

47. A field coil must carry current and survive its own force

A magnetic field stores energy. In vacuum its energy density is

$$u_B = \frac{B^2}{2\mu_0}. \quad (91)$$

One way to obtain this expression is to build a solenoid gradually. Its magnetic energy is $U = \frac{1}{2}LI^2$; dividing by volume and substituting the ideal-solenoid relations $L = \mu_0 N^2 A/\ell$ and $B = \mu_0 NI/\ell$ gives $U/(A\ell) = B^2/(2\mu_0)$.

The same scale appears as a magnetic pressure. If a field-filled region expands by a small volume dV at fixed flux, the released magnetic energy supplies mechanical work $p_B dV$. The resulting stress scale is

$$p_B = \frac{B^2}{2\mu_0}. \quad (92)$$

At high field this is enormous: the dependence is quadratic, so doubling field quadruples this stress scale. A TF coil is therefore not just superconducting cable. It needs a structural load path that prevents the winding pack from being crushed, pulled apart, or driven beyond allowable strain.

For a transparent thickness estimate, let a representative magnetic force be $F \sim p_B A_{\text{load}}$ and let structural material of cross-sectional area A_{str} be limited to stress σ_{allow} . Force balance requires

$$\sigma_{\text{allow}} A_{\text{str}} \gtrsim p_B A_{\text{load}}.$$

If the geometry converts these areas into a representative support length ℓ_s , then the structural contribution obeys the scaling

$$c_{\text{str}} \sim \frac{B_{\text{max}}^2}{2\mu_0 \sigma_{\text{allow}}} \ell_s \times (\text{geometry factor}). \quad (93)$$

The geometry factor is not a nuisance to be discarded: detailed TF-coil engineering calculates it from tensile, centering, bending, and wedging loads. Equation (93) is the load-bearing physical reason that the parameter c exists at all.

Model and limits. This section establishes the stress scaling, not a certified coil design. Real designs use three-dimensional force analysis, material allowables at operating temperature and fluence, insulation and cooling channels, joints where applicable, fatigue limits, and safety margins.

48. The other part of coil thickness: enough current-carrying area

Stress is not the only reason a coil occupies space. Ampère’s law requires a total enclosed current approximately

$$NI \simeq \frac{2\pi R_{\text{in}} B_{\text{max}}}{\mu_0}. \quad (94)$$

The product NI , called ampere-turns, cannot be carried through zero area. If J_{wp} is the allowed current density averaged over the complete winding pack (conductor, stabilizer, insulation, cooling paths, and structural allocation), the required current-carrying area scales as

$$A_{\text{wp}} \gtrsim \frac{NI}{J_{\text{wp}}}. \quad (95)$$

For a representative coil height h_{wp} , this yields a winding-pack thickness scale

$$c_{\text{wp}} \sim \frac{2\pi R_{\text{in}} B_{\text{max}}}{\mu_0 J_{\text{wp}} h_{\text{wp}}}. \quad (96)$$

This is a scaling estimate, not a prescription for the cross-sectional shape of a real coil. Its purpose is to expose the independent current-density reason that coil thickness cannot be wished away.

The full primary parameter is therefore a radial ledger:

$$c = c_{\text{str}} + c_{\text{wp}} + c_{\text{ins}} + c_{\text{cool}} + c_{\text{margin}}. \quad (97)$$

The final three terms acknowledge insulation, cooling, manufacturing tolerances, and design margin. Their values must come from an explicit technology model. The first two terms, Eqs. (93) and (96), are the two irreducible physical demands: survive the field and produce the field.

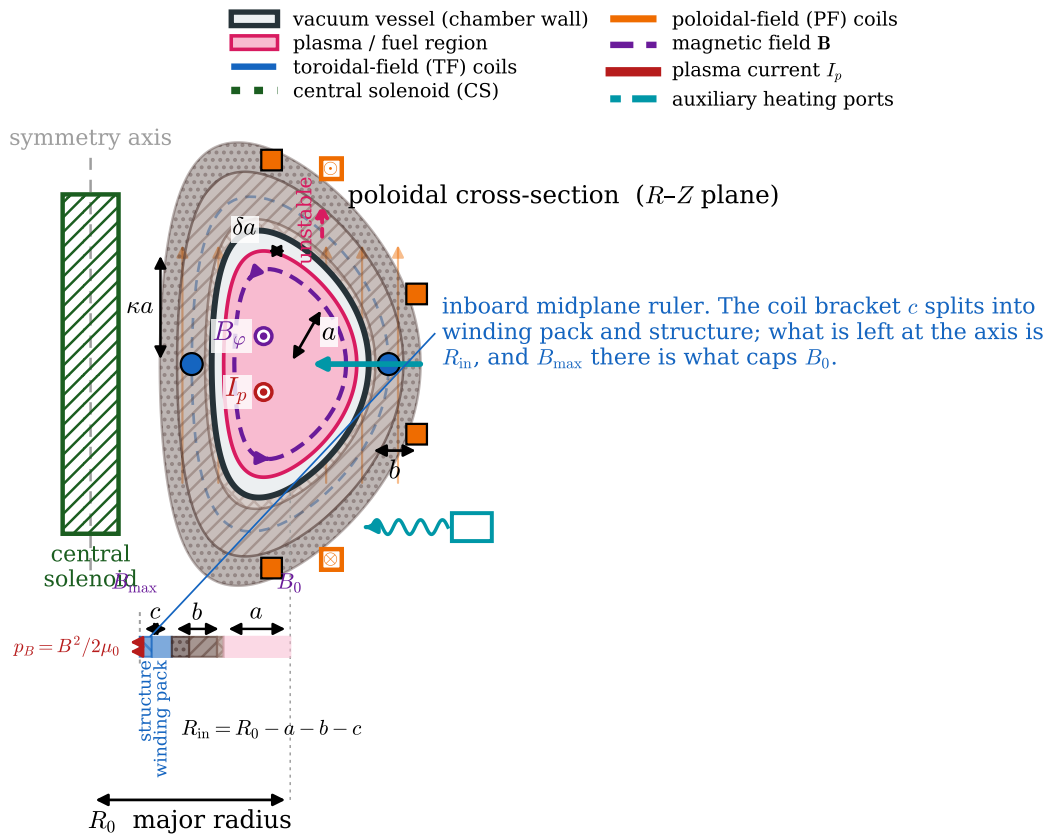
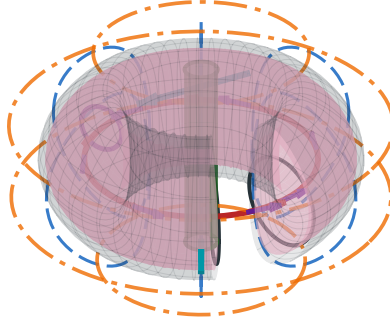
Primary Freidberg parameter — c (TF-magnet thickness). The toroidal-field magnet needs radial space both for structure that carries magnetic stress and for a winding pack that carries the ampere-turns required by Ampère’s law. Its total radial build c therefore couples mechanical strength and current density to B_{max} , B_0 , R_0 , a , and the nuclear-island thickness b .

PLACE IMAGE HERE — Cumulative construction state 13: the complete radial build. Keep the already-earned TF coil solid. In a labelled inboard midplane cut, retain plasma radius a and nuclear-island bracket b ; replace the open c annotation with a derived split into “winding pack” and “structure/support.” Add the field labels B_0 at R_0 and B_{max} at R_{in} . Use a small force arrow or stress arrows on the inboard coil. This is still a conventional tokamak, not an ST.

Stage 14 — the complete radial build

$R_{\text{in}} = R_0 - a - b - c$ is what the inboard side leaves for the coil, and that is what caps B_0

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 28: Machine state: the complete radial build. $R_{\text{in}} = R_0 - a - b - c$ is what the inboard side leaves for the coil, and that is what caps B_0 .

49. The high-field door is now visible, but not yet opened

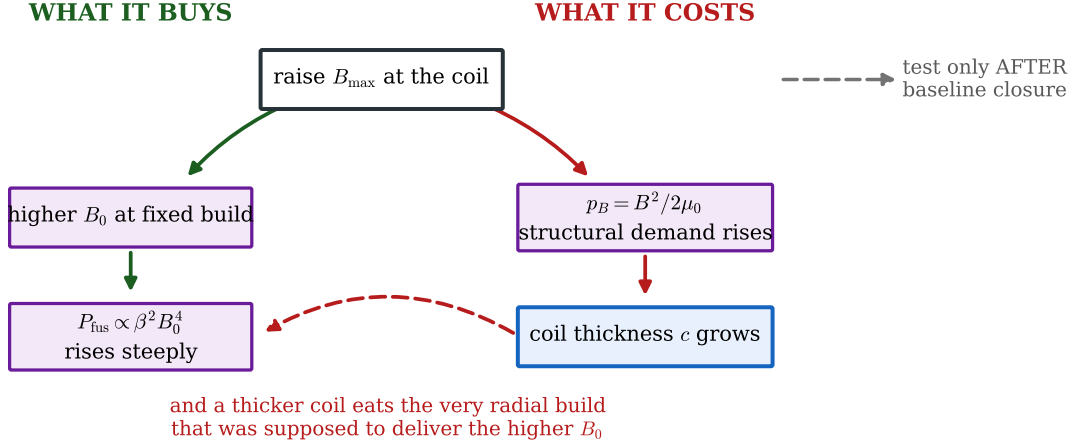
We can now say precisely what a higher-field magnet changes. At fixed radial geometry, Eq. (90) makes a higher allowable $B_{\max}$ available as a higher central field B_0 . But Eqs. (93) and (96) also show the price: field increases magnetic stress quadratically and imposes a technology-dependent current-density requirement. "Use a stronger magnet" is not a solution until both sides of this trade are analysed.

This is the correct logical location for high-temperature superconductors (HTS). They are not needed to explain why a tokamak has TF coils, and they should not appear as an unexplained early feature. They matter later if the baseline, conventional field limit makes the plasma-current closure fail. At that point a conductor able to operate at the required field and current density can raise $B_{\max}$; the rest of the thesis must test whether the resulting improvement in current, stability, and current drive outweighs the extra magnet build and cost.

Design consequence. HTS is an enabling option for a specified high-field reactor strategy, not a universal first-principles requirement for every conceivable tokamak. The thesis will earn the stronger design claim only after the classical baseline has been reconstructed and shown to require a field beyond its assumed conventional-magnet limit.

Supporting quantity — material field limit $B_{\max}$. The maximum inboard coil field is an engineering constraint rather than a Table-1 reactor parameter. It governs the primary parameters B_0 and c through the radial build. Later, a technology model will compare a conventional superconducting limit with an HTS-enabled high-field limit, including stress, protection, joints, neutron tolerance, and maintenance.

PLACE IMAGE HERE — Reset trade-off sketch: field is performance and burden. Create a two-sided causal diagram, not a sales graphic. Left: higher $B_{\max}$ permits higher B_0 at fixed radial build. Right: $p_B = B^2/(2\mu_0)$ raises structural demand and can increase c . Add a dashed arrow labelled "test only after baseline closure" to the later plasma-current/stability analysis. Do not yet draw an HTS brand, a spherical tokamak, or a numerical performance claim.



Explanatory sketch. No magnet technology, vendor or numerical performance claim is implied; this is the shape of the trade-off only.

Figure 29: Reset trade-off sketch: field is performance and burden. *Idealized schematic; not a fabrication drawing.*

50. The primary design vocabulary is complete

The tokamak has now revealed every member of the primary parameter set:

$$a, R_0, \kappa, b, c, T, n, p, \tau_E, B_0, \beta, I, q_*, f_B.$$

They do not have the same logical status. Some describe geometry; some are plasma state variables; some record magnetic/current balance; and b, c are radial builds forced by nuclear transport and magnet engineering. Nor are they independent. For example, the causal relations already derived include

$$p = 2nT, \quad q_* \propto \frac{a^2 B_0}{R_0 I}, \quad B_0 = B_{\max} \left(1 - \frac{a + b + c}{R_0} \right).$$

The purpose of writing them together is not to solve the reactor yet. It is to make clear that a proposed change in one component propagates through the whole machine.

The next part of the thesis can therefore enter Freidberg’s 2015 argument on honest terms. It will select explicit engineering and nuclear assumptions, express the primary parameters as coupled functions rather than isolated knobs, and then compare the demanded plasma state with four operational tests: Greenwald density, Troyon beta, kink safety factor, and achievable bootstrap fraction. The core question will be whether all four inequalities can hold simultaneously for a conventional baseline.

Design consequence. Only now is it legitimate to begin the numerical closure. If the baseline fails, the later high-field/HTS discussion will be an earned response to a specific contradiction, not an imported preference.

Model and limits. The open manuscript by J. P. Freidberg, F. J. Mangiarotti, and J. Minervini (2015), especially its Sections 3–6, is used here as a freely available reference design. Its one-dimensional blanket and analytic TF-coil models motivate the present derivation; the thesis will state their assumptions before reproducing any numerical result.

51. Entering the 2015 argument: a model, not a verdict

We now change tasks. The first fifty pages asked, "Why must the machine contain these parts?" The next task is, "If a particular conventional-reactor model is required to make a specified amount of power, what plasma state does that model demand?" This is the calculation made analytically by Freidberg, Mangiarotti, and Minervini in 2015.

Their reference design is intentionally simplified. It specifies an electric output, an allowed neutron wall loading, a thermal conversion efficiency, a maximum inboard coil field, magnet stress and winding-pack limits, and a cap on recirculating power. It then asks whether a conventional plasma can simultaneously meet the resulting density, beta, kink-safety-factor, and bootstrap-current requirements.

Design consequence. The following conclusion, if reproduced, will be conditional: there is no self-consistent reactor within the stated baseline assumptions. It will not say that no tokamak can ever work. That distinction is where honest engineering begins.

Model and limits. Every numerical input in this reconstruction will be visibly tagged as either an economic choice, an engineering limit, a nuclear-data/model choice, or an empirical plasma-physics fit. Changing one of them defines a different reactor case; it is not a mathematical error.

52. The reference design asks the machine to do a specific job

For the 2015 baseline, the primary imposed mission choices are

$$P_E = 1000 \text{ MW}_e, \quad W_n = 4 \text{ MW m}^{-2}, \quad \eta_T = 0.40.$$

P_E is desired electric output, W_n is the chosen upper bound on average neutron wall loading, and η_T converts fusion heat into electric power:

$$P_E = \eta_T P_F. \tag{98}$$

The subscript F denotes the total thermal power credited to fusion and blanket reactions in the paper's accounting model. The point of displaying these numbers is not to bless them. A 1000 MW_e plant, a 4 MW m⁻² wall limit, and 40% conversion efficiency are assumptions to be audited.

The reference magnet model also takes

$$B_{\max} = 13 \text{ T}, \quad \sigma_{\text{allow}} = 600 \text{ MPa}, \quad J_{\text{wp}} = 20 \text{ MA m}^{-2}.$$

These are a baseline conventional-superconductor and structural model, not claims about the best magnets available today. They are precisely the assumptions later challenged by the high-field option.

Supporting quantity — reference inputs. The quantities P_E , W_n , η_T , $B_{\max}$, σ_{allow} , and J_{wp} are inputs to a chosen design case. They guide the determination of the primary parameter set; they are not additional Table-1 parameters.

53. Power and wall loading turn two dimensions into one family

For the shaped boundary, the source uses the standard mean-square ellipse-perimeter approximation. Revolving it around the major radius gives the first-wall area

$$S_{\text{wall}} \simeq 4\pi^2 R_0 a \left(\frac{1 + \kappa^2}{2} \right)^{1/2}. \quad (99)$$

This agrees with the circular result $4\pi^2 R_0 a$ when $\kappa = 1$. It is an analytic shape approximation, not a replacement for an actual first-wall surface.

The D–T neutron carries $E_n = 14.1$ MeV of the fusion energy $E_{\text{DT}} = 17.6$ MeV. In the reference energy accounting,

$$P_n = \frac{E_n}{E_F} P_F = \frac{E_n}{E_F} \frac{P_E}{\eta_T}.$$

Equating neutron power to the allowed wall loading times the wall area, $P_n = W_n S_{\text{wall}}$, gives

$$R_0 a = \frac{E_n P_E}{4\pi^2 E_F W_n \eta_T} \left(\frac{2}{1 + \kappa^2} \right)^{1/2}. \quad (100)$$

All quantities on the right are chosen inputs or previously earned parameters. Thus the design does not choose R_0 and a independently:

$$R_0(a) = \frac{C_{\text{wall}}}{a}, \quad C_{\text{wall}} \equiv \frac{E_n P_E}{4\pi^2 E_F W_n \eta_T} \left(\frac{2}{1 + \kappa^2} \right)^{1/2}. \quad (101)$$

This is why Freidberg can make a one-parameter scan. After fixing the mission and the blanket/magnet model, each trial minor radius implies a corresponding major radius.

Model and limits. Equation (99) uses a simple perimeter approximation and an average wall loading. A final reactor analysis needs local neutron transport and the actual first-wall surface. The one-parameter logic survives those refinements.

54. The radial build makes the field a coupled result

The preceding page makes R_0 a function of a . For its particular analytic TF-coil calculation, Freidberg defines the inboard field-build fraction

$$\epsilon_B = \frac{a + b}{R_0}$$

and writes

$$B_0 = B_{\max}(1 - \epsilon_B).$$

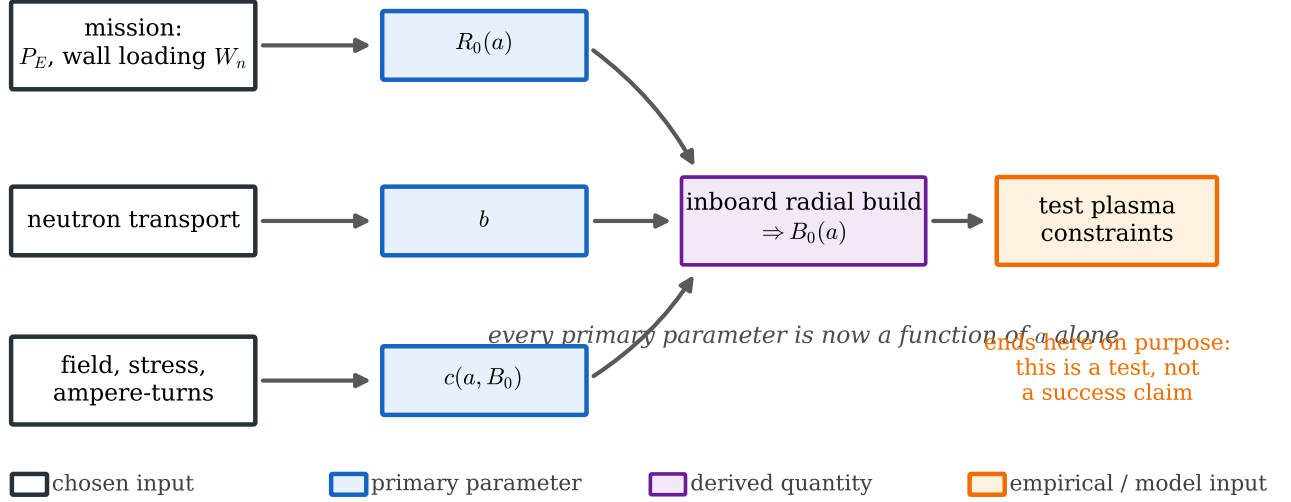
This is the source model's field convention: c is then found from its stress and winding-pack equations, rather than inserted into ϵ_B . The physical radial ledger on pages 46–48 remains useful for understanding why c exists; reproducing the source's numerical scan requires us not to mix that fuller ledger with its analytic convention. Its logical structure is more important than its algebra:

$$\begin{aligned} \text{mission and wall loading} &\implies R_0(a), \\ \text{neutron transport} &\implies b, \\ \text{field, stress, and ampere-turns} &\implies c(a, B_0), \\ \text{inboard radial build} &\implies B_0(a). \end{aligned}$$

The central field is therefore neither a free wish nor merely a property of the superconductor. It is the outcome of a coupled geometric and mechanical problem.

This is also the first place to be alert to a future HTS comparison. If a different conductor technology changes the credible $B_{\max}$ or J_{wp} , it changes the solution for c and therefore the accessible B_0 . We will not assume that this improves the reactor until the baseline plasma constraints have been tested.

PLACE IMAGE HERE — Causal ledger diagram: how the reference design reduces to a scan in a . Make a compact flow diagram of the four arrows displayed above. Use a distinct style for chosen inputs, primary parameters, and derived supporting quantities. End at "test plasma constraints," not at a success claim. This is a reset explanatory figure, not a cumulative machine image.



Explanatory sketch. The arrows are the manuscript's own causal chain; no numerical claim is made here.

Figure 30: Causal ledger diagram: how the reference design reduces to a scan in a . *Idealized schematic; not a fabrication drawing.*

55. Fusion power fixes the required pressure

The D–T reaction rate in an equal fuel mixture is

$$\mathcal{R} = \frac{n^2}{4} \langle \sigma v \rangle (T),$$

because $n_D = n_T = n/2$. Multiplying by the energy released per reaction and integrating over plasma volume gives

$$P_{\text{fus}} = \frac{\langle \sigma v \rangle (T) E_{\text{DT}}}{4} \int n^2 dV. \quad (102)$$

For a first transparent estimate, use volume-averaged n and T :

$$P_{\text{fus}} \simeq \frac{\langle \sigma v \rangle (T) E_{\text{DT}}}{4} n^2 V.$$

The equal-temperature quasineutral model gave $p = 2nT$, hence

$$P_{\text{fus}} \simeq \frac{\langle \sigma v \rangle (T) E_{\text{DT}}}{16T^2} p^2 V. \quad (103)$$

Power balance therefore determines a required pressure:

$$p_{\text{req}} \simeq \left[\frac{16T^2 P_{\text{fus,req}}}{\langle \sigma v \rangle (T) E_{\text{DT}} V} \right]^{1/2}. \quad (104)$$

This is a crucial conceptual reversal. The plasma physicist does not simply choose a pleasantly low pressure; the requested electric output and available volume demand one.

The 2015 reference model selects $T \simeq 14 \text{ keV}$ because the ratio $\langle \sigma v \rangle / T^2$ is near its favorable broad maximum for D–T fuel. Later we will reproduce that statement from a cited reactivity fit and replace the uniform approximation with the paper's declared density and temperature profiles.

56. Ignition turns pressure into a required confinement time

Fusion power does not remain as useful plasma heat. The neutron escapes; for D–T, the alpha fraction is

$$\frac{E_\alpha}{E_{\text{DT}}} = \frac{3.5}{17.6} \simeq \frac{1}{5}.$$

For an ignited plasma, alpha heating must replace the energy conducted and transported out of the core. Let W be thermal energy stored in the plasma and define

$$\tau_E = \frac{W}{P_{\text{loss}}}.$$

The ignition balance is $P_\alpha = P_{\text{loss}} = W/\tau_E$, or

$$\tau_{E,\text{req}} = \frac{W}{P_\alpha}. \quad (105)$$

In the equal-temperature model, the thermal energy density is $w = \frac{3}{2}(n_e + n_i)T = 3nT = \frac{3}{2}p$. Thus $W \simeq \frac{3}{2}pV$. With $P_\alpha \simeq P_{\text{fus}}/5$,

$$\tau_{E,\text{req}} \simeq \frac{15pV}{2P_{\text{fus}}}. \quad (106)$$

Substituting the pressure demanded by Eq. (104) gives the next causal chain:

$$P_E, W_n, \eta_T, a \implies R_0, V, p \implies \tau_{E,\text{req}}.$$

The remaining, decisive question is whether an empirical tokamak confinement law can deliver this τ_E without demanding an unacceptable plasma current. That is where the one indispensable plasma-physics input enters the 2015 design.

57. Before testing a limit: what kind of statement is it?

The four tests ahead do not all have the same epistemic status. It would be a serious error to place them in one box labelled "first principles." We will use three labels.

Derived relation	follows from stated equations and stated approximations; for example, $p = 2nT$ and $q_* \propto a^2 B_0 / (R_0 I)$.
Physics model	follows after adopting a specified equilibrium, profile, or kinetic model; it may be calculated but is not universal.
Empirical boundary	summarizes experimental operation or a regression; its coefficient and range are evidence, not a theorem.

Greenwald density is an empirical boundary. The practical Troyon beta coefficient is an experiment- and model-informed stability boundary. The kink threshold is a result of a specified MHD stability model, with profile and wall dependence. The bootstrap fraction requires a neoclassical kinetic model plus a current profile. Their physical motivation can be derived; their reactor-grade numerical coefficients cannot honestly be manufactured from the motivation alone.

The test itself is nevertheless elementary. For every constraint define a demand-to- allowance ratio,

$$\mathcal{R} \equiv \frac{\text{quantity demanded by the reference design}}{\text{quantity allowed by the stated constraint}}. \quad (107)$$

A baseline is admissible only if each relevant $\mathcal{R} \leq 1$. This notation forces us to name both numerator and denominator before announcing pass or failure.

Design consequence. The goal is not to make empirical plasma physics look more certain than it is. The goal is to expose exactly which evidence a conventional reactor would need to satisfy, and whether the same design point can satisfy all of it at once.

58. The one empirical input that determines the required current

The 2015 reference case uses an ELMy H-mode confinement regression. H-mode is a high-confinement tokamak regime with a strong edge transport barrier. ELMy means that the edge undergoes recurring edge-localized-mode (ELM) relaxations: short bursts that expel a small amount of edge energy and particles. The word identifies the experiment class behind the regression; it does not add an independently derived law.² In the declared units, the regression is

$$\tau_E = 0.145H \frac{I^{0.93} R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}}{P_\alpha^{0.69}} \text{ s.} \quad (108)$$

Here I is in MA, R_0 and a in m, $\bar{n}$ in 10^{20} m^{-3} , B_0 in T, P_α in MW, $A = 2.5$ is the mean ion mass used by the fit, and H is the confinement multiplier. The coefficient is meaningless if these units or the database/regime are changed.

Equation (108) is not deduced from the Lorentz force. It is the particular empirical closure selected by the paper. Its physical direction is intelligible: larger current and machine size correlate with better confinement; greater heating power correlates with more difficult power exhaust. But the exponents are measured regression exponents, not universal laws of nature.

Supporting quantity — reference confinement fit. Equation (108) is the explicit version of $\tau_E = H\tau_{E,\text{ref}}$ used in this reconstruction. It is the one direct plasma-performance input that turns the ignition requirement into a required current. The freely available source is Freidberg, Mangiarotti, and Minervini (2015), Eq. (28), linked on page 1 of this thesis.

Model and limits. The reference paper assumes $H = 1$ for its conventional baseline. We will retain that assumption only while reproducing that baseline, then vary it openly when testing the advanced-confinement option.

²A freely accessible ITER explanation describes H-mode's steep edge pressure region and the associated edge-localized modes: <https://www.iter.org/node/20687/progress-elm-physics-and-elm-control>.

59. Solving the confinement fit for the current

The ignition calculation has already supplied $\tau_{E,\text{req}}$. Set it equal to the right side of Eq. (108) and isolate the only unknown in this step, I :

$$I^{0.93} = \frac{\tau_{E,\text{req}} P_{\alpha}^{0.69}}{0.145 H R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}}.$$

Raising both sides to the power $1/0.93$ gives

$$I_{\text{req}} = \left[\frac{\tau_{E,\text{req}} P_{\alpha}^{0.69}}{0.145 H R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}} \right]^{1/0.93} \text{ MA}. \quad (109)$$

There is no mystery step here: the current is whatever value makes the selected empirical confinement law equal the confinement time required by ignition.

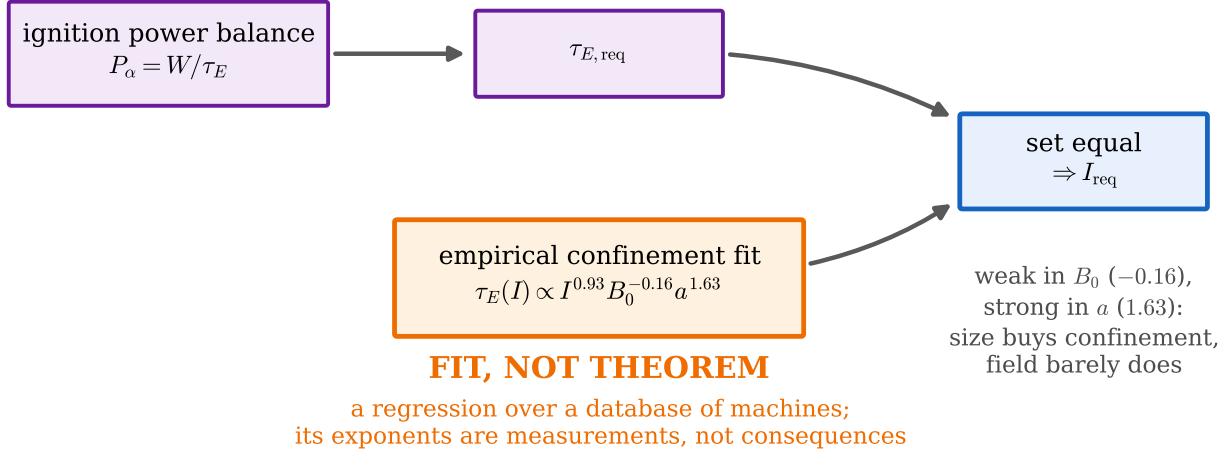
After substituting the reference design's previously determined relations, Freidberg obtains the compact scaling

$$I_{\text{req}} \simeq 12.1 \frac{a^{1.63}}{B_0^{0.16}} \text{ MA}, \quad (110)$$

where the remaining fixed baseline assumptions and the stated units are absorbed into the coefficient. The striking feature is not the exact decimal exponents. It is the strong approximately $a^{5/3}$ current demand and the weak $B_0^{-0.16}$ dependence.

For the reference point $a = 1.34 \text{ m}$ and $B_0 = 6.83 \text{ T}$, the paper finds $I_{\text{req}} = 14.3 \text{ MA}$. This is a demanded current, not a performance target selected because it is easy to sustain.

PLACE IMAGE HERE — Reset derivation sketch: confinement requirement becomes current demand. Show a left-to-right chain: ignition power balance gives $\tau_{E,\text{req}}$; the explicit empirical fit gives $\tau_E(I)$; equating them produces I_{req} . Put the words "fit, not theorem" directly beneath the confinement-law block. Include the weak $B_0^{-0.16}$ and strong $a^{1.63}$ reference scaling.



Explanatory sketch. Exponents are those of the reference regression quoted in the text; they are empirical.

Figure 31: Reset derivation sketch: confinement requirement becomes current demand. *Idealized schematic; not a fabrication drawing.*

60. High required current immediately tests field-line twist

The safety factor was introduced as the number of toroidal circuits made by a field line per poloidal circuit. At the plasma edge, more plasma current produces more poloidal field, so it reduces this twist count. For the elongated reference geometry, the paper uses

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I} \left(\frac{1 + \kappa^2}{2} \right). \quad (111)$$

The first factor is the Ampère-law estimate already derived from $B_\theta \sim \mu_0 I / (2\pi a)$; the shape factor records the paper's chosen elongation correction. Thus substituting I_{req} , not an independently chosen current, produces the required q_* .

Why does a lower q_* matter? A current-carrying plasma column contains free energy. A helical displacement can bend and reconnect the field in a way that releases some of that current-associated magnetic energy. The exact stability boundary depends on current profile, pressure profile, wall response, rotation, and mode number. The reference model uses the practical external-kink/disruption condition

$$q_* \geq q_K, \quad q_K \simeq 2. \quad (112)$$

Here a disruption means an abrupt loss of controlled plasma equilibrium and confinement, commonly accompanied by a rapid current termination and large transient loads. The mechanism is MHD current-driven instability; the numerical threshold is a stated model boundary, not a number obtained by setting two elementary equations equal.

Supporting quantity — kink ratio. The transparent test is $\mathcal{R}_K = q_K/q_*$. It is below one only when the demanded current leaves enough field-line twist to meet the adopted kink-safety criterion.

61. The four tests are now motivated, not imported

At the same reference design point, evaluate four distinct ratios:

$$\mathcal{R}_G = \frac{n}{n_G}, \quad n_G = \frac{I}{\pi a^2} \times 10^{20} \text{ m}^{-3}, \quad (113)$$

$$\mathcal{R}_T = \frac{\beta}{\beta_T}, \quad \beta_T = \beta_N \frac{I}{a B_0}, \quad (114)$$

$$\mathcal{R}_K = \frac{q_K}{q_*}, \quad q_K \simeq 2, \quad (115)$$

$$\mathcal{R}_B = \frac{f_B}{f_{\text{NC}}}. \quad (116)$$

Units in the Greenwald and normalized-beta forms are the conventional MA–m–T units stated earlier. Each ratio has a different physical origin:

- $\mathcal{R}_G$ checks the experimentally observed density boundary introduced on page 35;
- $\mathcal{R}_T$ checks pressure against the practical MHD beta allowance motivated on pages 33–34;
- $\mathcal{R}_K$ checks the current-driven kink boundary just explained;
- $\mathcal{R}_B$ checks whether non-inductive current sources can replace the required current in steady state.

The notation makes the decision rule uniform but does not erase the different kinds of evidence behind the four denominators. A claim that the reactor ”passes plasma physics” will be permitted only after every one of these ratios is shown, with its assumptions, at the same design point.

Design consequence. The current required by confinement appears in every test: directly in n_G , β_T , and q_* , and indirectly in the current balance. This is why the conventional-tokamak difficulty has a common root rather than four unrelated problems.

62. First audit: the density requirement

The fusion-power calculation demands the reference average density

$$n = 1.43 \times 10^{20} \text{ m}^{-3}.$$

The Greenwald reference density, using the confinement-demanded $I = 14.3 \text{ MA}$ and $a = 1.34 \text{ m}$, is

$$n_G = \frac{14.3}{\pi(1.34)^2} \times 10^{20} \text{ m}^{-3} = 2.54 \times 10^{20} \text{ m}^{-3}.$$

Therefore

$$\mathcal{R}_G = \frac{1.43}{2.54} = 0.56 < 1. \tag{117}$$

The baseline passes this particular test with margin. That result is valuable: it shows the later failure is not created by declaring every well-known limit violated.

Model and limits. The use of the Greenwald relation at reactor scale remains an extrapolation. The conclusion here is only that the 2015 baseline lies below the boundary assumed by that model. It is not a first-principles proof that such a density is attainable in a burning reactor.

63. Second audit: pressure is barely allowed

The same reference design requires

$$\beta = 4.13\%.$$

Its adopted normalized-beta coefficient is $\beta_N = 2.8$. With $I = 14.3$ MA, $a = 1.34$ m, and $B_0 = 6.83$ T,

$$\beta_T = 2.8 \frac{14.3}{(1.34)(6.83)}\% = 4.39\%.$$

Thus

$$\mathcal{R}_T = \frac{4.13}{4.39} = 0.94 < 1. \quad (118)$$

The comparison passes, but only narrowly. It is not a reserve that can be spent silently on profile changes, control margin, or modelling error.

Model and limits. The 2.8 coefficient is the reference model's Troyon-normalized-beta assumption. A different equilibrium, wall, profile, or stability-control model can change it. The algebra above is exact once that assumption and its conventional units are stated; the coefficient itself is not derived from the algebra.

64. Third audit: the confinement-driven current fails the kink test

Insert the same demanded current into Eq. (111). The reference calculation gives

$$q_* = 1.56.$$

The adopted disruption-avoidance threshold is $q_K \simeq 2$, so

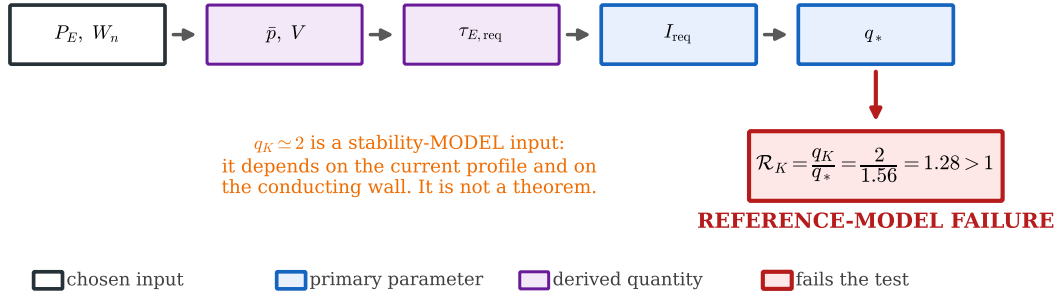
$$\mathcal{R}_K = \frac{2}{1.56} = 1.28 > 1. \quad (119)$$

This is a failure of the stated baseline. The required current produces insufficient edge field-line twist for the adopted kink-safety margin.

Notice the causal chain. The kink problem was not inserted after the reactor was chosen. Electric output and wall loading demanded pressure and volume; ignition demanded τ_E ; the confinement fit demanded I ; that I forced q_* below the adopted stability boundary. Reducing current would improve q_* , but would then fail to provide the empirical confinement time used in the baseline.

PLACE IMAGE HERE — Causal failure sketch: why the kink ratio exceeds one.

Draw the chain $P_E, W_n \rightarrow p, V \rightarrow \tau_{E,\text{req}} \rightarrow I_{\text{req}} \rightarrow q_* \rightarrow \mathcal{R}_K$. Mark the last comparison $1.28 > 1$ as a reference-model failure. Include a note that $q_K \simeq 2$ is a profile/wall-dependent stability-model input.



Explanatory sketch. The failure is a failure OF THE SELECTED MODEL SET at the reference point, not a statement about tokamaks in general.

Figure 32: Causal failure sketch: why the kink ratio exceeds one. *Idealized schematic; not a fabrication drawing.*

65. Fourth audit begins with a current ledger, not a magic bootstrap number

A steady-state tokamak cannot rely on a transformer indefinitely. Its current ledger must be

$$I = I_{\text{CD}} + I_{\text{bs}},$$

where I_{CD} is externally driven current and I_{bs} is bootstrap current. Divide by the required total current:

$$1 = \frac{I_{\text{CD}}}{I} + f_B, \quad f_B = 1 - \frac{I_{\text{CD}}}{I}. \quad (120)$$

This equation is only charge accounting. It does not claim that the plasma can supply the required f_B . For the reference recirculating-power and lower-hybrid current-drive model, the paper obtains $I_{\text{CD}} = 2.25$ MA. With $I = 14.3$ MA,

$$f_B = 1 - \frac{2.25}{14.3} = 0.84. \quad (121)$$

The required bootstrap fraction is therefore very large because the demanded current is large while economically permitted external drive is limited.

The next task is not to announce a bootstrap limit from nowhere. We must derive the physical source of bootstrap current from trapped-particle collisional asymmetry, state the selected neoclassical current-density model, and integrate it over the declared profiles to obtain the achievable fraction f_{NC} . Only then can we evaluate $\mathcal{R}_B = f_B/f_{\text{NC}}$.

Design consequence. Three audited comparisons now exist: density passes, beta is marginal, and kink safety fails. The remaining steady-state test will ask whether the pressure gradients that help make fusion power can naturally supply enough current to replace what the transformer cannot.

66. Where bootstrap current actually comes from

Page 36 established the first ingredient: a torus has trapped as well as passing particles. Page 37 stated the second: the pressure falls from the hot, dense core toward the edge. Put those facts together, but do not yet call the result a formula.

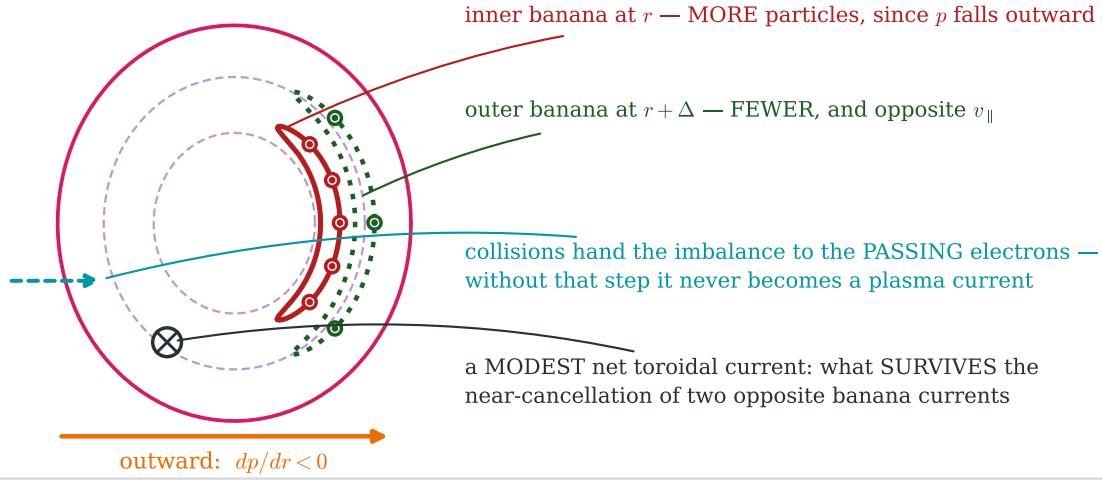
In a uniform straight field, particles moving parallel to $\mathbf{B}$ in the two directions can cancel one another's current. In a torus, trapped particles bounce between mirror points and do not complete a full parallel circuit. A pressure gradient means that the population arriving from the core side of a narrow radial shell is not identical to the population arriving from the edge side. Grad- B and curvature drifts then separate the two populations slightly. Collisions transfer part of that imbalance between trapped and passing particles. The cancellation of parallel electron motion is therefore incomplete: a toroidal current remains.

The energy source is not mysterious. It is the pressure profile itself. Flatten the profile so that $dp/dr = 0$, and the drive disappears. Make the profile steeper, and the available drive grows, while the pressure-gradient stability burden grows with it. This is precisely why bootstrap current belongs in the same closure problem as beta and kink safety.

PLACE IMAGE HERE — Physics sketch: from trapped particles to bootstrap current. Draw a small annular plasma shell. On the inboard side show stronger field and trapped electron bounce points; on the outboard side show passing electrons. Add an outward pressure-gradient arrow and a collision arrow transferring the trapped-particle asymmetry to passing electrons. End with a modest toroidal-current arrow labelled “incomplete cancellation,” not “free current.”

Design consequence. The pressure gradient that is required to make fusion power also creates a possible self-current. Whether it is large enough is not an independent choice: it must be calculated from a kinetic model, the same profiles, and the magnetic geometry already demanded by the reactor.

FROM TRAPPED PARTICLES TO A CURRENT — and why it is not free



1. At a given point the two adjacent bananas carry OPPOSITE $v_{||}$, so their currents nearly cancel.
2. Because $dp/dr < 0$, slightly more particles sit on the inner orbit — the cancellation is INCOMPLETE.
3. Collisions transfer that residue to the passing electrons. That step is what makes it a plasma current.
4. Scaling: $j_{bs} \sim -C \frac{\sqrt{\epsilon}}{B_0} \frac{dp}{dr}$, the $\sqrt{\epsilon}$ inherited from the trapped fraction $\sim \sqrt{2\epsilon/(1+\epsilon)}$.

The coefficient C is left unwritten ON PURPOSE. It is where the kinetic model enters — collisionality regime, profile shape, impurity content — and the primer's own audit of the 2015 baseline turns on exactly that number. Quoting one here would hide the modelling choice the audit later argues about.

This is an INCOMPLETE CANCELLATION, not free current. It exists only while the pressure gradient is maintained, and maintaining that gradient is what the heating system is for.

Explanatory sketch; schematic, not to scale. The banana width and the orbit separation are exaggerated for legibility.

Figure 33: Physics sketch: from trapped particles to bootstrap current. *Idealized schematic; not a fabrication drawing.*

67. The first irreducibly kinetic input is declared, not smuggled in

The mathematical object behind the preceding paragraph is a drift-kinetic equation. In schematic form for a distribution function $f(\mathbf{x}, \mathbf{v})$ it is

$$(v_{\parallel} \mathbf{b} + \mathbf{v}_d) \cdot \nabla f + \frac{qE_{\parallel}}{m} \frac{\partial f}{\partial v_{\parallel}} = C[f]. \quad (122)$$

Here $\mathbf{b} = \mathbf{B}/B$, $\mathbf{v}_d$ represents the guiding-center drifts, and $C[f]$ is the Coulomb collision operator. Solving Eq. (122) with trapped orbits is beyond a first course in differential equations; writing a number for it without acknowledgement would be worse than leaving a difficult step visible.

Freidberg, Mangiarotti, and Minervini therefore choose a particular large-aspect-ratio, low-collisionality neoclassical closure. They assume equal ion and electron temperatures, $T_i = T_e = T$, one effective ion charge, $Z = 1$, and a circular area-equivalent radius

$$\hat{a} = \kappa^{1/2} a, \quad \rho = \frac{r}{\hat{a}}. \quad (123)$$

Their local bootstrap-current model is their Eq. (41):

$$J_B(\rho) = -2.44 \left(\frac{r}{R_0} \right)^{1/2} \frac{p}{B_{\theta}} \left(\frac{1}{n} \frac{dn}{dr} + 0.055 \frac{1}{T} \frac{dT}{dr} \right). \quad (124)$$

The coefficient 2.44 and the relative temperature-gradient weight 0.055 are the kinetic closure, expressed in the paper's conventional working units; they are not deducible from Maxwell's equations and elementary calculus alone. Everything from this point onward—the gradients, the poloidal field, the area integral, and the comparison with the required fraction—can and will be shown explicitly.

Model and limits. Equation (124) is a declared neoclassical model, not a universal bootstrap law. Its regime assumptions, effective charge, collisionality treatment, and chosen profiles are part of the result. The accessible source is Freidberg, Mangiarotti, and Minervini (2015), Eq. (41).

68. The selected profiles make the pressure-gradient term calculable

The same paper does not leave $n(r)$ and $T(r)$ implicit. Its standard-profile choice (its Eq. (20)) is

$$T = 2\bar{T}(1 - \rho^2), \quad p = 2.5\bar{p}(1 - \rho^2)^{3/2}, \quad n = 1.5\bar{n}(1 - \rho^2)^{1/2}. \quad (125)$$

The numerical factors make the bars cross-sectional averages. For example, $\int_0^1 2\rho [2\bar{T}(1 - \rho^2)] d\rho = \bar{T}$.

Because $d\rho/dr = 1/\hat{a}$, ordinary differentiation gives

$$\frac{1}{n} \frac{dn}{dr} = \frac{1}{1.5\bar{n}(1 - \rho^2)^{1/2}} \left[1.5\bar{n} \frac{1}{2} (1 - \rho^2)^{-1/2} (-2\rho) \frac{1}{\hat{a}} \right] = -\frac{\rho}{\hat{a}(1 - \rho^2)}, \quad (126)$$

$$\frac{1}{T} \frac{dT}{dr} = \frac{1}{2\bar{T}(1 - \rho^2)} \left[2\bar{T} (-2\rho) \frac{1}{\hat{a}} \right] = -\frac{2\rho}{\hat{a}(1 - \rho^2)}. \quad (127)$$

Thus the bracket in Eq. (124) is

$$\frac{1}{n} \frac{dn}{dr} + 0.055 \frac{1}{T} \frac{dT}{dr} = -\frac{(1 + 0.110)\rho}{\hat{a}(1 - \rho^2)}. \quad (128)$$

Substitution of Eqs. (125) and (128) into Eq. (124) then gives, with $2.44(1.110)(2.5) = 6.77 \simeq 6.8$,

$$J_B(\rho) = 6.8 \frac{\bar{p}}{\hat{a}^{1/2} R_0^{1/2}} \frac{\rho^{3/2}(1 - \rho^2)^{1/2}}{B_\theta(\rho)}. \quad (129)$$

The sign convention is now carried by the direction chosen for B_θ and plasma current. The useful content is transparent: this model predicts a current density which vanishes at the axis and edge and is driven in the finite-gradient interior.

69. The poloidal field is obtained from the assumed total-current profile

Equation (129) still contains $B_\theta(\rho)$. It cannot be chosen after seeing the desired answer: it is produced by the total plasma-current profile. The paper selects

$$J_T(\rho) = \frac{I}{\pi \hat{a}^2} \frac{9\rho^{1/4}}{8} \frac{\alpha^2(1-x)e^{\alpha x}}{e^\alpha - 1 - \alpha}, \quad x = \rho^{9/4}, \quad \alpha = 2.53. \quad (130)$$

This is another profile assumption, chosen to put appreciable current away from the axis. It is normalized correctly: using $\rho = x^{4/9}$ gives

$$\rho^{1/4} \rho d\rho = \frac{4}{9} dx, \quad \frac{I_{\text{enc}}}{I} = \int_0^x \frac{\alpha^2(1-u)e^{\alpha u}}{e^\alpha - 1 - \alpha} du. \quad (131)$$

The antiderivative is

$$\int \alpha^2(1-u)e^{\alpha u} du = (1 + \alpha - \alpha u) e^{\alpha u}, \quad (132)$$

so evaluation between 0 and x yields

$$\frac{I_{\text{enc}}}{I} = \frac{(1 + \alpha - \alpha x)e^{\alpha x} - 1 - \alpha}{e^\alpha - 1 - \alpha}. \quad (133)$$

Ampere's law $2\pi r B_\theta = \mu_0 I_{\text{enc}}$ now gives

$$b_\theta(\rho) \equiv \frac{B_\theta}{\mu_0 I / (2\pi \hat{a})} = \frac{1}{\rho} \left[\frac{(1 + \alpha - \alpha x)e^{\alpha x} - 1 - \alpha}{e^\alpha - 1 - \alpha} \right]. \quad (134)$$

This is the missing magnetic denominator in the bootstrap calculation. No additional machine component has appeared; we have only made the previously required current profile explicit.

70. Integrate the local current, then test the steady-state requirement

In the area-equivalent circular cross-section, $dA = 2\pi\hat{a}^2\rho d\rho$ after integrating over the polar angle. Thus

$$I_B = 2\pi\hat{a}^2 \int_0^1 J_B(\rho)\rho d\rho \quad (135)$$

$$= 2\pi\hat{a}^2 \int_0^1 \left[6.8 \frac{\bar{p}}{\hat{a}^{1/2}R_0^{1/2}} \frac{\rho^{3/2}(1-\rho^2)^{1/2}}{B_\theta} \right] \rho d\rho. \quad (136)$$

Replace B_θ using Eq. (134), divide by I , and collect constants. Since $4\pi^2(6.8) = 268.5$, the result is the paper's Eq. (44):

$$f_{\text{NC}} \equiv \frac{I_B}{I} = 268 \frac{\hat{a}^{5/2}\bar{p}}{\mu_0 R_0^{1/2} I^2} \int_0^1 \frac{\rho^{5/2}(1-\rho^2)^{1/2}}{b_\theta(\rho)} d\rho. \quad (137)$$

This integral has no elementary antiderivative. That is not a gap in the argument: it is a fully specified one-dimensional numerical quadrature. With $\alpha = 2.53$, its dimensionless value is 0.318 (three significant figures). For the reference point,

$$\kappa = 1.70, \quad \hat{a} = \sqrt{1.70}(1.34 \text{ m}) = 1.747 \text{ m}, \quad \bar{p} = 7.57 \text{ atm}, \quad R_0 = 5.34 \text{ m}, \quad I = 14.3 \text{ MA}, \quad (138)$$

where 1 atm = 101325 Pa and 1 MA = 10^6 A. Equation (137) then gives

$$f_{\text{NC}} = 0.44. \quad (139)$$

The required value from Eq. (121) was $f_B = 0.84$. The audited comparison is therefore

$$\mathcal{R}_B = \frac{f_B}{f_{\text{NC}}} = \frac{0.84}{0.44} = 1.91 > 1. \quad (140)$$

This is a failure of the selected standard-profile neoclassical model: its calculated self-current is about half the required self-current. It is not a proof that no tokamak can ever operate steadily.

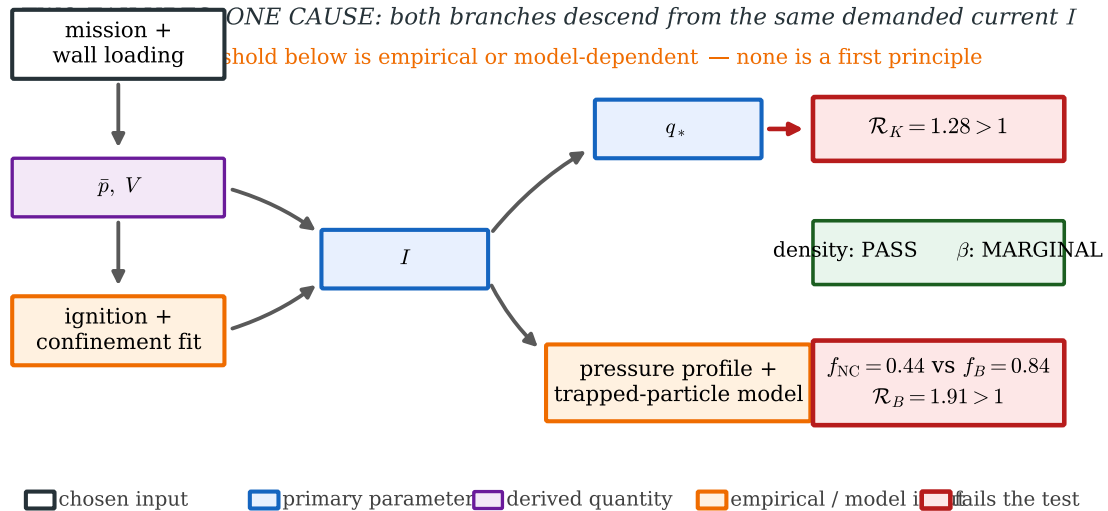
71. What the four tests have, and have not, established

The baseline can now be read as one causal argument rather than a bag of named limits:

Test	ratio	status	meaning
Greenwald density	$\mathcal{R}_G = 0.56$	passes	density is below the adopted empirical boundary
Troyon beta	$\mathcal{R}_T = 0.94$	marginal	pressure is near the adopted MHD boundary
Kink safety	$\mathcal{R}_K = 1.28$	fails	required current gives too little q_* margin
Bootstrap closure	$\mathcal{R}_B = 1.91$	fails	calculated self-current is below required current share

The first two boundaries are empirical or model-dependent operating margins. The last two failures are coupled: fusion mission and wall loading demand p and volume; ignition and the confinement fit demand I ; that current lowers q_* , while the same pressure profile can supply only a limited bootstrap share under the declared kinetic model.

PLACE IMAGE HERE — Reset sketch: the conventional baseline’s two coupled closure failures. Draw one concise logic map. Mission and wall loading lead to p, V ; ignition and the confinement fit lead to I . Split into two branches: $I \rightarrow q_* \rightarrow \mathcal{R}_K = 1.28 > 1$, and pressure profile plus trapped-particle kinetic model $\rightarrow f_{\text{NC}} = 0.44$ compared with $f_B = 0.84$, hence $\mathcal{R}_B = 1.91 > 1$. Keep density and beta as side checks marked pass and marginal. Label every threshold or regression as model/empirical rather than first principle.



Explanatory sketch. Both ratios are properties of the selected 2015 model set at its reference point, not universal tokamak limits.

Figure 34: Reset sketch: the conventional baseline’s two coupled closure failures. *Idealized schematic; not a fabrication drawing.*

Design consequence. The conventional reference point is not rejected because a named limit appeared from the left field. It fails two explicit closure tests after their physical origin, profile assumptions, and calculation have been exposed. The next question is which previously declared demands can be changed together without simply moving the failure elsewhere.

72. A reference point is not a design-space conclusion

The reference point $a = 1.34$ m was chosen in the source by imposing $R_0/a = 4$. It is a useful check, but it cannot establish that the conventional baseline fails at every size. The wall-loading calculation fixes instead

$$R_0(a) = \frac{7.16}{a} \text{ m} \quad (\kappa = 1.70, \text{ 2015 baseline inputs}), \quad (141)$$

so a is the remaining geometric trial variable.

For every trial a , the source's analytic radial-build convention supplies

$$\epsilon_B(a) = \frac{a+b}{R_0(a)}, \quad B_0(a) = B_{\max}[1 - \epsilon_B(a)], \quad b = 1.20 \text{ m}, \quad B_{\max} = 13 \text{ T}. \quad (142)$$

Its TF-coil stress and winding-pack model then determines $c(a)$; the plasma calculation does not get to prescribe c afterward. This is an important bookkeeping point. The physical build developed earlier explains why the magnet has finite radial thickness. Equation (142) is the particular reduced convention required to reproduce this paper's numerical scan.

The already-derived power and ignition relations become compact functions of a :

$$\bar{p}(a) = \frac{8.76}{a^{1/2}} \text{ atm}, \quad \bar{n}(a) = \frac{1.66}{a^{1/2}} \times 10^{20} \text{ m}^{-3}, \quad \tau_E(a) = 0.81a^{1/2} \text{ s}. \quad (143)$$

These are not new laws of nature. They are the stated mission, profile, and wall-loading model expressed as a one-parameter family.

73. The current-drive allowance must move with the trial design

In the reference case, $I_{\text{CD}} = 2.25 \text{ MA}$ was obtained from a fixed recirculating-power allowance. A scan must not hold that number fixed when density, field, and size change. The power allocated to waves is

$$P_{\text{CD}} = \eta_{\text{RF}} \eta_{\text{kly}} f_{\text{RP}} P_E = (0.80)(0.50)(0.10) P_E = 40 \text{ MW}. \quad (144)$$

The source's lower-hybrid model writes the local efficiency in its conventional units as

$$\eta_{\text{CD}} \equiv \frac{R_0 \bar{n} I_{\text{CD}}}{P_{\text{CD}}} \simeq \frac{1.2}{n_{\parallel}^2}. \quad (145)$$

Here $\bar{n}$ is measured in 10^{20} m^{-3} , I_{CD} in MA, and P_{CD} in MW. The refractive index is calculated at the intended deposition surface, $\rho_m \simeq 0.8$, from

$$n_{\parallel} \simeq \frac{\omega_{pe}}{\Omega_e} + \left(1 + \frac{\omega_{pe}^2}{\Omega_e^2}\right)^{1/2} \left(1 - \frac{\omega_{LH}^2}{\omega^2}\right)^{1/2}, \quad \omega \simeq 2\omega_{LH}(\rho_m). \quad (146)$$

The frequencies follow from the local density and field:

$$\omega_{pe}^2 = \frac{ne^2}{\epsilon_0 m_e}, \quad \Omega_e = \frac{eB}{m_e}, \quad \omega_{LH}^2 = \frac{\omega_{pi}^2}{1 + \omega_{pe}^2/\Omega_e^2}. \quad (147)$$

Thus each trial design gives $n_{\parallel}(a)$, then $\eta_{\text{CD}}(a)$, then $I_{\text{CD}}(a) = P_{\text{CD}} \eta_{\text{CD}} / [R_0(a) \bar{n}(a)]$, and finally

$$f_B(a) = 1 - \frac{I_{\text{CD}}(a)}{I(a)}. \quad (148)$$

This is why the bootstrap-demand curve cannot be drawn from the single reference fraction alone.

Model and limits. Equations (145)–(147) are a specified lower-hybrid current-drive model, not a universal efficiency theorem. They are included because the scan needs its assumptions exposed as clearly as the bootstrap model did.

74. Four ratio functions, one admissibility question

For each permitted a , use the confinement regression and the relations above to obtain $I(a)$, then calculate

$$\mathcal{R}_G(a) = \frac{\bar{n}(a)}{I(a)/[\pi a^2]}, \quad (149)$$

$$\mathcal{R}_T(a) = \frac{\beta(a)}{2.8 I(a)/[aB_0(a)]}, \quad (150)$$

$$\mathcal{R}_K(a) = \frac{2}{q_*(a)}, \quad (151)$$

$$\mathcal{R}_B(a) = \frac{f_B(a)}{f_{\text{NC}}(a)}. \quad (152)$$

The denominator in the first line is in the usual Greenwald units. The 2.8 in the second line is the stated Troyon-normalized-beta model input, in percent convention. The 2 in the third line is the adopted kink-safety threshold. The final denominator is the profile-integrated neoclassical result derived on pages 66–70.

The calculation has a clean pass rule:

$$\mathcal{R}_{\text{max}}(a) = \max\{\mathcal{R}_G, \mathcal{R}_T, \mathcal{R}_K, \mathcal{R}_B\}, \quad \text{admissible only if} \quad \mathcal{R}_{\text{max}}(a) \leq 1. \quad (153)$$

The scan used here is implemented in the accompanying transparent arithmetic file:

`freidberg_baseline_scan.py`

It evaluates the published analytic coil, confinement, lower-hybrid, and bootstrap closures directly; it does not fit a curve to the source’s published figure.

Design consequence. A claimed “operating window” is an intersection, not four separate checks. Changing a is acceptable only if it reduces the largest violation without raising another ratio above one.

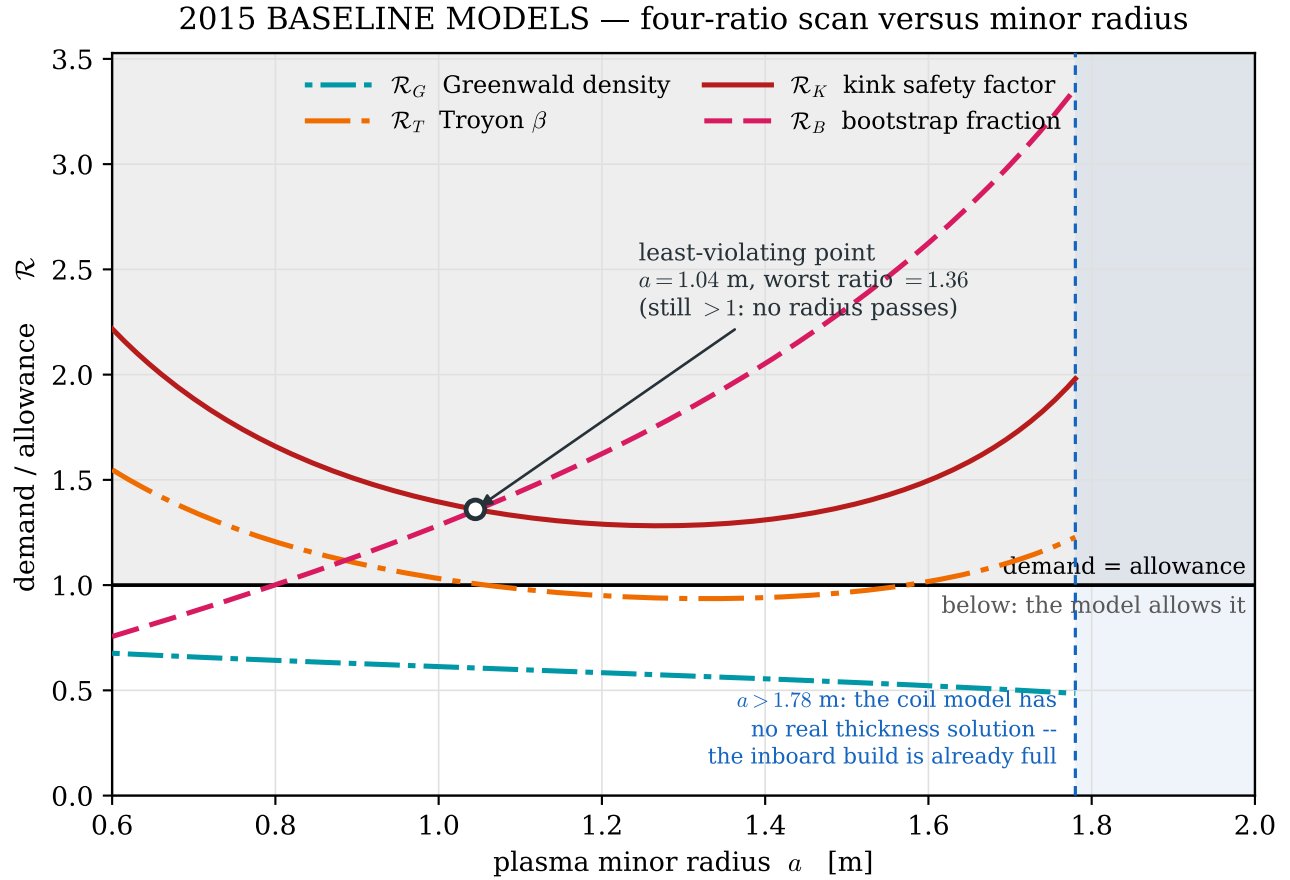
75. The conventional baseline has no simultaneous window in the scan

The table below gives representative outputs of the fully disclosed calculation. The valid high- a end is limited by the source’s TF-coil stress/current-density model: with the stated inputs, its last real radial-build solution is $a \simeq 1.782$ m. Beyond that point the model has no real coil thickness, so the inboard build has failed before a plasma ratio can be evaluated. The small- a end is already excluded by the rapidly worsening kink and beta ratios.

a (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$	$\mathcal{R}_{\max}$
0.40	0.72	2.31	3.56	0.53	3.56
0.80	0.64	1.21	1.66	1.00	1.66
1.00	0.61	1.03	1.40	1.28	1.40
1.05	0.61	1.00	1.36	1.36	1.36
1.20	0.58	0.95	1.29	1.62	1.62
1.34	0.56	0.94	1.28	1.91	1.91
1.60	0.52	1.02	1.50	2.62	2.62

The least-violating sampled point is near $a = 1.05$ m, but it still has $\mathcal{R}_K \simeq 1.36$ and $\mathcal{R}_B \simeq 1.36$. No row makes all four ratios at most one. This independently reproduces the logical content of the source’s Fig. 5: shrinking the machine makes kink and beta worse; enlarging it eventually makes the required bootstrap share worse. Greenwald density is not the bottleneck in this baseline family.

PLACE IMAGE HERE — Reset plot: reproducible four-ratio scan versus minor radius. Plot the four calculated ratios from `freidberg_baseline_scan.py` against a . Add a horizontal pass line at one, shade only the region above one, and mark the least-violating point near $a = 1.05$ m. Label the plot “2015 baseline models,” not a universal tokamak limit. Do not add ST or HTS hardware.



Reproduced from primer/freidberg_baseline_scan.py. These are the ratios implied by ONE declared model set — an analytic coil build, one empirical confinement fit, one stability threshold, one kinetic bootstrap coefficient. This is NOT a universal tokamak limit: it is the behaviour of that model set, and each ingredient is a modelling choice that can be argued with.

Figure 35: Reset plot: reproducible four-ratio scan versus minor radius. *Idealized schematic; not a fabrication drawing.*

76. The first conventional conclusion is conditional and specific

We can now state precisely what has been demonstrated. Within the chosen mission, wall-loading, blanket, TF-coil, confinement, lower-hybrid-current-drive, standard-profile, and neoclassical-bootstrap models, changing conventional tokamak size alone does not close both the kink and steady-state current requirements.

This is a strong result, but it is not a slogan about fusion. It does *not* prove that conventional tokamaks cannot operate, that advanced scenarios cannot improve profiles, or that a modern magnet system cannot alter the trade. It identifies the two demands that any proposed remedy must address: lower the confinement-driven total current, and/or increase the allowed current and field-line-twist margin without merely transferring the burden to beta, wall loading, or recirculating power.

The next section of the 2015 paper tests remedies in a disciplined order. First it asks what improved confinement ($H > 1$) buys and what beta margin it costs. Second it raises the engineering field ceiling $B_{\max}$ —the point at which a modern HTS-enabled magnet discussion becomes relevant. Only after those conventional trade-offs are visible will the thesis introduce alternative aspect-ratio geometry.

Design consequence. The conventional baseline has now earned the question “what changes the current closure?” High field and HTS enter next as a response to a calculated failure, not as an unmotivated technology preference.

77. Remedy one: what an H factor does and does not mean

The first remedy in the 2015 paper does not add a new piece of hardware. It asks whether the plasma can lose energy more slowly than the reference H-mode regression predicts. Write that possibility honestly as

$$\tau_E = H\tau_{E,\text{ref}}, \quad H > 1. \quad (154)$$

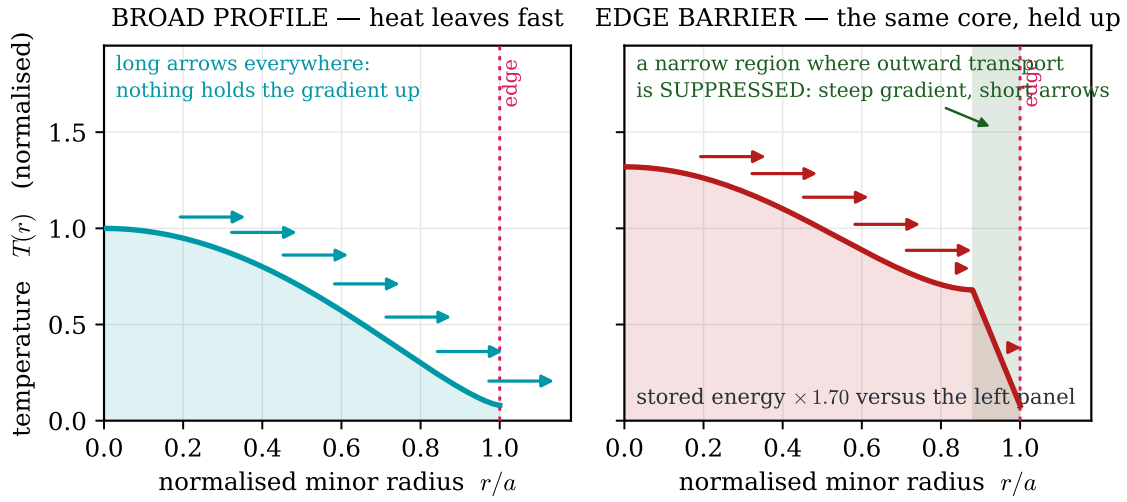
The symbol H is a measured-performance multiplier relative to the particular regression on page 58. It is not a new conservation law, a material property, or a guarantee that a reactor can be run at the stated value.

There is nevertheless a physical picture behind trying it. Turbulent eddies carry heat and particles down their gradients. Carefully shaped pressure and current profiles can, in favorable regimes, change the magnetic shear and flow environment felt by those eddies; an internal or edge transport barrier can then reduce the effective outward transport. Maintaining such a state requires profile control and must be demonstrated in the relevant operating regime. In an ignited plasma, alpha heating also makes that control problem less independent than it is in an externally heated experiment.

PLACE IMAGE HERE — Reset physics sketch: why an H factor is not a knob on the machine.

Draw a toroidal plasma cross-section only, with a broad radial temperature profile on the left and a steeper, protected-gradient region on the right. Show small outward turbulent heat arrows in the first case and shorter suppressed arrows in the second. Label the second as a transport-barrier and profile-control empirical performance change, not a new coil or reactor component. Do not add ST, HTS, divertor, or a spherical plasma.

Model and limits. Raising H means assuming a better-than-reference confinement state and then testing its consequences. It is a scenario assumption with experimental burden of proof, not an available design margin.



$H \equiv \tau_E / \tau_E^{\text{scaling}}$ is a RATIO TO A FIT, and therefore an OUTPUT. There is no dial marked H on the machine: nothing here is a new coil, a new supply or a new component. What changed is a TRANSPORT STATE, reached empirically and held only while its conditions hold — an input-power threshold, a shape, a divertor configuration — and paid for with edge-localised relaxations and impurity retention. The $\times 1.70$ above is the integral of the two curves drawn here, so even that number is a property of these two shapes, not a design input. Assuming a value of H and then quoting the performance it implies is assuming the conclusion.

Figure 36: Reset physics sketch: why an H factor is not a knob on the machine. *Idealized schematic; not a fabrication drawing.*

78. Better confinement lowers current—and moves every coupled test

Equation (109) contains H only in the denominator inside a power $1/0.93$. Holding the other inputs of that equation fixed gives

$$I_{\text{req}}(H) = I_{\text{req}}(1) H^{-1/0.93}. \quad (155)$$

Thus a larger H lowers the current that the empirical fit demands. Since $q_* \propto 1/I$ at fixed geometry,

$$q_*(H) = q_*(1) H^{1/0.93}. \quad (156)$$

This is the intended benefit: less current produces more field-line twist and hence more margin in the adopted kink test.

But the same reduced current tightens two allowance ratios. The Greenwald allowance is proportional to I , and the adopted Troyon pressure allowance is also proportional to I . At fixed geometry and fusion pressure,

$$\mathcal{R}_G(H) = \mathcal{R}_G(1) H^{1/0.93}, \quad \mathcal{R}_T(H) = \mathcal{R}_T(1) H^{1/0.93}. \quad (157)$$

In words: lowering current helps kink safety but makes a fixed density and pressure look larger relative to current-based operating boundaries.

Bootstrap closure shifts in the helpful direction at fixed geometry. From Eq. (137), $f_{\text{NC}} \propto p/I^2$, while $f_B = 1 - I_{\text{CD}}/I$. Lower I both raises the calculated bootstrap fraction and reduces the required bootstrap share if the current-drive allowance is held fixed. That is why increasing H can remove the two original current-closure failures. It cannot be called a solution until the density and beta checks are repeated.

Design consequence. Improved confinement is not a free improvement. It trades a current-and-kink problem for reduced current-based density and beta headroom. The only fair test is a simultaneous recalculation.

79. The H scan: close bootstrap first, then inspect the remaining margins

For each trial H , the 2015 procedure chooses the radius that closes the steady-state current ledger:

$$f_B(a, H) = f_{\text{NC}}(a, H). \quad (158)$$

This does not make the design acceptable; it simply removes one degree of freedom before the other ratios are tested. The same transparent calculation file used for the size scan now evaluates this condition, using the changed current in the defining safety-factor relation rather than reusing a baseline q_* .

H	a (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
1.00	0.799	0.64	1.21	1.66	1.00
1.10	0.990	0.68	1.15	1.26	1.00
1.20	1.167	0.72	1.17	1.06	1.00
1.26	1.263	0.74	1.20	1.00	1.00
1.30	1.323	0.75	1.24	0.97	1.00
1.40	1.459	0.78	1.37	0.93	1.00

The table exhibits the trade directly. Near $H = 1.26$, bootstrap and kink closure are both obtained and the Greenwald ratio remains below one, but the Troyon ratio is about 1.20: the selected 2.8% normalized-beta allowance is exceeded. Larger H only increases that beta demand in this family.

Audit note. The paper reports the same qualitative conclusion and quotes a still larger required normalized beta, 3.6% rather than 2.8%, at its threshold case. The printed analytic ledger reproduced here gives approximately 3.37% on a 10^{-3} m radius grid. Both exceed 2.8%. We retain the discrepancy rather than silently tune a coefficient to reproduce a prose number: the published paper does not supply the full numerical implementation behind its figure.

80. What the advanced-confinement remedy has established

The H exercise has sharpened the diagnosis. If an advanced transport state were reliably available, it could lower the confinement-driven current enough to close the baseline kink and bootstrap ledgers. In the declared conventional family, it does so by asking the plasma to operate above the adopted normalized-beta allowance. The missing quantity is not an unexplained new component; it is MHD pressure margin.

This is a useful result even if future experiments improve that margin. It tells us exactly what must be supplied together: reproducible confinement enhancement, profile control, and an MHD-stability basis for the higher normalized beta. Calling only the first item the solution would conceal the coupled demand.

The next remedy changes a different primitive constraint: the maximum field that the inboard TF system can bear. Higher field changes the required current, the current drive efficiency, and the MHD ratios without assuming $H > 1$. That is the logical place to introduce high-temperature superconductors: first establish what a higher $B_{\max}$ buys in this model, then ask whether an HTS magnet can provide it while surviving its structural, thermal, and manufacturing constraints.

Design consequence. The advanced-confinement route is a conditional plasma-physics remedy, not a completed reactor design. The high-field route will be tested next as a distinct engineering remedy, with its cost and stress penalties kept visible.

81. Freidberg’s second remedy: raise the field ceiling, not the confinement claim

The preceding remedy changed the empirical confinement multiplier H . Freidberg’s second remedy holds $H = 1$ and changes an engineering input instead: the maximum magnetic field allowed at the inboard TF coil, $B_{\max}$. This is a clean counterfactual. It asks what the already-derived conventional machine would demand if its coil system could credibly tolerate a higher peak field.

At a fixed trial geometry, the immediate result is already known from Eq. (90), or from the source convention Eq. (142):

$$B_0 = B_{\max}(1 - \epsilon_B). \quad (159)$$

The central field consequently rises in direct proportion to the inboard field ceiling if the fractional radial build is held fixed. The plasma pressure required by the mission does not disappear. By the definition of beta,

$$\beta = \frac{2\mu_0 p}{B_0^2}. \quad (160)$$

Thus at fixed pressure and geometry, doubling B_0 quarters β . This is the first reason high field can relieve a pressure-stability problem: it increases the magnetic pressure available to balance the same plasma pressure.

The reference compact expression for beta had 13 T absorbed into its numerical coefficient. To vary $B_{\max}$ honestly, restore that dependence:

$$\beta(\%) = \frac{1.31}{a^{1/2}(1 - \epsilon_B)^2} \left(\frac{13 \text{ T}}{B_{\max}} \right)^2. \quad (161)$$

Equation (161) is not a new plasma law; it is Eq. (160) after the 2015 power-and-size relations have been substituted. Writing the factor explicitly prevents a common audit error: one must not use a coefficient calibrated at 13 T unchanged in a high-field scan.

Supporting quantity — raised field ceiling $B_{\max}$. This is an engineering input, not a new primary Freidberg parameter. It changes the primary central field B_0 and the coil build c , then indirectly changes the current, stability, and current-drive calculations.

82. What field can help—and what it cannot help by itself

First hold geometry fixed as a thought experiment. The selected confinement regression gave the compact current requirement

$$I_{\text{req}} \propto B_0^{-0.16}.$$

The exponent is weak: a field increase alone reduces the regression’s current demand, but not dramatically. The more consequential changes occur after the current, safety factor, current-drive, bootstrap, and radial-build relations are recalculated together.

The safety-factor relation makes one helpful direction visible without any empirical claim:

$$q_* \propto \frac{a^2 B_0}{R_0 I}. \quad (162)$$

At unchanged geometry and current, a higher toroidal field increases field-line twist. If the confinement relation also permits a lower current, the two effects both raise q_* . Thus high field can address the same kink failure that improved confinement addressed, but it does so through a different physical lever.

There is no free magnetic pressure. The field energy density and representative mechanical stress scale are

$$u_B = p_B = \frac{B_{\text{max}}^2}{2\mu_0}. \quad (163)$$

This is the same result derived on page 47, now applied to the proposed remedy. Higher field means larger Lorentz loads and, for a fixed allowable stress, more structure. The source’s analytic coil model calculates this response as a larger c , rather than assuming that a better superconductor can make the support material vanish.

Design consequence. Raising B_{max} is physically promising because it lowers beta and tends to raise q_* . It is physically expensive because the very field that helps the plasma makes the coil’s magnetic pressure grow as B_{max}^2 . The scan must retain both statements.

PLACE IMAGE HERE — Reset physics sketch: a stronger inboard field changes two ledgers. Show a conventional inboard midplane radial cut, not a new machine. Left branch: $B_{\text{max}} \uparrow \Rightarrow B_0 \uparrow \Rightarrow \beta = 2\mu_0 p / B_0^2 \downarrow$ and $q_* \propto B_0 / I \uparrow$. Right branch: $B_{\text{max}} \uparrow \Rightarrow p_B = B_{\text{max}}^2 / (2\mu_0) \uparrow \Rightarrow$ greater support demand and coil build c . Mark the right branch as a mechanical consequence, not a drawback that superconductivity erases. No ST or divertor.

This brief asks for the two-sided field diagram that Figure 29 already carries; its right branch is identical and its left differs only by two further nodes. Under the standing rule that redundancy is fixed rather than expanded, no second figure is drawn — the extra nodes belong on that figure, and this brief should be retired.

83. Why field also changes the selected current-drive calculation

The reference current ledger limits externally driven current because wave power is limited. Its lower-hybrid model does not treat current-drive efficiency as a constant:

$$\eta_{\text{CD}} \simeq \frac{1.2}{n_{\parallel}^2}, \quad I_{\text{CD}} = \frac{P_{\text{CD}} \eta_{\text{CD}}}{R_0 \bar{n}}. \quad (164)$$

Here $n_{\parallel} \equiv ck_{\parallel}/\omega$ is the parallel refractive index of the launched wave. In the selected outside-launch approximation it contains the ratio

$$\frac{\omega_{pe}}{\Omega_e}, \quad \omega_{pe}^2 = \frac{ne^2}{\epsilon_0 m_e}, \quad \Omega_e = \frac{eB}{m_e}. \quad (165)$$

The first expression measures the electron-plasma oscillation rate; the second is the electron cyclotron rate. At fixed density, increasing local B increases Ω_e and decreases ω_{pe}/Ω_e . In this particular lower-hybrid launch model, the resulting change in $n_{\parallel}$ improves the available current-drive efficiency. That improvement reduces the bootstrap share required by the ledger

$$f_B = 1 - \frac{I_{\text{CD}}}{I}.$$

This is a model-dependent bridge between magnet field and steady-state current. It must not be generalized into the claim that every wave system becomes more efficient at higher field. Launch geometry, accessibility, damping, parasitic absorption, current profile, and the power plant's recirculating-power budget all require a separate calculation.

Model and limits. For Section 6.2 we keep Freidberg's 40 MW wave-power allocation, its lower-hybrid accessibility model, its profiles, and $H = 1$. We change only B_{max} and then repeat the whole coupled calculation. This isolates what the high-field proposal actually claims.

84. The high-field scan: one closure condition, then four independent tests

For every trial $B_{\max}$, retain the mission and wall-loading relations, compute $R_0(a)$, $B_0(a, B_{\max})$, the analytic TF build c , pressure, density, confinement-demanded current, lower-hybrid drive, and bootstrap fraction. As in the H scan, choose the remaining radius by the steady-state closure condition

$$f_B(a, B_{\max}) = f_{\text{NC}}(a, B_{\max}). \quad (166)$$

Only after that bookkeeping condition is met do we inspect the Greenwald, Troyon, kink, and bootstrap ratios. The procedure is implemented directly in `code/calculations/freidberg_baseline_scan.py`; its radius grid is 0.001 m, so the last displayed digit should not be read as a precision claim.

$B_{\max}$ (T)	a (m)	B_0 (T)	c (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
13.0	0.799	10.10	1.02	0.64	1.21	1.66	1.00
15.0	0.875	11.20	1.27	0.65	1.00	1.30	1.00
17.0	0.944	12.19	1.55	0.65	0.85	1.06	1.00
17.6	0.964	12.47	1.64	0.65	0.82	1.00	1.00
18.0	0.976	12.66	1.70	0.65	0.80	0.97	1.00
20.0	1.036	13.53	2.05	0.65	0.70	0.83	1.00

The table has a simple reading. Greenwald changes little along this bootstrap-matched family. Troyon becomes comfortable before the threshold. The kink ratio is the last ratio to fall to one. On this transparent grid the first all-pass point is $B_{\max} = 17.65$ T; rounding to the precision of the underlying model gives about 17.6 T, the threshold reported in the 2015 paper.

Audit note. This is a conditional result, not a prediction of a built reactor. It says that the selected 2015 confinement regression, current-drive model, bootstrap model, and four boundaries overlap when the inboard field ceiling is raised to roughly 17.6 T. Changing those models may move the number; it does not remove the need to test their overlap.

85. The price is measurable: more engineered volume and more exhaust burden

The increasing c column is already a direct mechanical cost of high field. The source also records two simple engineering figures of merit. Its engineered-volume proxy is

$$\frac{V_I}{P_E} = \frac{V_B + V_{\text{TF}}}{P_E}, \quad (167)$$

where V_B is blanket-region volume and V_{TF} is TF-coil volume in its specified geometry. It is not a literal construction-price quotation. It is a transparent proxy for how much highly engineered blanket and magnet material must be provided per electric megawatt.

The source's parallel-heat-flux proxy follows the alpha power along the field:

$$Q_{\parallel} = \frac{P_{\alpha} B_0}{R_0}. \quad (168)$$

Its units are MW T m^{-1} . The relation is a scaling measure, not a divertor-surface heat-flux calculation; a later chapter must introduce scrape-off layer width, magnetic expansion, radiation, and material limits before asking what a plate can survive.

For the source's compared cases, moving from its advanced-confinement case at 13 T to the high-field all-constraint case near 17.6 T increases V_I/P_E from 1.07 to $1.87 \text{ m}^3 \text{ MW}^{-1}$ (about a factor 1.75) and reports $Q_{\parallel} = 657 \text{ MW T m}^{-1}$, about 1.30 times its advanced-confinement comparison case. These are source-reported model outputs; we do not reverse-engineer extra precision from a plotted calculation whose full implementation was not published.

Design consequence. The high-field branch closes the selected plasma ratios only by moving difficulty into magnet structure, engineered volume, and exhaust. That is not a refutation of high field. It is the reason magnet technology and heat exhaust must be developed as part of the same reactor, not treated as a footnote to plasma performance.

86. HTS is earned here: an enabler with a mechanical bill

A superconductor is useful because below its operating critical surface—set by temperature, magnetic field, current density, field angle, strain, and material state—it can carry a large current with negligible DC electrical resistance. The important engineering quantity is therefore not the slogan zero resistance. It is a measured, margin-bearing critical-current surface

$$I_c = I_c(B, T, \theta, \varepsilon, \text{conductor architecture}).$$

The conductor must remain below that surface throughout normal operation and credible transients, while its stabilizer, insulation, joints where present, cryogenic system, and protection system handle disturbances.

The 13 T reference model was deliberately a conventional-magnet assumption. The preceding scan has now made a more precise statement: *within this conventional $H = 1$ route and these four declared plasma tests*, its $B_{\max} = 13$ T ceiling is insufficient, whereas a ceiling near 17.6 T makes the selected constraints overlap. A credible high-field conductor is therefore necessary for this particular remedy. It is not a theorem that HTS is necessary for every fusion strategy.

Rare-earth barium copper oxide (REBCO), commonly called an HTS conductor, is a serious candidate because it can retain useful current capability in fields and temperatures where a conventional low-temperature-superconductor design is much more constrained. This is no longer purely aspirational: a representative-scale REBCO TF model coil was built and tested to 20.1 T peak field on conductor at 20 K. ³

That last fact is the corrective to magical thinking. HTS can open a higher $B_{\max}$ window; it does not repeal Eq. (163). A fusion magnet program must still close stress and strain, quench detection and energy extraction, radiation tolerance, joint resistance where demountability is desired, manufacturing yield, cryogenic power, inspection, and replacement. The next sections remain in Freidberg’s conventional design space; only after its closure problem is fully mapped will the thesis ask whether a changed aspect ratio can solve a different geometric problem.

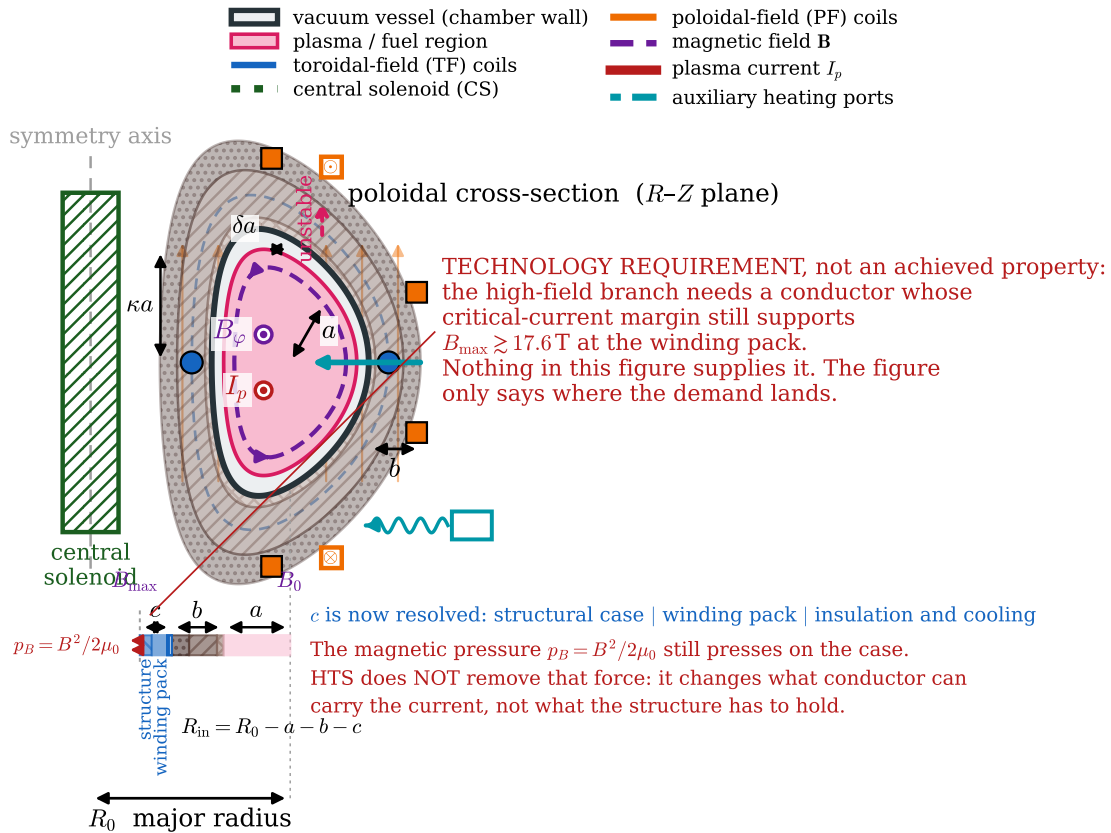
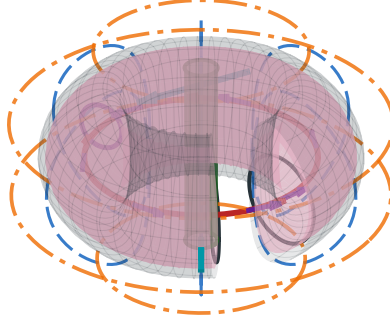
PLACE IMAGE HERE — Cumulative high-field design state: the inboard TF system now has a technology requirement. Use cumulative state 13, preserving its conventional tokamak geometry. In the inboard TF-coil bracket c , distinguish winding pack, structural case, insulation/cooling, and the nuclear island b . Add a callout: high-field branch requires a conductor technology whose operating critical-current margin supports $B_{\max} \gtrsim 17.6$ T. Put force arrows on the structural case and state explicitly that HTS does not remove them. Do not introduce ST geometry or a divertor.

³Z. S. Hartwig *et al.*, The SPARC Toroidal Field Model Coil Program, open preprint, 2023: <https://arxiv.org/abs/2308.12301>. The report also records 40.5 kA terminal current, large Lorentz loading, and nearly 1 GPa structural-case stress.

High-field design state — the inboard coil acquires a technology requirement

the same machine as stage 14. Only the coil bracket c is resolved further, and the demand it now carries is named

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 37: The same machine as the previous state; only the coil bracket c is resolved further, into structural case, winding pack and insulation/cooling. The field threshold is drawn as a REQUIREMENT on a conductor, not as a property the machine now has, and the force arrows stay: HTS changes what carries the current, not what the case must hold.

87. Freidberg's third remedy: change the plant scale

The high-field route changed a magnet limit. The next route changes a mission input: the requested electric output P_E . The reference case chose $P_E = 1000 \text{ MW}_e$. Raising that number may sound like making the problem harder, because a larger plant must make more fusion power. At fixed allowed neutron wall loading, however, it first forces a larger wall area:

$$R_0 a = 7.16 \left(\frac{P_E}{1000 \text{ MW}} \right) \text{ m}^2, \quad (169)$$

where the coefficient retains the 2015 blanket, shape, wall-loading, and thermal conversion assumptions.

The equation has an immediate geometrical meaning. For a chosen a , raising output raises R_0 in proportion. The machine gains toroidal length and plasma volume rather than being asked to generate arbitrarily more power per square metre of first wall. This is the physical content behind the familiar phrase economy of scale. It is not an accounting trick and it is not free: the absolute amount of plant, inventory, construction, and grid-connected power also grows.

For this test Freidberg holds the original plasma and magnet assumptions:

$$H = 1, \quad B_{\text{max}} = 13 \text{ T}, \quad W_n = 4 \text{ MW m}^{-2},$$

and again uses $f_B = f_{\text{NC}}$ to select the compatible a for each trial output. The purpose is narrow: discover whether scale alone can make the four chosen plasma-model constraints overlap without asserting better confinement or a higher-field magnet.

88. More output at fixed wall loading means more volume, not higher pressure

At the fixed temperature and profiles of the reference model, fusion power has the form

$$P_{\text{fus}} \propto p^2 V. \quad (170)$$

For a torus of fixed elongation, $V \propto R_0 a^2$. Insert Eq. (169):

$$V \propto (R_0 a) a \propto P_E a.$$

The thermal fusion power is proportional to P_E , so

$$P_E \propto p^2 (P_E a) \implies p \propto a^{-1/2}.$$

The factor P_E cancels. At fixed a , the mission asks for a proportionally larger volume at proportionally larger power, leaving the required average pressure unchanged. With the fixed $T \simeq 14$ keV model and $p = 2nT$, the same is true of the average density:

$$n \propto a^{-1/2}.$$

The confinement time makes the cancellation visible a second way. The stored thermal energy scales as $W \sim pV \propto P_E a^{1/2}$, while alpha heating scales as $P_\alpha \propto P_E$. Thus

$$\tau_{E,\text{req}} = \frac{W}{P_\alpha} \propto a^{1/2}, \quad (171)$$

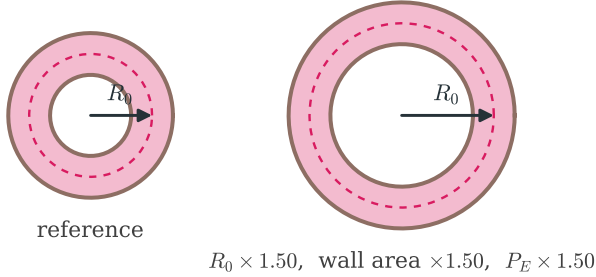
again with no direct P_E dependence at fixed a .

Design consequence. Higher output at fixed wall loading does not buy lower pressure by itself. It buys a larger volume in which to produce the larger total power. The benefit, if any, must enter through the way that larger R_0 changes the confinement-driven current and the current ledger.

PLACE IMAGE HERE — Reset physics sketch: scale at fixed neutron wall loading. Draw two conventional torus silhouettes at the same labelled wall loading W_n . The right machine has larger R_0 , larger wall area, and proportionally larger P_E ; keep the plasma cross-section and pressure shading comparable. Beside it show the short chain $P_E \uparrow \Rightarrow R_0 a \uparrow \Rightarrow V \uparrow$ and $P_E \propto p^2 V \Rightarrow p$ unchanged at fixed a . This is a reset scaling sketch, not a claim that large plants are automatically economical. No ST, HTS, or divertor.

TWO WAYS TO GROW THE SAME MACHINE — and what each one exports

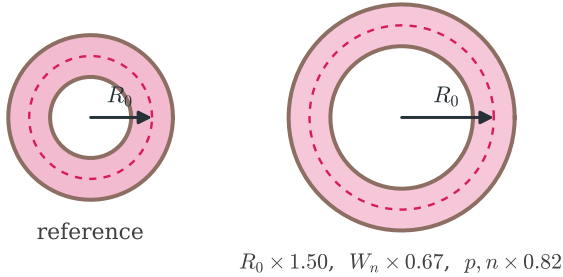
A. fixed wall loading W_n — the plant gets bigger because P_E did



$P_E \uparrow \Rightarrow R_0 a \uparrow \Rightarrow V \uparrow$
 $P_E \propto p^2 V$, and V grew by the
 same factor, so at fixed a :
 $p \times 1.00$ — UNCHANGED

Exported: capital scale — nothing about
 the plasma itself got any easier.

B. fixed P_E , lower W_n — the plant gets bigger to spread the SAME neutron power



$W_n \downarrow \Rightarrow V \uparrow \Rightarrow p, n \downarrow$
 same P_E over $\times 1.50$ the volume:
 $p \times 0.82$, $W_n \times 0.67$

Exported, and not small: MORE blanket,
 MORE shield, MORE TF-coil volume — all
 competing for the inboard radius \$46
 already showed is full.

Both rows are the SAME drawing with a different quantity held fixed — which is why they are one figure, not two.

Every ratio is computed from $V = 2\pi^2 R_0 a^2$ and $A_w = 4\pi^2 R_0 a$ at fixed a ; none is estimated. Row A's
 "p unchanged" is an identity, not an approximation. A scaling sketch only: it says nothing about economics.

Explanatory sketch; schematic, not to scale. Plan view: the annulus is the first wall, its shading the pressure.
 Not a cumulative machine state.

Figure 38: Two ways to grow the same machine. Row A is this brief (fixed W_n); row B is the brief on lower wall loading. They are one figure because they are the same drawing with a different quantity held fixed — and only side by side can the two chains be compared.

89. The enlarged machine changes the current requirement

The selected confinement relation, solved for current on page 59, has the dependence

$$I^{0.93} \propto \frac{\tau_{E,\text{req}} P_\alpha^{0.69}}{R_0^{1.39} a^{0.58} n^{0.41} B_0^{0.15}}.$$

At fixed a , the preceding page showed that $\tau_{E,\text{req}}$ and n do not carry a direct power-output factor. In contrast,

$$P_\alpha^{0.69} \propto P_E^{0.69}, \quad R_0^{1.39} \propto P_E^{1.39}.$$

Ignoring for one line the smaller change in B_0 caused by the changed radial geometry, their ratio is $P_E^{-0.70}$. Raising both sides to $1/0.93$ gives

$$I_{\text{req}} \propto P_E^{-0.70/0.93} \simeq P_E^{-0.75}. \quad (172)$$

This is the central mechanism of the power-output remedy. The larger major radius helps the empirical confinement scaling more strongly than the increased alpha power hurts it, so the regression asks for less current at a fixed cross-section.

The current-drive budget must be updated at the same time. If the selected recirculating-power fraction remains fixed, $P_{\text{CD}} \propto P_E$. But

$$I_{\text{CD}} = \frac{P_{\text{CD}} \eta_{\text{CD}}}{R_0 n}$$

contains the same factor $R_0 \propto P_E$ in its denominator. At fixed a its direct power dependence approximately cancels, leaving field and wave-accessibility changes to determine the remainder. The required total current, however, falls according to Eq. (172). Therefore the bootstrap fraction required by the ledger falls:

$$f_B = 1 - \frac{I_{\text{CD}}}{I}.$$

This is why a larger plant can ease both the kink test and steady-state closure in the stated model. It is not because a large reactor creates bootstrap current from nothing.

Model and limits. The exponent -0.75 is a consequence of the selected empirical confinement fit, fixed wall loading, fixed temperature/profiles, and a fixed recirculating-power fraction. It is a transparent model implication, not a size-scaling theorem for all tokamaks.

90. The power-output scan: the model closes near 1.55 GW_e

For each P_E , the calculation holds $H = 1$ and $B_{\max} = 13$ T, imposes $f_B = f_{\text{NC}}$, and then tests all four ratios. The output and radius grids are 1 MW and 0.001 m respectively.

P_E (MW)	a (m)	R_0 (m)	B_0 (T)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
1000	0.799	8.96	10.10	0.64	1.21	1.66	1.00
1200	1.030	8.34	9.52	0.71	1.09	1.31	1.00
1400	1.263	7.94	8.97	0.77	1.02	1.10	1.00
1550	1.436	7.73	8.57	0.81	0.99	1.00	1.00
1600	1.493	7.67	8.44	0.82	0.99	0.97	1.00
1800	1.717	7.51	7.95	0.87	0.97	0.89	1.00

The kink and Troyon ratios are the last two to close. The script's first all-pass point is $P_E = 1556$ MW on its displayed grids; Freidberg reports 1550 MW, an agreement appropriate to a simplified model and discrete scan. Greenwald remains below one with appreciable margin.

Audit note. This result does not say that a 1.55 GW_e reactor is preferred, financeable, grid-compatible, licensable, or buildable. It says only that raising the output is a third way to make the stated four plasma-model inequalities overlap while retaining $H = 1$ and the paper's 13 T magnet assumption.

91. A larger plant trades plasma closure for absolute scale and heat exhaust

The source’s engineered-volume proxy at the 1550 MW_e case is

$$\frac{V_I}{P_E} = 1.17 \text{ m}^3 \text{ MW}^{-1},$$

close to its 1.07 m³ MW^{−1} advanced-confinement comparison. That is the limited sense in which the source identifies an economy of scale: the engineered volume *per megawatt* need not rise sharply. It does not mean the total engineering volume or capital required for a 1550 MW_e plant is small; both are necessarily larger in absolute terms.

The same source reports its parallel-heat-flux proxy rising to

$$Q_{\parallel} = 673 \text{ MW T m}^{-1}$$

at the high-output point, compared with 507 MW T m^{−1} in its advanced-confinement comparison case. The proxy is a warning that core plasma-constraint closure does not settle power exhaust. It is not a divertor design calculation, and it must not be translated directly into a material surface limit.

Design consequence. The third remedy establishes a genuine alternative within the 2015 model: conventional field and ordinary confinement can meet the four selected plasma tests at a sufficiently large output. It also leaves us with a larger, more-concentrated generating station and a more demanding exhaust proxy. A reactor is not complete merely because one family of plasma ratios reaches one.

92. Freidberg's fourth remedy: lower the permitted wall loading

The last single-knob remedy changes the allowed average neutron wall loading W_n . The reference value was 4 MW m^{-2} . Lowering it means demanding a larger first-wall area for the same neutron power:

$$R_0 a = 7.16 \left(\frac{4 \text{ MW m}^{-2}}{W_n} \right) \text{ m}^2 \quad (P_E = 1000 \text{ MW}_e). \quad (173)$$

This is not an arbitrary way to make the plasma easier. It represents a conservative nuclear-material and first-wall design choice: less neutron power is assigned to each square metre, so more square metres must be built.

The immediate trade is visible before any plasma fit is used. At fixed a , $R_0 \propto W_n^{-1}$ and $V \propto R_0 a^2 \propto W_n^{-1}$. The same fusion output is produced in a larger volume. That lowers the required power density, pressure, and density, but it makes the reactor physically larger. Freidberg holds

$$P_E = 1000 \text{ MW}_e, \quad H = 1, \quad B_{\text{max}} = 13 \text{ T},$$

and again imposes bootstrap closure to select $a(W_n)$ before testing the four ratios.

Design consequence. A lower wall-loading limit is not a plasma improvement supplied for free. It is a decision to purchase lower power density with more wall, blanket, shielding, magnets, and building-scale volume. The next pages quantify how that engineering choice reaches the current problem.

93. Lower wall loading reduces the required pressure

At fixed output and temperature, use again

$$P_{\text{fus}} \propto p^2 V.$$

From Eq. (173), $V \propto W_n^{-1} a$ at fixed elongation. Holding total fusion power fixed therefore gives

$$p \propto \left(\frac{W_n}{a} \right)^{1/2}, \quad n \propto \left(\frac{W_n}{a} \right)^{1/2}. \quad (174)$$

Unlike the higher-output case, there is no cancellation: reducing W_n genuinely reduces the average pressure and density required to make the same total power.

The stored energy scales as

$$W \sim pV \propto \left(\frac{W_n}{a} \right)^{1/2} \left(\frac{a}{W_n} \right) = \left(\frac{a}{W_n} \right)^{1/2}.$$

The alpha power is unchanged because the total fusion output is unchanged. Hence

$$\tau_{E,\text{req}} \propto \left(\frac{a}{W_n} \right)^{1/2}. \quad (175)$$

This longer required confinement time may look like a penalty, but it must be placed beside the larger R_0 , lower density, and changed field geometry in the empirical confinement law before judging the net current demand.

PLACE IMAGE HERE — Reset physics sketch: lower wall loading trades density for machine scale. Draw two conventional torus silhouettes at the same P_E and temperature. The right machine has a larger R_0 , more first-wall area, lower labelled W_n , and lighter pressure/density shading. Show $W_n \downarrow \Rightarrow V \uparrow \Rightarrow p, n \downarrow$, then the explicit warning: more blanket, shield, and TF-coil volume. This is a reset scaling sketch, not a cumulative machine state. No ST, HTS, or divertor.

This brief and the previous one ask for the same drawing with a different quantity held fixed, so they share one figure: see row B of Figure 38.

94. The net current direction follows from all terms together

Use the same selected confinement dependence:

$$I^{0.93} \propto \frac{\tau_{E,\text{req}} P_\alpha^{0.69}}{R_0^{1.39} a^{0.58} n^{0.41} B_0^{0.15}}.$$

At fixed a , Eqs. (174) and (175) give

$$\tau_{E,\text{req}} \propto W_n^{-1/2}, \quad R_0^{-1.39} \propto W_n^{1.39}, \quad n^{-0.41} \propto W_n^{-0.205}.$$

Multiplying the three explicit wall-loading powers gives

$$W_n^{-0.500+1.390-0.205} = W_n^{0.685}.$$

Consequently,

$$I_{\text{req}} \propto W_n^{0.685/0.93} \simeq W_n^{0.74}, \tag{176}$$

apart from the smaller self-consistent change in B_0 . Lowering wall loading lowers the current demanded by the selected regression. It also lowers beta directly through Eq. (174), while the larger R_0 and lower current improve the kink ratio in the full calculation.

External current drive deserves the same caution as before. With fixed output, P_{CD} is fixed. As W_n falls, $R_0 n \propto W_n^{-1/2}$ grows, which tends to reduce the externally driven current in the selected efficiency model. Yet I_{req} falls more rapidly in this family; the required bootstrap share still decreases after all quantities are recalculated. The conclusion is a coupled result, not a consequence of holding I_{CD} fixed by hand.

95. The wall-loading scan closes near 2.1 MW m^{-2}

The calculation now scans W_n while retaining $P_E = 1000 \text{ MW}_e$, $H = 1$, and $B_{\max} = 13 \text{ T}$. At each value it first sets $f_B = f_{\text{NC}}$ and then tests the same four ratios.

$W_n \text{ (MW m}^{-2}\text{)}$	$a \text{ (m)}$	$R_0 \text{ (m)}$	$B_0 \text{ (T)}$	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
4.00	0.799	8.96	10.10	0.64	1.21	1.66	1.00
3.50	0.890	9.19	10.04	0.65	1.11	1.49	1.00
3.00	1.009	9.46	9.96	0.67	1.00	1.32	1.00
2.50	1.171	9.78	9.85	0.68	0.89	1.14	1.00
2.10	1.349	10.11	9.72	0.69	0.80	1.00	1.00
2.00	1.403	10.21	9.68	0.70	0.78	0.97	1.00

The kink criterion is again the final one to close. The transparent script finds the largest all-pass wall loading as 2.09 MW m^{-2} on its displayed grids; the source reports $W_n \lesssim 2.1 \text{ MW m}^{-2}$. This agreement is a useful check on the algebra, not validation of every physical assumption beneath the 2015 model.

Audit note. Lower wall loading can close the selected plasma constraints without advanced confinement or a higher-field magnet. It does so by selecting a larger, lower-power-density machine. Any claim that this is the best reactor strategy must also compare its blanket, shield, coil, construction, maintenance, and financing burden.

96. The reward is lower exhaust proxy; the price is a much larger plant

The source reports the low-wall-loading case's engineered-volume proxy as

$$\frac{V_I}{P_E} = 2.59 \text{ m}^3 \text{ MW}^{-1}$$

near $W_n = 2.1 \text{ MW m}^{-2}$, compared with $1.07 \text{ m}^3 \text{ MW}^{-1}$ in its advanced-confinement comparison: about a factor 2.42 increase. This is qualitatively expected from the enlarged radial and toroidal dimensions, though the exact number belongs to its specified blanket and TF-coil geometry.

The reported parallel-heat-flux proxy moves in the favorable direction, from 507 to 374 MW T m^{-1} in those compared cases. Lower average power density makes an exhaust problem less severe, but it does not solve it. Divertor and first-wall design still require a separate calculation of power sharing, scrape-off-layer transport, radiation, magnetic expansion, detachment, and material response.

Design consequence. The four single-variable remedies have now revealed four distinct currencies: empirical confinement margin, magnet field and structure, plant output, and neutron-wall area. Each can close the selected plasma ledger under stated assumptions; each exports difficulty into another part of the reactor. The next task is to compare the remedies on one honest ledger before moving beyond Freidberg's conventional baseline.

97. One comparison ledger: which changes actually close the four tests?

The four options are not four versions of the same physical claim. Each changes one declared input while retaining the others as closely as the coupled calculation permits:

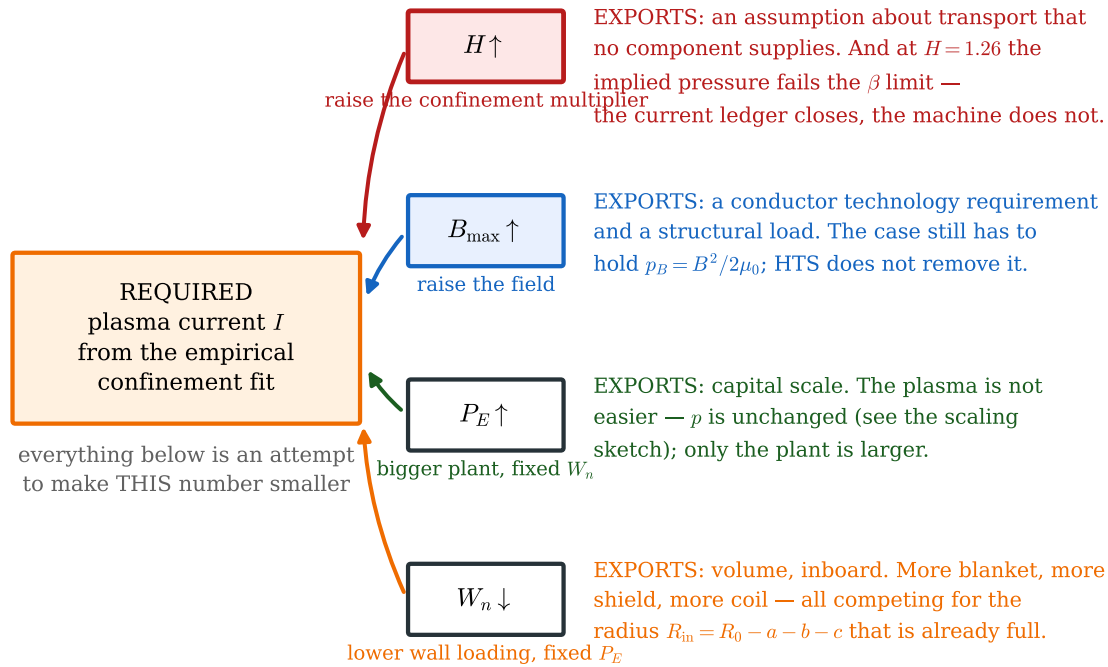
Option	Changed demand	Result in the stated model
Improved confinement	$H : 1 \rightarrow 1.26$	Kink and bootstrap close, but the Troyon ratio is about $1.20 > 1$; not a full closure.
High field	$B_{\max} : 13 \rightarrow 17.6 \text{ T}$	All four ratios close; magnet stress, coil build, and engineered-volume proxy rise.
Higher output	$P_E : 1000 \rightarrow 1550 \text{ MW}_e$	All four ratios close; absolute plant size and heat-exhaust proxy rise.
Lower wall loading	$W_n : 4 \rightarrow 2.1 \text{ MW m}^{-2}$	All four ratios close; engineered volume per megawatt rises strongly.

The first row is deliberately retained. It teaches the method: finding one simultaneous equality, such as $f_B = f_{\text{NC}}$, is not enough. Every inequality must be retested at the same state. The remaining three routes pass this particular four-ratio screen only because the changed input alters the confinement-driven current, the current ledger, and the pressure/stability ledger together.

PLACE IMAGE HERE — Reset comparison sketch: four ways to move the same current closure. Draw one compact causal map with a central block labelled required plasma current from the empirical confinement fit. Four incoming branches: $H \uparrow$ (empirical transport assumption); $B_{\max} \uparrow$ (field and structural demand); $P_E \uparrow$ at fixed W_n (larger plant); and $W_n \downarrow$ at fixed P_E (larger lower-power-density plant). Each branch must show a distinct exported burden. Mark $H = 1.26$ as beta failure, and the other three as four-ratio closure only within the stated 2015 models. This is a reset causal graphic, not a new machine drawing.

FOUR WAYS TO MOVE THE SAME NUMBER — and where each one puts the cost

 empirical / model input
 primary parameter
 chosen input
 fails the test



Only the top branch is marked as a FAILURE. $H = 1.26$ closes the current ledger and then violates the β limit — it is not a fourth option, it is the same difficulty moved where the ledger cannot see it. The other three DO close the four ratios, but only inside the stated 2015 model set, whose confinement fit, stability threshold, coil build and bootstrap coefficient are each a modelling choice. None of the four is a result about tokamaks — all four are results about that model set, and each hands its cost to a different engineer.

Explanatory sketch; a causal map, not a machine drawing. Branch positions carry no ordering.

Figure 39: Reset comparison sketch: four ways to move the same current closure. *Idealized schematic; not a fabrication drawing.*

98. The source’s final cases, stated without pretending to measure reality

For orientation, the 2015 paper reports the following rounded states. The values are not independent targets; they are outputs of its coupled equations after the listed input change. The reference row is shown because it makes the original mismatch visible.

	reference	$H = 1.26$	high field	high output	low wall load
$B_{\max}$ (T)	13	13	17.6	13	13
P_E (MW)	1000	1000	1000	1554	1000
W_n (MW m ⁻²)	4.0	4.0	4.0	4.0	2.1
a (m)	1.34	1.26	0.97	1.44	1.35
R_0 (m)	5.34	5.57	7.39	7.70	10.2
I (MA)	14.3	10.0	7.64	11.2	8.52
q_K/q_*	1.28	1.00	1.00	1.00	1.00
f_B/f_{NC}	1.90	1.00	1.00	1.00	1.00
β/β_T	0.94	1.21	0.82	1.00	0.80

The table confirms three conceptual lessons. First, the reference failure is current-centered: its q_K/q_* and bootstrap ratio exceed one. Second, reducing current is not sufficient if the way it is reduced also tightens the beta allowance, as the $H = 1.26$ row shows. Third, high field, plant scale, and low wall loading are not interchangeable technologies; they reach current closure through different geometric and engineering commitments.

Model and limits. This table reproduces rounded values reported in the freely available 2015 source. Our companion script independently reproduces the three closure thresholds to the resolution stated in the relevant scans. Neither result turns the source’s profile choices, empirical boundaries, or coil model into universally valid reactor predictions.

99. What Freidberg’s calculation has rigorously established

The word rigorously must be used carefully. The following implication has been established by transparent substitution and numerical quadrature:

2015 mission, geometry, coil, profile, confinement, current-drive,
bootstrap, and four-limit assumptions

⇓

the reference conventional state fails kink and
steady-state bootstrap closure simultaneously.

The size scan shows that merely choosing another minor radius within that baseline does not make all four tests pass. The remedy scans then identify which changed assumptions can restore overlap, and what new burden each imports.

What has *not* been established is equally important. The calculation is not a proof that every conventional tokamak fails, that any one high-field plant can be built, or that a ratio below one makes a reactor safe. The Greenwald relation and the chosen normalized-beta allowance are operational boundaries, not theorems. The confinement and current-drive laws are empirical/model closures. The bootstrap result depends on the chosen equilibrium and profiles. The analytic coil model is not a detailed magnet design.

Design consequence. The real achievement of the reconstruction is not a slogan about tokamaks. It is a map of conditional causal failures: which physics assumption or engineering choice causes each failure, what intervention can remove it in the model, and what new design obligation is thereby created.

100. Four ratios are not a complete reactor acceptance test

The 2015 paper intentionally keeps several major reactor questions outside its four-ratio screen. The most immediate is power exhaust. $Q_{\parallel}$ is only a parallel-transport proxy; no scrape-off-layer width, divertor geometry, impurity radiation, detachment window, transient load, or material lifetime has yet been calculated. A passing value of q_K/q_* is likewise not a complete disruption-risk assessment. Vertical stability, control authority, error fields, energetic particles, resistive walls, and nonlinear evolution are not exhausted by one edge safety factor.

On the engineering side, the radial-build calculation has identified why blanket, shield, winding pack, and structure consume space, but it has not produced a maintainable nuclear island. Neutron damage, tritium inventory, remote handling, cryoplat electricity, joint reliability where relevant, fatigue, and component replacement remain design problems. On the plant side, net electric output and recirculating power must eventually be joined to availability, construction schedule, licensing, fuel cycle, and grid integration.

None of these omissions invalidates the four-ratio argument. They prevent us from misrepresenting it as a completed reactor design. The correct next move is to retain the conventional baseline as a reference and ask whether a different aspect-ratio geometry can improve the current-and-field trade *without* hiding the radial build and central-stack obligations it creates.

101. The conventional baseline is now complete enough to earn the next question

We began without a tokamak and let the conventional machine reveal itself: a hot charged fusion plasma required magnetic confinement; the escape direction required a torus and TF coils; toroidal drifts required current, poloidal field, and shaping; steady state required a current ledger; fusion neutrons required the nuclear island; and the field required a finite inboard magnet build. The primary parameter set

$$a, R_0, \kappa, b, c, T, n, p, \tau_E, B_0, \beta, I, q_*, f_B$$

has acquired physical meaning before it was used as engineering input.

Freidberg's reference calculation then exposed the conventional current closure: under its declared assumptions, an ordinary reference machine needs too much current for both its adopted kink margin and its available steady-state bootstrap share. The high-field result gives HTS its correct causal position: it is a technology that can open a needed field window in one closure route, while intensifying structural and protection demands.

Only now is it legitimate to consider low-aspect-ratio, spherical-tokamak geometry. It will not be introduced as a compact machine-shaped answer. It must first be asked what aspect ratio changes in the same equations, which conventional failures it can relieve, and why its reduced central space creates the central-stack problem. The next chapter begins that comparison from the already-earned radial-build ledger.

102. A spherical tokamak changes one geometric ratio, not the laws of plasma

Define the plasma aspect ratio by

$$A \equiv \frac{R_0}{a}. \quad (177)$$

A conventional tokamak has A of order 3–4 or larger. A low-aspect-ratio tokamak, commonly called a spherical tokamak (ST), has A of order 1–2. The word spherical describes the compact toroidal shape; the plasma remains a torus, not a filled sphere. It still needs TF field, poloidal field, equilibrium control, heating, exhaust, a vacuum boundary, and a nuclear island.

At a fixed fusion mission and wall-loading product $R_0 a = C_{\text{wall}}$, the two geometric variables can instead be written in terms of A :

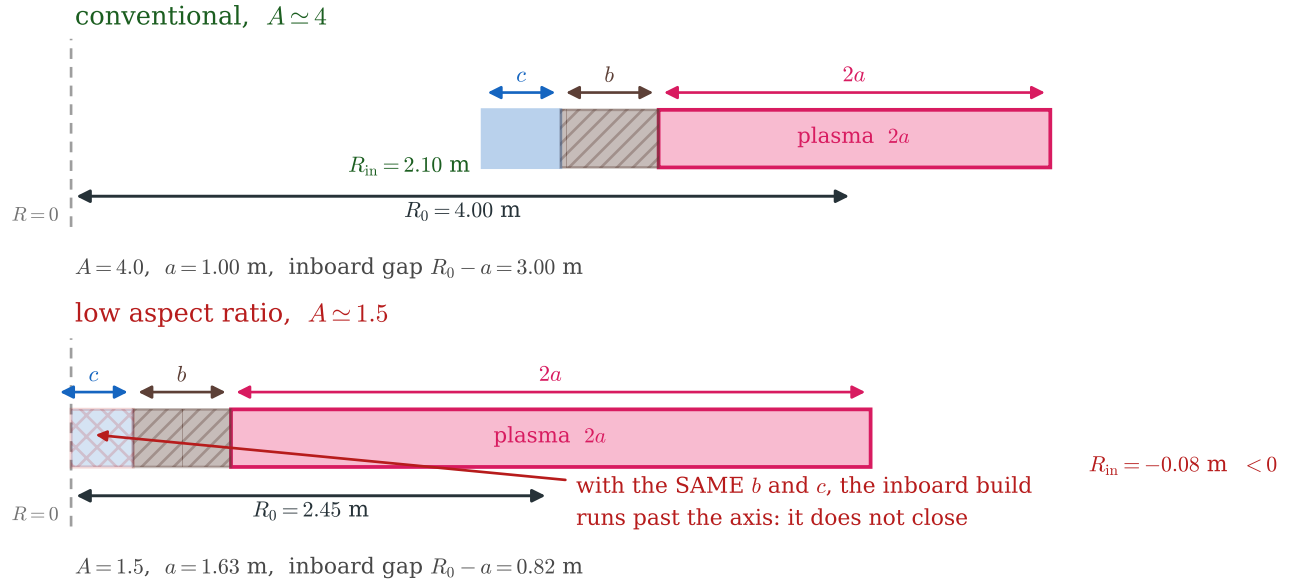
$$R_0 = (A C_{\text{wall}})^{1/2}, \quad a = \left(\frac{C_{\text{wall}}}{A} \right)^{1/2}. \quad (178)$$

Thus lowering A makes the cross-section larger while bringing the plasma centre closer to the symmetry axis. It is not simply a uniformly shrunken conventional machine.

Design consequence. Aspect ratio is a causal geometric variable. Lowering it can alter field-line twist, trapped-particle physics, volume, and radial space at once. We must derive each direction separately before calling low aspect ratio an advantage.

PLACE IMAGE HERE — Reset geometry sketch: aspect ratio changes the torus, not the physics inventory. Show two midplane cuts with the same labelled $R_0 a = C_{\text{wall}}$: a conventional $A \simeq 4$ plasma and a low- $A \simeq 1.5$ plasma. Mark R_0 , a , and the symmetry axis. Retain only the already-earned plasma, nuclear-island bracket, and TF-system location as simplified outlines. Make the low- A inboard gap visibly smaller. Do not label either design as automatically superior and do not yet add an HTS brand or an ST-specific divertor.

THE SAME INVENTORY, TWO ASPECT RATIOS — at fixed $R_0 a = C_{\text{wall}}$



Both rows hold $R_0 a = 4.00 \text{ m}^2$, so the first-wall area is the same. $b = 0.50 \text{ m}$ and $c = 0.40 \text{ m}$ are fixed IN METRES: they are set by neutron attenuation and by conductor and structure, not by how round the plasma is—scaling them with a would assume the answer. Everything above is then arithmetic: $a = \sqrt{C_{\text{wall}}/A}$, $R_0 = Aa$.

NEITHER design is superior here, and the figure does not say one is. It says something narrower: low A cannot be adopted while holding b and c fixed — the inboard allocations have to be renegotiated. That is exactly what the following sections do, and why they quote their own shielding allocation and centre-column envelope instead of inheriting the conventional ones. The physics inventory has not changed; the room it fits into has.

Explanatory sketch; midplane cuts at a common radial scale. No HTS brand and no ST divertor: neither is derived yet.

Figure 40: Reset geometry sketch: aspect ratio changes the torus, not the physics inventory. *Idealized schematic; not a fabrication drawing.*

103. Low aspect ratio strengthens toroidal field variation and trapping

The toroidal-field relation remains

$$B_\varphi(R)R = B_0R_0.$$

At the inboard and outboard plasma edges, $R = R_0 - a$ and $R = R_0 + a$. Hence

$$\frac{B_{\text{out}}}{B_{\text{in}}} = \frac{R_0 - a}{R_0 + a} = \frac{A - 1}{A + 1}. \quad (179)$$

When A falls, this ratio falls: the inboard field becomes much stronger than the outboard field. The mirror argument from page 36 then says that a larger fraction of particles can be trapped. In the simple low-collisionality estimate used earlier,

$$f_t \simeq \left(1 - \frac{B_{\text{min}}}{B_{\text{max}}}\right)^{1/2} \simeq \left(\frac{2}{A + 1}\right)^{1/2}. \quad (180)$$

For orientation, this expression gives $f_t \simeq 0.63$ at $A = 4$ and $f_t \simeq 0.89$ at $A = 1.5$.

More trapped particles make a larger bootstrap response physically plausible because bootstrap current begins with trapped-particle/passing-particle collisional asymmetry. But Eq. (180) is not a bootstrap fraction. The large-aspect-ratio local closure used on pages 67–70 is no longer controlled at $A \sim 1$ –2. A finite-aspect-ratio kinetic calculation must replace it before claiming a numerical f_{NC} for an ST.

Model and limits. The direction of the trapping effect follows from the toroidal-field geometry and magnetic-moment argument. The resulting bootstrap fraction, profile, and collisionality dependence are finite-aspect-ratio neoclassical physics, not a number that can be extrapolated from Freidberg’s large-aspect-ratio coefficient.

104. Aspect ratio also changes the kink–beta coupling

The shaped edge safety-factor estimate was

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I} \left(\frac{1 + \kappa^2}{2} \right) = \frac{2\pi}{\mu_0} \left(\frac{1 + \kappa^2}{2} \right) \frac{a B_0}{A I}. \quad (181)$$

Normalized beta contains the same dimensional group $a B_0 / I$:

$$\beta_N \equiv \beta(\%) \frac{a B_0}{I_{\text{MA}}}. \quad (182)$$

Here $I_{\text{MA}} = I / (10^6 \text{ A})$ and $\beta(\%) = 100\beta$. Substitute

$$\frac{a B_0}{I} = \frac{\beta_N}{10^8 \beta}$$

into Eq. (181) to obtain

$$q_* = \frac{0.05}{A} \left(\frac{1 + \kappa^2}{2} \right) \frac{\beta_N}{\beta}. \quad (183)$$

The numerical 0.05 comes only from $\mu_0 = 4\pi \times 10^{-7}$ and the conventional percent–MA units in β_N .

Equation (183) is a geometric insight, not a stability limit. At fixed β , β_N , and shape, lower A raises q_* . Equivalently, at a required q_* it permits a larger normalized beta within this relation. This is the first-principles reason low aspect ratio is a plausible response to the Freidberg kink–beta coupling: it changes the geometry of the current that produces the stabilizing poloidal field.

Audit note. The equation does *not* prove that an ST is stable at a chosen β_N . The allowable β_N remains an equilibrium, wall, profile, rotation, and kinetic-MHD question. Low aspect ratio changes the relevant stability physics, so the conventional Troyon coefficient 2.8 cannot simply be reused as an ST theorem.

105. What an ST can plausibly relieve in the Freidberg closure

The conventional reference failed because its required current simultaneously made q_* too small and made steady-state replacement too demanding. Low aspect ratio attacks both mechanisms in direction, though not yet in certified magnitude:

- Equation (183) shows that smaller A increases q_* for the same normalized-beta and shape inputs. This can relax the kink-versus-pressure coupling that was restrictive in the conventional scan.
- Equation (180) shows stronger toroidal trapping. Combined with the pressure gradients already required by fusion, this creates a physical route to larger bootstrap current and a smaller externally driven current share.
- At fixed R_0a , Eq. (178) gives a larger minor radius. That changes volume, current profile, wall area distribution, shaping, and power-exhaust geometry; none may be held fixed without calculation.

This is enough to formulate a responsible test, not to announce its result. The question immediately next is narrow and mathematical: does a finite-aspect-ratio ST calculation satisfy the two ratios that failed in Freidberg's conventional baseline, namely the kink ratio and the non-inductive current ratio? We will use a published low- A equilibrium and current-drive model, state every replacement assumption, and evaluate those two ratios. The centre-stack problem is *not* a premise of that test; it is the separate engineering problem revealed only after the two Freidberg constraints have been answered.

Integrated ST studies make the same methodological point. Menard *et al.* report low-aspect-ratio pilot-plant configurations only after free-boundary equilibrium, detailed device layout, and three-dimensional neutronics are coupled; their configurations must also meet tritium breeding, divertor, and maintenance requirements.⁴

⁴J. E. Menard *et al.*, Fusion nuclear science facilities and pilot plants based on the spherical tokamak, *Nuclear Fusion* 56 (2016) 106023, freely accessible at <https://iopscience.iop.org/article/10.1088/0029-5515/56/10/106023/pdf>.

106. The ST question is exactly Freidberg's two failed ratios

Freidberg's conventional reference failed two current-centred tests:

$$\mathcal{R}_K = \frac{q_K}{q_*} > 1, \quad \mathcal{R}_B = \frac{1 - I_{CD}/I}{f_{NC}} > 1.$$

The first says that the confinement-demanded current leaves too little safety factor; the second says that the calculated self-current is smaller than the self-current share required after permitted external current drive is subtracted. For the conventional reference, pages 64 and 70 obtained 1.28 and 1.91, respectively.

The ST test uses the same accounting, but it must replace the two large-aspect-ratio inputs rather than reuse them carelessly:

$$\mathcal{R}_{K,ST} = \frac{q_{K,ST}}{q_{*,ST}}, \quad \mathcal{R}_{B,ST} = \frac{1 - I_{CD,allow}/I_{ST}}{f_{BS,ST}}.$$

Here $q_{K,ST}$ is an ST-specific kink boundary, $f_{BS,ST}$ is obtained from an ST finite-aspect-ratio bootstrap model, and $I_{CD,allow}$ is the external current that the plant can afford to supply under an independently stated power and efficiency constraint. The logic has not changed. A candidate closes Freidberg's bootstrap obstruction only when those two inputs are independently obtained and the ratio is at most one.

Design consequence. The centre-stack question is not included in Eq. (). It must not be used to weaken or postpone the answer. The task first is to calculate the ST kink ratio and to compare its finite- A bootstrap prediction with an independently justified current-drive allowance.

107. The finite-aspect-ratio bootstrap model is declared

Menard *et al.* use a low-aspect-ratio equilibrium-informed bootstrap closure of the form

$$f_{\text{BS}} = C_{\text{BS}} \sqrt{\epsilon} \beta_{\text{pol,th}}, \quad \epsilon \equiv \frac{a}{R_0} = \frac{1}{A}, \quad (184)$$

where $\beta_{\text{pol,th}}$ is the thermal poloidal beta and C_{BS} is a profile/safety-factor coefficient. Their stated parameterization is

$$C_{\text{BS}} = \max \left(1.2 - 0.2 \frac{q_*}{q_{\min}}, 0.6 \right). \quad (185)$$

Here $q_{\min}$ is the minimum safety factor of the assumed equilibrium/current profile. It is an input to this finite- A closure, not a new universal constant. ⁵

The definition of poloidal beta makes the physical content visible:

$$\beta_{\text{pol,th}} \equiv \frac{2\mu_0 \langle p_{\text{th}} \rangle}{B_{\text{pol}}^2}.$$

It increases when thermal pressure rises relative to the poloidal magnetic field. Equation (184) then combines the pressure-gradient source of bootstrap current with the low- A orbit geometry through $\sqrt{\epsilon}$. This is the finite- A replacement for Freidberg's large-aspect-ratio local integral on pages 67–70.

Model and limits. The new coefficient and the equilibrium-derived poloidal beta make Eq. (184) a model closure. That is acceptable and necessary: Freidberg's own numerical bootstrap result also required a declared neoclassical closure. What would be unacceptable is to use the conventional large- A coefficient at $A \simeq 1.7$ and call the extrapolation a proof.

⁵J. E. Menard *et al.*, cited on page 105, Sec. 2.1.5, Eqs. and surrounding text on bootstrap/current-drive modeling. They state that this form is consistent with low-aspect-ratio equilibrium calculations; it is not deduced from Maxwell's equations alone.

108. A published finite- A candidate supplies the kink test and current split

The following is not a generic ST slogan. It is a specified low-aspect-ratio systems calculation from Menard *et al.*: $A = 1.7$, $\kappa = 2.75$, $B_{T0} = 3$ T, ITER- $H_{98} = 1.25$, and 80 MW of neutral-beam heating/current-drive power. Their density scan at $R_0 = 1.6$ m reports

$$\beta_N = 4.0\text{--}4.7, \quad q_* \geq 3, \quad I = 10.8\text{--}12 \text{ MA}.$$

At normalized density near 0.8, their current model gives a bootstrap fraction of about $f_{BS} = 0.76$ and a beam-driven share of about 0.24.⁶

These inputs give an ST safety factor and an ST bootstrap/current-drive distribution at the same operating point. They demonstrate that the authors can construct a fully non-inductive scenario in their model. They do *not*, by themselves, state the independently derived current-drive allowance required in the numerator of Eq. (). That distinction is the difference between constructing a current balance and testing Freidberg's economic current-drive constraint.

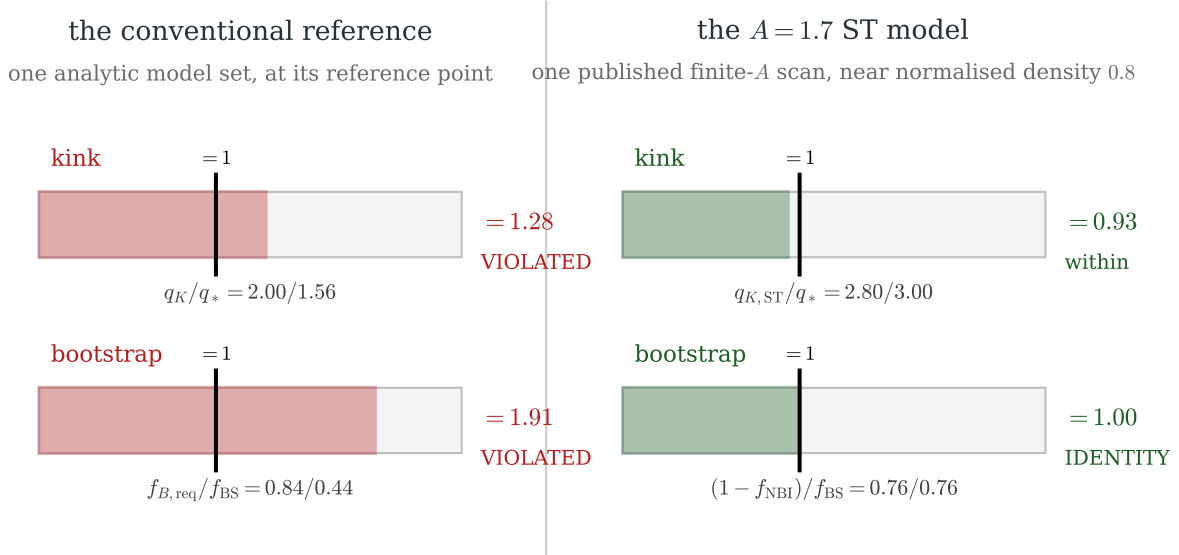
Audit note. The displayed $\beta_N = 4.0\text{--}4.7$ range characterizes this candidate, but it is not silently used as a passed stability test in the present two-constraint comparison. Menard *et al.* provide a separate normalized-beta/stability assessment for their ST equilibria. That analysis must be evaluated explicitly before this thesis makes any broader high- β_N stability claim.

PLACE IMAGE HERE — Freidberg two-constraint closure: published low- A ST operating point. Use a two-column, deliberately non-engineering graphic. Left: the conventional Freidberg reference with $q_* = 1.56 < 2$ and required/calculated bootstrap 0.84/0.44. Right: the specified $A = 1.7$ ST model with $q_* \geq 3$ against $q_{K,ST} = 2.8$, and current fractions $f_{BS} \simeq 0.76$ plus $f_{NBI} \simeq 0.24$. Label the right-hand current sum as an identity imposed by a fully non-inductive scenario, not as an independently passed Freidberg ratio. State the resulting required external current explicitly. Do not draw magnets, centre stacks, HTS, or reactor hardware; this image separates the resolved kink result from the still-conditional bootstrap question.

⁶Menard *et al.*, cited on page 105, Secs. 2.1.3–2.1.6 and Figs. 8–9. Values read from the published scan are shown to two significant figures. The paper calculates total non-inductive current as the sum of bootstrap and neutral-beam current.

FREIDBERG'S TWO OBSTRUCTIONS — one model set each side, read the same way

every cell is written DEMAND / ALLOWANCE, so > 1 means violated in all four



WHAT THIS DOES SAY. In the stated finite- A model the kink obstruction closes on its own terms: $q_* \geq 3$ against ST-specific $q_{K, ST} = 2.8$ gives 0.93, and that IS a test that could have failed. It is a model-backed existence result — one configuration in which the number that failed on the left does not fail.

READ THE BOTTOM-RIGHT CELL CAREFULLY. Its 1.00 is an IDENTITY, not a measurement. The scenarios are fully non-inductive — total non-inductive current is computed as bootstrap + NBI — so $f_{BS} + f_{NBI} = 1$ at every point of the scan (60+40 and 84+16 at its two ends). $(1 - f_{NBI})/f_{BS}$ is therefore exactly one everywhere and could not have come out otherwise. The real question the right column raises is the PRICE of that closure: 80 MW of 0.5 MeV negative-ion NBI supplying 0.24 of a 10.8–12 MA plasma. Margin lives there, not in the ratio. And nothing here says every ST is stable or that this one is feasible: the centre-stack question is untouched.

Explanatory sketch; a comparison of declared model results. No magnet, centre stack, HTS or reactor hardware appears: this figure answers the plasma question only.

Figure 41: Freidberg's two obstructions, read with their required logical inputs visible. The ST kink comparison is an actual threshold test. The ST 0.76/0.24 current split is an identity of Menard's fully non-inductive scenario, not an independent bootstrap-ratio pass: it identifies the external current the scenario must provide. Both columns are declared model results, not properties of tokamaks or of STs.

109. First answer: the ST candidate closes the kink obstruction

The relevant ST disruption evidence used by Menard *et al.* is more demanding than Freidberg's conventional reference threshold: NSTX disruption data show a marked increase in disruptivity below $q_* \simeq 2.8$. Their stated ST design requirement is therefore

$$q_{K,ST} = 2.8.$$

⁷ The same candidate scan gives $q_* \geq 3$. Substitute these published values:

$$\mathcal{R}_{K,ST} = \frac{2.8}{q_*} \leq \frac{2.8}{3.0} = 0.93 < 1. \quad (186)$$

This is an affirmative answer to Freidberg's kink obstruction within the declared finite- A model. It does not claim that every ST is kink stable; it demonstrates that the low- A candidate can satisfy a stricter ST-specific safety-factor criterion than the $q_K = 2$ criterion that the conventional reference failed. The geometrical reason anticipated on page 104 is now joined to an explicit operating-point test.

⁷Menard *et al.*, cited on page 105, Sec. 2.1.3. The numerical boundary comes from an NSTX data-based stability/disruption criterion and remains profile, wall, and operating-regime dependent.

110. The bootstrap result identifies the price; it does not yet test Freidberg's constraint

The current ledger does not change with aspect ratio:

$$I_{ST} = I_{BS} + I_{CD}.$$

Divide by I_{ST} . The Menard candidate gives, to the precision supported by the published scan,

$$\frac{I_{BS}}{I_{ST}} \simeq 0.76, \quad \frac{I_{CD}}{I_{ST}} \simeq 0.24,$$

and the two fractions sum to one. This equality is not an independent test. The Menard scenario is constructed to be fully non-inductive, so

$$\frac{I_{BS}}{I} + \frac{I_{CD}}{I} = 1$$

by its current-accounting definition. Dividing its own complement by its own bootstrap fraction produces 1 identically; it could not reveal a failure.

What the published point does give us is the physical and engineering burden that an independent Freidberg test must meet:

$$I_{CD,needed} = (1 - f_{BS})I \simeq 0.24I \simeq (0.24)(10.8 \text{ MA}) \simeq 2.6 \text{ MA}. \quad (187)$$

Here $I \simeq 10.8 \text{ MA}$ is paired with $f_{BS} \simeq 0.76$ at normalized density near 0.8; the full scan's current extrema are not combined with this selected fraction. The scenario assumes 80 MW of 0.5 MeV neutral-beam heating/current-drive power to obtain this fully non-inductive balance. It therefore predicts a substantial bootstrap fraction in its equilibrium-informed model and states the associated external-drive burden; it does not independently establish that a plant can afford that burden.

The proper Freidberg test remains conditional until an economic/recirculating-power analysis supplies $I_{CD,allow}$ independently:

$$\mathcal{R}_{B,ST} = \frac{1 - I_{CD,allow}/I}{f_{BS}} \leq 1 \quad \Longleftrightarrow \quad I_{CD,allow} \geq I_{CD,needed}. \quad (188)$$

This is the question the next research pass must answer. It is more rigorous than calling the identity $0.76/0.76 = 1$ a measured closure.

The condition can be priced with published scalars alone. Freidberg's allowance was 100 MW of wall electricity for current drive, of which 40 MW reaches the plasma (a 50% wall-to-klystron and 80% absorption chain); Menard's FNSF systems analysis assumes a neutral-beam wall-plug efficiency of 0.4.⁸ Keeping the published 80 MW beam state would therefore draw $80/0.4 = 200 \text{ MW}_e$, twice that allowance; fitting under it would need a wall-plug efficiency of 0.8, or, if only the allowance's 40 MW were injected, a drive efficacy of about $2.6 \text{ MA}/40 \text{ MW} \simeq 65 \text{ kA/MW}$ at that state. These are bounds on the published state, not on a redesigned ST: lowering the beam power changes heating, fast-ion pressure, fusion power, current, and bootstrap fraction together, and the FNSF point is a $Q_{DT} \simeq 2$ neutron-source mission rather than a 1 GW_e plant. The affordability question can only be settled on a power-plant-mission ST scan carrying its own recirculating-power budget.

⁸Freidberg *et al.*, Sec. 4.6; Menard *et al.*, Sec. 2.1.6, where $\eta_{aux} = 0.4$ is described as higher than presently achievable. The arithmetic is reproduced in `thesis/code/calculations/st_drive_affordability.py`.

111. Verdict: ST closes the kink obstruction; bootstrap closure has a stated condition

The published finite-aspect-ratio ST model answers one of Freidberg’s two current-centred obstructions affirmatively:

$$\mathcal{R}_{K,ST} \leq 0.93 < 1.$$

It also predicts a large bootstrap fraction, $f_{BS} \simeq 0.76$, and makes the missing condition quantitative: the operating point needs approximately 2.6 MA of independently affordable current drive.

We must not call that second result closure until its current-drive allowance has been obtained independently. Freidberg’s conventional comparison was meaningful because the permitted drive came from a separate recirculating-power calculation while the bootstrap fraction came from a neoclassical calculation. Reusing the complement of a fully non-inductive ST current balance would erase that distinction.

Design consequence. Low aspect ratio has now been shown to close the kink obstruction in a published finite- A model and to raise the bootstrap fraction to a quantified, promising level. Whether it closes Freidberg’s bootstrap obstruction depends on the separate question $I_{CD,allow} \gtrsim 2.6$ MA.

112. Only after that answer does the separate centre-stack problem begin

The conclusion on page 111 does not say that the ST has no further problems. It says the kink question is answered affirmatively and the bootstrap question has been reduced to an explicit independent current-drive requirement. Lowering A simultaneously brings the plasma closer to the symmetry axis and strengthens the inboard toroidal field. That creates a new engineering question: can a D–T reactor fit shielding, breeding, TF conductor, support, cooling, protection, and perhaps current-initiation function inside the remaining central radius?

That question is logically separate. It cannot retroactively turn the resolved kink ratio into a failure, any more than it can replace the independent current-drive analysis still required for bootstrap closure. The remainder of this chapter changes its subject explicitly: from *what does low- A plasma physics change?* to *what engineering debt does the low- A solution create, and can HTS help pay it?*

113. The same geometry takes radial space away from the centre stack

The inboard radial ledger is the unavoidable counterargument. Let c_{cs} denote the radial build occupied by the centre-stack TF conductor, its structural case, insulation, cooling, and any allocated central-solenoid function. The innermost radius available to the toroidal field is

$$R_{cs} = R_0 - a - b - c_{cs} = R_0 \left(1 - \frac{1}{A} \right) - b - c_{cs}. \quad (189)$$

The same $BR = \text{constant}$ relation now gives

$$B_{cs} = \frac{B_0 R_0}{R_{cs}}, \quad B_0 = B_{cs} \left[1 - \frac{1}{A} - \frac{b + c_{cs}}{R_0} \right]. \quad (190)$$

Every term in the square bracket has a physical meaning. Lowering A subtracts geometric room; b is demanded by neutrons; c_{cs} is demanded by current, structure, insulation, cooling, and possibly start-up flux. The bracket must be positive even before a usable central field can exist.

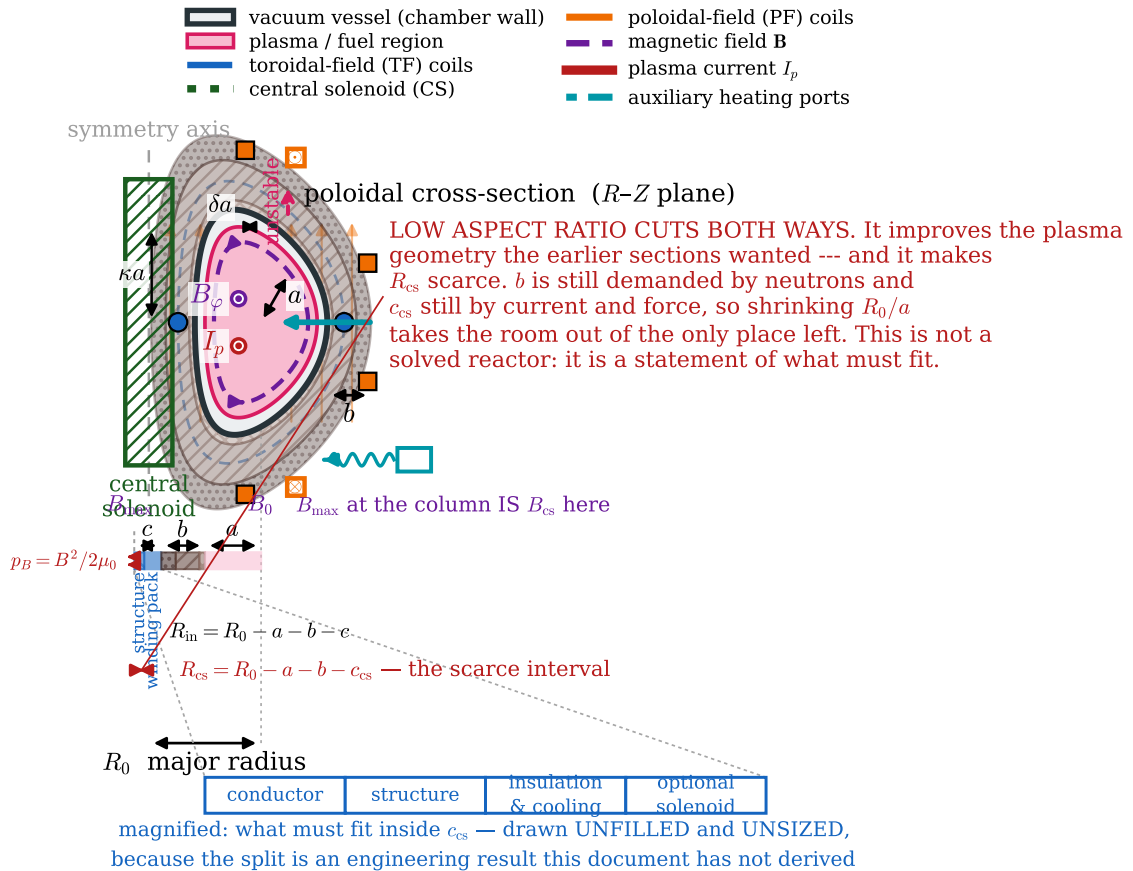
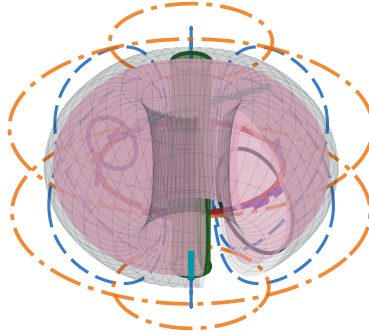
As $A \rightarrow 1$, the purely geometric factor $1 - 1/A$ tends to zero. No improved plasma stability can reverse that subtraction. A low- A reactor must instead choose among a larger R_0 , thinner-but-credible inboard nuclear and magnet systems, a higher permissible B_{cs} , different start-up/current-drive architecture, or some combination. This is the centre-stack problem in its most elementary form.

PLACE IMAGE HERE — Cumulative construction state: low aspect ratio reveals the centre stack. Start with the complete conventional radial-build state, then reduce A while preserving the plasma, first wall/nuclear island b , and TF function. On the inboard side show the shrinking interval $R_{cs} = R_0 - a - b - c_{cs}$. Split c_{cs} into conductor, structure, insulation/cooling, and optional central- solenoid allocation. Label B_0 at the plasma and B_{cs} at the central column. The image must show that low A improves some plasma geometry while making this physical interval scarce; do not depict a solved reactor.

Low aspect ratio — the centre stack becomes the scarce interval

the same inventory as stage 14, at $A=2$: the plasma, the nuclear island b and the TF function are all preserved

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.

$R_0/a = 2.0$, $\kappa = 1.70$, $\delta = 0.40$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 42: The same inventory as the complete radial build, at $A = 2$: the plasma, the nuclear island b and the TF function are all preserved — only the aspect ratio changes. What changes with it is the room left for R_{cs} . The magnified strip is drawn unfilled and unsized on purpose: that split is an engineering result this document has not derived.

114. HTS is a centre-stack lever, not a centre-stack eraser

Equation (190) explains why HTS becomes especially valuable in a low-aspect-ratio branch. For fixed desired B_0 , reducing A raises the required central-column field B_{cs} . A conductor with credible operating current margin at higher field can make a smaller or more capable centre stack possible. It may therefore preserve some of the low- A gain that an ordinary conductor field ceiling would otherwise erase.

But the same equation returns us to page 47:

$$p_B = \frac{B_{cs}^2}{2\mu_0}.$$

The central column must carry that magnetic loading while remaining manufacturable, cooled, electrically insulated, protected, and maintainable in a neutron environment. HTS does not turn c_{cs} into zero; its conductor and support still occupy space, and higher field can increase the structural allocation. The central column also competes with shielding and tritium-breeding volume, which cannot be dismissed for a D-T plant.

Design consequence. HTS can raise the credible field ceiling in the precise radial interval that low aspect ratio makes scarce. It is a powerful enabling lever for an ST, but it cannot by itself close the centre-stack equation.

115. What remains after the ST answers Freidberg's two questions

The narrowly posed Freidberg result is now established on page 111: the kink obstruction is closed within the declared finite- A model, and the bootstrap obstruction has been reduced to an explicit current-drive condition that still awaits an independent affordability calculation. That result does not settle every physical or engineering demand placed on a reactor. The following questions are separate and still require an integrated calculation:

1. a finite-aspect-ratio equilibrium and stability calculation, rather than the large-aspect-ratio kink and Troyon closures;
2. a finite-aspect-ratio neoclassical bootstrap and current-drive calculation, with explicit profiles and collisionality;
3. a transport and confinement basis applicable to the intended ST regime;
4. a radial centre-stack design that makes the bracket in Eq. (190) positive with credible stress, current density, shielding, cooling, protection, and maintenance margins;
5. a divertor, first-wall, blanket, and tritium-breeding calculation. This includes the exhaust problem Freidberg's four ratios intentionally did not close.

The centre-stack calculation begins next. Its task is not to reopen the two Freidberg ratios; one has been passed and the other has been reduced to a stated condition. Its task is to measure the exact geometric and mechanical debt created by the low- A solution. Only then can HTS, reduced shielding, larger radius, alternative current initiation, or modular maintenance be discussed as quantified responses rather than wishful labels.

116. The central column cannot evade Ampere's law

The toroidal field outside an axis-enclosing current path obeys

$$B_\varphi(R) = \frac{\mu_0 NI}{2\pi R}.$$

Evaluate it at the plasma centre:

$$NI = \frac{2\pi R_0 B_0}{\mu_0}. \quad (191)$$

This is a striking centre-stack fact. The required TF ampere-turns are set by the desired plasma product $R_0 B_0$, not by the small radius of the central column. Moving the current path inward raises the local field so that RB_φ remains constant; it does not make the required enclosed current disappear.

Let $J_{\text{wp,cs}}$ be the allowed winding-pack current density after conductor, stabilizer, insulation, cooling, joints where relevant, and allocated structural fraction are counted honestly. The current-carrying cross-sectional area has the necessary lower bound

$$A_{\text{wp,cs}} \geq \frac{NI}{J_{\text{wp,cs}}} = \frac{2\pi R_0 B_0}{\mu_0 J_{\text{wp,cs}}}. \quad (192)$$

It is a lower bound because a real column also needs support and non-conductor functions. The equation makes the danger of vague compactness claims visible: low aspect ratio shrinks the available column, while the TF current-carrying area is controlled by the plasma field target.

117. A first necessary radius inequality for the centre stack

Even in the unrealistically generous case that the entire circular column were available to the TF winding pack,

$$\pi R_{\text{cs}}^2 \geq A_{\text{wp,cs}}.$$

Substitute Eq. (192):

$$R_{\text{cs}} \geq \left(\frac{2R_0 B_0}{\mu_0 J_{\text{wp,cs}}} \right)^{1/2}. \quad (193)$$

This is not a design formula. It is a necessary condition with every real allocation set to zero except current-carrying area. A credible design needs a stronger inequality,

$$\pi R_{\text{cs}}^2 \geq A_{\text{wp,cs}} + A_{\text{str}} + A_{\text{ins/cool}} + A_{\text{sol}} + A_{\text{margin}}, \quad (194)$$

where A_{str} carries magnetic load, $A_{\text{ins/cool}}$ provides electrical and thermal function, A_{sol} is any central-solenoid allocation, and the final term represents manufacturing, protection, and operational margin.

Combine Eq. (189) with Eq. (194). Lowering A reduces the left-side geometric radius, while the right side is set by field, materials, and operating functions. The centre-stack problem is therefore a geometric inequality test. It can fail before we ask whether the plasma is stable.

Design consequence. A proposed ST must publish a centre-stack area ledger, not merely an aspect ratio and a peak field. Without the conductor, structure, shielding, and start-up allocations, there is no check that the proposed central column exists.

118. Higher central field helps the plasma only by increasing mechanical demand

At the centre-stack radius, the local magnetic-pressure scale is

$$p_{B,cs} = \frac{B_{cs}^2}{2\mu_0}.$$

If a representative structural load path of area A_{load} is carried by material limited to σ_{allow} , force balance requires

$$A_{\text{str}} \gtrsim \frac{p_{B,cs}}{\sigma_{\text{allow}}} A_{\text{load}}. \quad (195)$$

The detailed geometry determines A_{load} and the actual stress distribution; the quadratic field dependence is already inescapable. As low A reduces R_{cs} , Eq. (190) pushes B_{cs} upward for a fixed B_0 . The winding pack and its supporting structure then compete for the same shrinking area.

HTS can improve the conductor part of the ledger by providing useful current margin at higher field. It does not make the structural fraction vanish. In fact, if a design uses the higher field that HTS enables, Eq. (195) can require more support. That is why a high-field ST must report both conductor performance and a credible mechanical case.

Model and limits. Equation (195) is a lower-bound scaling, not a finite-element stress certification. Real centre columns need three-dimensional load paths, strain limits for the superconductor, cyclic-fatigue analysis, joints and terminations where used, electrical insulation, quench or fault protection, and neutron-degraded material properties.

119. The transformer also loses flux swing as the column shrinks

The transformer argument from page 20 gives a second, independent centre-stack constraint. A solenoid of effective radius R_{sol} changes linked magnetic flux by the scale

$$\Delta\Psi_{\text{sol}} \sim \pi R_{\text{sol}}^2 \Delta B_{\text{sol}}. \quad (196)$$

Faraday's law converts this flux swing into the volt-seconds available to initiate and ramp plasma current. Since R_{sol} must fit inside the same central column, $R_{\text{sol}} < R_{\text{cs}}$, the available inductive flux falls at least quadratically as the column becomes narrow.

This fact connects directly to the Freidberg current ledger. A pulsed ST cannot claim steady operation simply because it can make an initial plasma current; it needs enough flux swing for its chosen pulse or a non-inductive start-up and sustainment route. Conversely, a steady-state ST with high bootstrap fraction has a second incentive to develop non-inductive current initiation: it can release central-column area otherwise reserved for a large solenoid.

The choice is an engineering-plasma co-design decision. Removing solenoid allocation from Eq. (194) may help the radial ledger, but transfers an obligation to radio-frequency, neutral-beam, helicity-injection, or other start-up/current-drive systems. It cannot be deleted from the thesis by a diagram.

120. The central-stack test is now ready to be made quantitative

The equations above turn the phrase centre-stack problem into five checkable questions:

1. What B_0 , R_0 , and A are demanded by the plasma mission?
2. Does Eq. (190) leave positive inboard radius after the required nuclear build b ?
3. Is πR_{cs}^2 sufficient for the full area ledger in Eq. (194)?
4. Can the structural allocation withstand the field load within stress and superconducting strain limits?
5. If solenoid area is reduced, what explicit non-inductive start-up and current maintenance system replaces its flux swing?

The next step is not to assign convenient answers. We will define a transparent reference ST radial ledger and vary its field, radius, shielding, conductor current density, and solenoid allocation. That will identify whether an HTS-enabled central stack can close geometrically, where it fails, and which missing technologies would have to carry the remaining burden.

121. A published ST study shows why the ledger must be literal

It is useful to test the preceding geometry against a published allocation, provided we do not mistake one study's choices for universal constants. Menard *et al.* examined an illustrative FNSF configuration with $A = 2$, $R_0 = 1.87$ m, and a 30 cm centre-column radius. Their stated inboard nuclear protection included approximately 60 cm of water-cooled tungsten-carbide shielding; the corresponding radial picture also includes a 2 cm first wall. Their magnet assessment used a 70 MA m^{-2} winding-pack current-density assumption and imposed both support-stress and superconducting-strain constraints.⁹

The first calculation needs no plasma model. At $A = 2$,

$$a = \frac{R_0}{A} = 0.935 \text{ m}, \quad R_0 - a = 0.935 \text{ m}.$$

Temporarily subtract only the stated shield and first-wall thickness:

$$0.935 - 0.600 - 0.020 = 0.315 \text{ m}. \quad (197)$$

This rough geometry check is close to the study's 0.30 m centre-column radius. It is not an argument that 15 mm is spare room: it says that even before every cable, support, coolant channel, tolerance, and current-initiation function is drawn, the low- A inboard budget is already measured in centimetres.

Design consequence. The centre stack is not an abstract penalty applied after plasma optimization. A realistic neutron shield can consume most of the inboard radius in an $A \simeq 2$ machine. A claimed compact ST must display its own radial ledger with the same honesty.

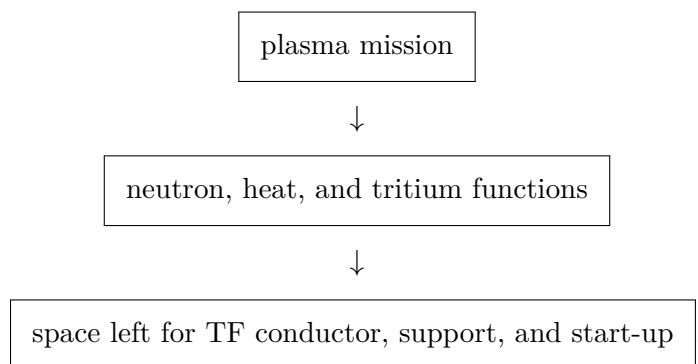
⁹Menard *et al.*, cited on page 105, Secs. 2.1 and 5.1, especially their Figs. 46–47. The numbers in this paragraph are inputs and results of that particular FNSF study, not universal requirements.

122. Why a D–T centre stack begins with shielding, not conductor

The inboard build b in Eq. (189) is not a decorative engineering margin. D–T fusion emits 14.1 MeV neutrons, which are uncharged. Magnetic confinement does not bend them. They stream into the first wall, blanket, shield, structural material, and magnets; a reactor must absorb their heat, breed tritium from lithium, and keep radiation damage to components within their allowed lifetime.

That is why the 60 cm shield in the illustrative calculation on page 121 cannot simply be traded for a better magnet. It was a neutron-protection choice for the study’s mission and material assumptions. Changing it demands a new neutronics, heating, lifetime, and tritium-breeding calculation. The same study found that at low aspect ratio the inboard side can also be important to a tritium-breeding solution.¹⁰

This gives the radial order of operations:



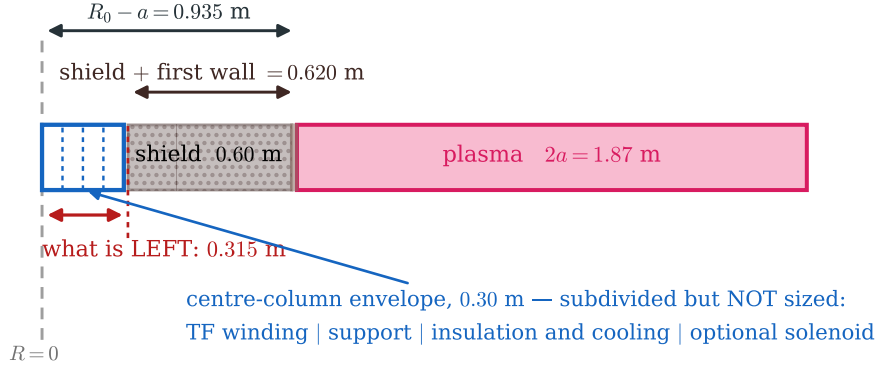
Reversing that order—choosing a desired tiny centre column first and assigning the nuclear functions whatever space survives—would be the same category error as assuming a finished tokamak at page 1. The nuclear build is a design demand that helps reveal the permitted machine shape.

PLACE IMAGE HERE — Cumulative ST radial ledger: a neutron first, magnet second construction. Draw one clean outboard-midplane to inboard-midplane cross-section for the illustrative $A = 2$, $R_0 = 1.87$ m state. On the inboard side add components in causal order: plasma edge, first wall, approximately 0.60 m shielding as a study-specific allocation, then the approximately 0.30 m central-column envelope. Inside that envelope use unfilled subdivisions for TF winding, support, insulation/cooling, and optional solenoid—do not assert their final sizes. Include the arithmetic of Eq. (197) and a clear illustrative published-allocation, not-universal-reactor-template note.

¹⁰Menard *et al.*, cited on page 105, discuss the inboard shielding and breeding allocation in their Sec. 5.3. Their conclusion is configuration-specific; the general point here is only that breeding and shielding claim radial space before the magnet is sized.

THE ST RADIAL LEDGER — neutron first, magnet second ($A = 2$, $R_0 = 1.87$ m)

one illustrative published allocation, laid out in the order the physics forces



the geometry check, evaluated

$$a = R_0 / A = 1.87 / 2 = 0.935 \text{ m}$$

$$R_0 - a = 0.935 \text{ m}$$

$$0.935 - 0.600 - 0.020 = 0.315 \text{ m}$$

against the study's stated 0.30 m centre column:
a difference of 15 mm.

That 15 mm is NOT spare room. It is the residue of a check that has not yet paid for one cable, one support, one coolant channel, one tolerance, or the current-initiation function. The low- A inboard budget is already measured in centimetres before engineering begins.

ILLUSTRATIVE PUBLISHED ALLOCATION, NOT A UNIVERSAL REACTOR TEMPLATE. The 0.60 m shield, the 20 mm first wall and the 0.30 m column are one study's choices, for one mission and one set of material assumptions. Change the mission and they change. What does NOT change is the ORDER: the neutron shield is sized by neutronics, not by what the magnet would prefer, so the magnet is designed into whatever is left. That ordering is the section's claim, and it is the only claim this figure makes.

Explanatory sketch; midplane cut drawn to scale in R . Subdivisions inside the centre-column envelope are indicative only.

Figure 43: The published illustrative allocation, laid out in the order the physics forces: shield first, magnet into what is left. The arithmetic is Eq. (197) evaluated. The 15 mm difference from the study's stated column radius is not spare room — it is the residue of a check that has not yet paid for one cable, support, coolant channel or tolerance.

123. What HTS changes in this ledger

HTS changes a specific term: the credible current density and operating field margin of the TF winding pack. In the necessary bound,

$$A_{\text{wp,cs}} \geq \frac{2\pi R_0 B_0}{\mu_0 J_{\text{wp,cs}}},$$

raising a credibly usable $J_{\text{wp,cs}}$ reduces the *minimum* current-carrying area for a prescribed $R_0 B_0$. It can therefore create room that a conventional conductor cannot. In the field relation

$$B_{\text{cs}} = \frac{B_0 R_0}{R_{\text{cs}}},$$

HTS can also make a larger central field technically plausible without immediately crossing the conductor's critical surface.

Those are real levers, but neither equation says that the whole right side of Eq. (194) scales down as $1/J_{\text{wp,cs}}$. The support, insulation, cooling, protection, radiation, and manufacturing terms remain. Moreover, raising B_{cs} raises $p_{B,\text{cs}} \propto B_{\text{cs}}^2$, which can enlarge the structural term. A good HTS design may close the equation; HTS alone is not the missing calculation.

The published low- A work gives a concrete architectural direction rather than a shortcut. A later $A = 2$, $R_0 = 3\text{m}$ HTS pilot-plant concept places all superconducting poloidal-field coils outside the TF system, uses continuous HTS TF coils, retains only a small central solenoid, and keeps inboard blanket/shield functions in the radial build.¹¹ Each choice answers one item in the ledger: outside PF coils avoid consuming central volume; a small solenoid makes non-inductive start-up more important; HTS improves the TF field option; blanket and shield remain non-negotiable.

Design consequence. HTS is the enabling material in a co-design, not an exemption from the co-design. Its success must be reported as a closed area, stress, radiation, and start-up ledger at the chosen R_0 , A , and B_0 .

¹¹Menard *et al.*, cited on page 105, Sec. 5.4 and Fig. 53. It is a conditional systems study, not a demonstration that every one of its performance targets has been experimentally achieved.

124. What remains after the Freidberg answer

The causal argument has reached a useful, deliberately narrow conclusion. The finite- A analysis in Sections 106–111 passed the stated kink screen and identified the independent condition still required for bootstrap closure. The earlier first-principles geometry explains why that answer was possible:

1. Eq. (183) changes the current–safety-factor geometry in the favorable direction for the kink/current coupling.
2. Eq. (180) strengthens the trapping physics that underlies a route to bootstrap current.

HTS then supplies a plausible material lever for the adverse inboard field and winding-area problem created by the same geometry. Published integrated ST studies show that these ingredients can be put into a serious systems calculation.

That establishes the result relevant here: a specified finite- A ST model closes the kink constraint that failed in Freidberg’s conventional baseline and reduces the bootstrap constraint to an explicit, quantified current-drive condition ($I_{\text{CD,allow}} \gtrsim 2.6$ MA, Section 111). Raising H in Freidberg’s conventional family could close the kink and bootstrap tests but still violated the Troyon bound; it did not close all four screens. It does not establish a theorem that every ST or every other reactor constraint is closed; exhaust, blanket, tritium, radiation, maintenance, and cost remain their own questions.

The next calculation must therefore have two ledgers, evaluated together:

finite- A plasma ledger	centre-stack reactor ledger
q, β_N , equilibrium/stability	$b, R_{\text{cs}}, A_{\text{wp}}, A_{\text{str}}$
bootstrap and driven current	shield, breeding, stress, start-up
transport and exhaust	HTS margin, protection, maintenance

The next section makes the centre-stack ledger into an explicit, auditable parameter scan. It does not relabel that separate engineering question as a failed Freidberg plasma constraint.

125. A dimensionless necessary test for a central TF winding

For a first scan, reserve the inboard nuclear build b before allocating the central column. The outer radius of the column envelope is then

$$R_{\text{env}} = R_0 \left(1 - \frac{1}{A} \right) - b. \quad (198)$$

This is deliberately more generous than a finished design: it gives every remaining square millimetre to the central column before deciding how much is conductor, structure, cooling, or solenoid.

Let f_{wp} be the fraction of the envelope's circular cross-sectional area that the designer has explicitly allocated to the TF winding pack. It is an accounting input, not a material constant. Equation (192) then requires

$$\frac{2\pi R_0 B_0}{\mu_0 J_{\text{wp,cs}}} \leq f_{\text{wp}} \pi R_{\text{env}}^2.$$

After cancelling π , define the dimensionless winding occupancy

$$\eta_{\text{wp}} \equiv \frac{2R_0 B_0}{\mu_0 J_{\text{wp,cs}} [R_0(1 - 1/A) - b]^2}. \quad (199)$$

The necessary test is simply

$$\eta_{\text{wp}} \leq f_{\text{wp}} < 1. \quad (200)$$

If $\eta_{\text{wp}} > 1$, even a column made entirely of winding pack cannot carry the required TF current. If $f_{\text{wp}} < \eta_{\text{wp}} < 1$, a central column exists only on paper because the current pack has consumed space explicitly promised to the other functions.

Model and limits. This is a necessary electromagnetic-area test. It treats the winding pack as uniformly current carrying and does not certify field quality, stress, strain, quench protection, neutron lifetime, joints, penetrations, or manufacturability. Those omissions make passing the test easier, never harder.

126. Solving the test exposes a lower-aspect-ratio boundary

The test can be solved for the smallest aspect ratio permitted by the chosen area allocation. Take the positive square root of Eq. (200):

$$R_0 \left(1 - \frac{1}{A} \right) - b \geq \left(\frac{2R_0 B_0}{\mu_0 J_{\text{wp,cs}} f_{\text{wp}}} \right)^{1/2}.$$

Divide by R_0 and rearrange:

$$A \geq A_{\text{min,wp}} \equiv \left[1 - \frac{b}{R_0} - \left(\frac{2B_0}{\mu_0 J_{\text{wp,cs}} f_{\text{wp}} R_0} \right)^{1/2} \right]^{-1}. \quad (201)$$

The bracket must be positive. If it is zero or negative, the stated R_0 , B_0 , nuclear build, winding-pack current density, and allocation cannot coexist at *any* aspect ratio in this simplified central-column arrangement.

Every trend in this equation has a physical reading:

- Increasing B_0 increases the required ampere-turns and raises the minimum allowed A .
- Increasing b/R_0 claims the inboard radius for nuclear function and raises the minimum allowed A .
- Increasing credible $J_{\text{wp,cs}}$ or giving the winding pack a larger honest fraction f_{wp} lowers the electromagnetic area demand.
- Increasing R_0 helps twice: the nuclear build occupies a smaller fraction of the radius and the required winding area is less crowded into the column.

The last two points identify exactly what HTS and a less compact plant can improve. They also show why neither word automatically guarantees a viable ST.

127. A transparent teaching scan: field makes the low- A limit move

Use the deliberately declared teaching inputs

$$R_0 = 3.0 \text{ m}, \quad b = 0.65 \text{ m}, \quad J_{\text{wp,cs}} = 70 \text{ MA m}^{-2}, \quad f_{\text{wp}} = 0.40.$$

The current-density scale is comparable to an assumption used in the illustrative Menard study on page 121; the radius, nuclear build, and 40% allocation here are *not* a proposed reactor. They merely make the algebra visible. Substitution in Eq. (201) gives:

B_0 (T)	$A_{\text{min,wp}}$	η_{wp} at $A = 2$
4	1.97	0.378
6	2.24	0.567
8	2.54	0.756

At $A = 2$, the available envelope radius is $R_{\text{env}} = 3(1 - 1/2) - 0.65 = 0.85 \text{ m}$. The first row barely clears the assigned 40% winding allocation. The latter two do not: they require more winding area than has been allocated even before support, insulation, cooling, or a solenoid are tested. The table is reproducible from `centre_stack_ledger.py`; every input is displayed at the top of that file.

This is the exact sense in which HTS is a boost mechanism. If it raises the *credible system-level* $J_{\text{wp,cs}}$ at the relevant field, temperature, strain, radiation, and protection margin, it can lower $A_{\text{min,wp}}$. If the design instead spends the HTS benefit to raise B_0 , the square-root field penalty pushes $A_{\text{min,wp}}$ upward. The net result is a co-design decision, not a one-directional law.

Audit note. Passing the 4 T row does not make it a reactor. The table omits the structure term that grows with central field, the non-winding allocations, the neutron and tritium calculation behind b , and all plasma/stability/exhaust tests. It is intentionally a failure detector, not a feasibility certificate.